FEMB QC Parameter Evaluation - Summary

Board version: FEMB-IO 1826-1L
Boards evaluated: 44 (latest passing test per board; excluded: 00046)
Sections: 9
Generated: 2026-10-16 16:38
Script: `summary.py` (`--rerun` regenerates every section first)

1. Judgement Schemes

Scheme	Used for	PASS	WARN	ALERT	FAIL
7σ	Board-level power, current and LDO readings (2.1, 3.2, 3.3, 4.2, 4.3)	≤ 3σ	3σ – 5σ	5σ – 7σ	> 7σ
5σ	Channel-level noise / pulse (3.1, 3.4, 4.1) and per-ASIC monitoring (5)	≤ 3σ	3σ – 5σ	–	> 5σ

A board's level in a section is the worst level over the section's core parameters (auxiliary parameters such as BIAS are listed but not counted). The overall level is the worst over all sections.

2. Overall Result

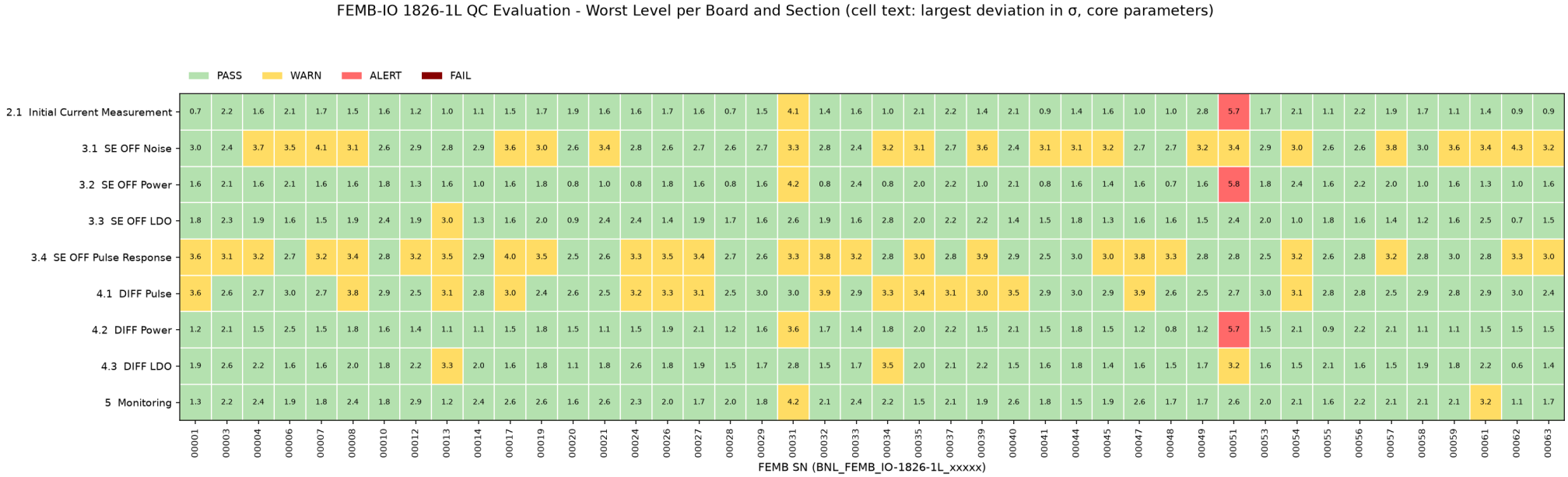
Overall level	Boards
PASS	9
WARN	34
ALERT	1
FAIL	0

Boards at ALERT / FAIL: BNL_FEMB_IO-1826-1L_00051

3. Sections

Section	Title	Scheme	Core params	PASS	WARN	ALERT	FAIL	Report
2.1	Initial Current Measurement	7σ	7	42	1	1	0	Report_2_1_Initial_Power.md
3.1	SE OFF Noise	5σ	2	22	22	–	0	Report_3_1_SE_OFF_Noise.md
3.2	SE OFF Power	7σ	7	42	1	1	0	Report_3_2_SE_OFF_Power.md
3.3	SE OFF LDO	7σ	11	43	1	0	0	Report_3_3_SE_OFF_LDO.md
3.4	SE OFF Pulse Response	5σ	3	20	24	–	0	Report_3_4_SE_OFF_Pulse.md
4.1	DIFF Pulse	5σ	3	29	15	–	0	Report_4_1_DIFF_Pulse.md
4.2	DIFF Power	7σ	7	42	1	1	0	Report_4_2_DIFF_Power.md
4.3	DIFF LDO	7σ	11	41	3	0	0	Report_4_3_DIFF_LDO.md
5	Monitoring	5σ	7	42	2	–	0	Report_5_Monitoring.md

4. Board × Section Overview



5. Boards Requiring Attention

Isolated WARN channels are expected by chance (≈ 15 per channel-level parameter over all boards), so a board is listed here only if at least one of these holds:

- **ALERT / FAIL** in any section;
- **repeated**: the same parameter at the same location (board-level reading, channel or chip) is flagged in more than one section (e.g. 2.1 / 3.2 / 4.2, or SE 3.4 and DIFF 4.1);
- **chip cluster**: ≥ 4 flagged items on one ASIC within one section.

BNL_FEMB_IO-1826-1L_00051 – overall ALERT

Reason: ALERT; repeated: `I_ColdDATA` board

Section	Parameter	Where	Value	Deviation	Level
2.1	`I_ColdDATA`	board	0.275 A	5.7σ	**ALERT**
3.1	`rms`	ch36 (chip 2)	12.73	3.4σ	WARN
3.2	`I_ColdDATA`	board	0.275 A	5.8σ	**ALERT**
4.2	`I_ColdDATA`	board	0.275 A	5.7σ	**ALERT**
4.3	`CDVDDA`	board	1179 mV	3.2σ	WARN

BNL_FEMB_IO-1826-1L_00032 – overall WARN

Reason: repeated: `NPK_mean` ch56, `PPK_mean` ch58, `PPK_mean` ch63; chip cluster: 3.4 chip 3 (4), 4.1 chip 3 (7)

Section	Parameter	Where	Value	Deviation	Level
3.4	NPK_mean	ch48 (chip 3)	4739.00	3.8σ	WARN
3.4	NPK_mean	ch56 (chip 3)	4730.00	3.7σ	WARN
3.4	PPK_mean	ch63 (chip 3)	12449.00	3.2σ	WARN
3.4	PPK_mean	ch58 (chip 3)	12458.00	3.1σ	WARN
4.1	PPK_mean	ch58 (chip 3)	12696.50	3.9σ	WARN
4.1	PPK_mean	ch55 (chip 3)	12718.50	3.5σ	WARN
4.1	NPK_mean	ch56 (chip 3)	4637.50	3.4σ	WARN
4.1	PPK_mean	ch52 (chip 3)	12743.75	3.2σ	WARN
4.1	PPK_mean	ch48 (chip 3)	12755.00	3.1σ	WARN
4.1	PPK_mean	ch49 (chip 3)	12755.50	3.0σ	WARN
4.1	PPK_mean	ch63 (chip 3)	12757.50	3.0σ	WARN

BNL_FEMB_IO-1826-1L_00008 – overall WARN

Reason: repeated: `PPK_mean` ch40; chip cluster: 4.1 chip 2 (6)

Section	Parameter	Where	Value	Deviation	Level
3.1	rms	ch66 (chip 4)	12.61	3.1σ	WARN
3.4	PPK_mean	ch40 (chip 2)	12427.00	3.4σ	WARN
3.4	NPK_mean	ch86 (chip 5)	4145.00	3.1σ	WARN
4.1	PPK_mean	ch40 (chip 2)	12700.75	3.8σ	WARN
4.1	PPK_mean	ch33 (chip 2)	12729.00	3.4σ	WARN
4.1	BL_mean	ch49 (chip 3)	8571.13	3.4σ	WARN
4.1	PPK_mean	ch34 (chip 2)	12737.00	3.3σ	WARN
4.1	PPK_mean	ch43 (chip 2)	12738.00	3.3σ	WARN
4.1	PPK_mean	ch39 (chip 2)	12741.75	3.2σ	WARN
4.1	PPK_mean	ch41 (chip 2)	12756.00	3.0σ	WARN

BNL_FEMB_IO-1826-1L_00031 – overall WARN

Reason: repeated: `I_ColdADC` board, `Total_Power` board

Section	Parameter	Where	Value	Deviation	Level
2.1	Total_Power	board	7.915 W	4.1σ	WARN
2.1	I_ColdADC	board	1.74 A	4.1σ	WARN
3.1	ped	ch64 (chip 4)	1253.36	3.3σ	WARN
3.2	I_ColdADC	board	1.74 A	4.2σ	WARN

Section	Parameter	Where	Value	Deviation	Level
3.2	Total_Power	board	7.862 W	4.1σ	WARN
3.4	NPK_mean	ch8 (chip 0)	4697.00	3.3σ	WARN
3.4	NPK_mean	ch5 (chip 0)	4682.00	3.2σ	WARN
3.4	BL_mean	ch5 (chip 0)	8794.00	3.1σ	WARN
4.2	I_ColdADC	board	1.709 A	3.6σ	WARN
4.2	Total_Power	board	8.463 W	3.1σ	WARN
5	ADC_VREFP	chip1	1922.80	4.2σ	WARN
5	ADC_VCMO	chip1	1213.70	3.1σ	WARN

BNL_FEMB_IO-1826-1L_00017 – overall WARN

Reason: repeated: PPK_mean ch110

Section	Parameter	Where	Value	Deviation	Level
3.1	rms	ch15 (chip 0)	12.84	3.6σ	WARN
3.1	ped	ch32 (chip 2)	1240.36	3.1σ	WARN
3.4	PPK_mean	ch96 (chip 6)	12375.00	4.0σ	WARN
3.4	PPK_mean	ch110 (chip 6)	12448.00	3.2σ	WARN
3.4	BL_mean	ch36 (chip 2)	8795.00	3.1σ	WARN
3.4	NPK_mean	ch107 (chip 6)	4672.00	3.0σ	WARN
4.1	PPK_mean	ch110 (chip 6)	12758.50	3.0σ	WARN

BNL_FEMB_IO-1826-1L_00024 – overall WARN

Reason: repeated: PPK_mean ch68

Section	Parameter	Where	Value	Deviation	Level
3.4	BL_mean	ch68 (chip 4)	8812.00	3.3σ	WARN
3.4	BL_mean	ch57 (chip 3)	8349.00	3.3σ	WARN
3.4	PPK_mean	ch68 (chip 4)	13039.00	3.1σ	WARN
3.4	PPK_mean	ch76 (chip 4)	13037.00	3.1σ	WARN
4.1	NPK_mean	ch40 (chip 2)	4624.50	3.2σ	WARN
4.1	PPK_mean	ch68 (chip 4)	13211.75	3.1σ	WARN

BNL_FEMB_IO-1826-1L_00001 – overall WARN

Reason: repeated: BL_mean ch38

Section	Parameter	Where	Value	Deviation	Level
3.4	BL_mean	ch38 (chip 2)	8330.00	3.6σ	WARN
4.1	BL_mean	ch38 (chip 2)	8559.92	3.6σ	WARN

Section	Parameter	Where	Value	Deviation	Level
4.1	BL_mean	ch49 (chip 3)	8569.29	3.4σ	WARN
4.1	PPK_mean	ch38 (chip 2)	12742.00	3.2σ	WARN

BNL_FEMB_IO-1826-1L_00013 – overall WARN

Reason: repeated: FELVDDP board

Section	Parameter	Where	Value	Deviation	Level
3.3	FELVDDP	board	1783 mV	3.0σ	WARN
3.4	PPK_mean	ch112 (chip 7)	12414.00	3.5σ	WARN
4.1	PPK_mean	ch88 (chip 5)	12752.50	3.1σ	WARN
4.3	FELVDDP	board	1656 mV	3.3σ	WARN

BNL_FEMB_IO-1826-1L_00047 – overall WARN

Reason: repeated: BL_mean ch11

Section	Parameter	Where	Value	Deviation	Level
3.4	BL_mean	ch11 (chip 0)	8848.00	3.8σ	WARN
3.4	PPK_mean	ch11 (chip 0)	13033.00	3.0σ	WARN
4.1	BL_mean	ch11 (chip 0)	8977.40	3.9σ	WARN

6. Per-Board Matrix

Cell: section level (largest core-parameter deviation).

FEMB SN	2.1	3.1	3.2	3.3	3.4	4.1	4.2	4.3	5	Overall
00001	PASS	PASS	PASS	PASS	WARN (3.6σ)	WARN (3.6σ)	PASS	PASS	PASS	WARN
00003	PASS	PASS	PASS	PASS	WARN (3.1σ)	PASS	PASS	PASS	PASS	WARN
00004	PASS	WARN (3.7σ)	PASS	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	WARN
00006	PASS	WARN (3.5σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN
00007	PASS	WARN (4.1σ)	PASS	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	WARN
00008	PASS	WARN (3.1σ)	PASS	PASS	WARN (3.4σ)	WARN (3.8σ)	PASS	PASS	PASS	WARN
00010	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00012	PASS	PASS	PASS	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	WARN
00013	PASS	PASS	PASS	WARN (3.0σ)	WARN (3.5σ)	WARN (3.1σ)	PASS	WARN (3.3σ)	PASS	WARN
00014	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00017	PASS	WARN (3.6σ)	PASS	PASS	WARN (4.0σ)	WARN (3.0σ)	PASS	PASS	PASS	WARN
00019	PASS	WARN (3.0σ)	PASS	PASS	WARN (3.5σ)	PASS	PASS	PASS	PASS	WARN
00020	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00021	PASS	WARN (3.4σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN

FEMB SN	2.1	3.1	3.2	3.3	3.4	4.1	4.2	4.3	5	Overall
00024	PASS	PASS	PASS	PASS	WARN (3.3σ)	WARN (3.2σ)	PASS	PASS	PASS	WARN
00026	PASS	PASS	PASS	PASS	WARN (3.5σ)	WARN (3.3σ)	PASS	PASS	PASS	WARN
00027	PASS	PASS	PASS	PASS	WARN (3.4σ)	WARN (3.1σ)	PASS	PASS	PASS	WARN
00028	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00029	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00031	WARN (4.1σ)	WARN (3.3σ)	WARN (4.2σ)	PASS	WARN (3.3σ)	PASS	WARN (3.6σ)	PASS	WARN (4.2σ)	WARN
00032	PASS	PASS	PASS	PASS	WARN (3.8σ)	WARN (3.9σ)	PASS	PASS	PASS	WARN
00033	PASS	PASS	PASS	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	WARN
00034	PASS	WARN (3.2σ)	PASS	PASS	PASS	WARN (3.3σ)	PASS	WARN (3.5σ)	PASS	WARN
00035	PASS	WARN (3.1σ)	PASS	PASS	WARN (3.0σ)	WARN (3.4σ)	PASS	PASS	PASS	WARN
00037	PASS	PASS	PASS	PASS	PASS	WARN (3.1σ)	PASS	PASS	PASS	WARN
00039	PASS	WARN (3.6σ)	PASS	PASS	WARN (3.9σ)	WARN (3.0σ)	PASS	PASS	PASS	WARN
00040	PASS	PASS	PASS	PASS	PASS	WARN (3.5σ)	PASS	PASS	PASS	WARN
00041	PASS	WARN (3.1σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN
00044	PASS	WARN (3.1σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN
00045	PASS	WARN (3.2σ)	PASS	PASS	WARN (3.0σ)	PASS	PASS	PASS	PASS	WARN
00047	PASS	PASS	PASS	PASS	WARN (3.8σ)	WARN (3.9σ)	PASS	PASS	PASS	WARN
00048	PASS	PASS	PASS	PASS	WARN (3.3σ)	PASS	PASS	PASS	PASS	WARN
00049	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN
00051	**ALERT (5.7σ)**	WARN (3.4σ)	**ALERT (5.8σ)**	PASS	PASS	PASS	**ALERT (5.7σ)**	WARN (3.2σ)	PASS	**ALERT**
00053	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00054	PASS	WARN (3.0σ)	PASS	PASS	WARN (3.2σ)	WARN (3.1σ)	PASS	PASS	PASS	WARN
00055	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00056	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00057	PASS	WARN (3.8σ)	PASS	PASS	WARN (3.2σ)	PASS	PASS	PASS	PASS	WARN
00058	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
00059	PASS	WARN (3.6σ)	PASS	PASS	PASS	PASS	PASS	PASS	PASS	WARN
00061	PASS	WARN (3.4σ)	PASS	PASS	PASS	PASS	PASS	PASS	WARN (3.2σ)	WARN
00062	PASS	WARN (4.3σ)	PASS	PASS	WARN (3.3σ)	PASS	PASS	PASS	PASS	WARN
00063	PASS	WARN (3.2σ)	PASS	PASS	WARN (3.0σ)	PASS	PASS	PASS	PASS	WARN

7. All Flagged Items

136 items beyond 3σ across all sections (`aux` = auxiliary parameter, not counted in the board level).

FEMB SN	Section	Parameter	Where	Value	Deviation	Level
00001	3.4	`BL_mean`	ch38 (chip 2)	8330.00	3.6σ	WARN
00001	4.1	`BL_mean`	ch38 (chip 2)	8559.92	3.6σ	WARN
00001	4.1	`BL_mean`	ch49 (chip 3)	8569.29	3.4σ	WARN

FEMB SN	Section	Parameter	Where	Value	Deviation	Level
00001	4.1	PPK_mean	ch38 (chip 2)	12742.00	3.2σ	WARN
00003	3.4	BL_mean	ch13 (chip 0)	8798.00	3.1σ	WARN
00004	3.1	ped	ch24 (chip 1)	1297.25	3.7σ	WARN
00004	3.1	ped	ch47 (chip 2)	1233.97	3.1σ	WARN
00004	3.1	ped	ch40 (chip 2)	1226.81	3.0σ	WARN
00004	3.4	NPK_mean	ch47 (chip 2)	4686.00	3.2σ	WARN
00006	3.1	rms	ch18 (chip 1)	12.78	3.5σ	WARN
00007	3.1	rms	ch15 (chip 0)	13.04	4.1σ	WARN
00007	3.1	ped	ch120 (chip 7)	1298.34	3.7σ	WARN
00007	3.4	PPK_mean	ch72 (chip 4)	12441.00	3.2σ	WARN
00007	3.4	PPK_mean	ch56 (chip 3)	12453.00	3.1σ	WARN
00008	3.1	rms	ch66 (chip 4)	12.61	3.1σ	WARN
00008	3.4	PPK_mean	ch40 (chip 2)	12427.00	3.4σ	WARN
00008	3.4	NPK_mean	ch86 (chip 5)	4145.00	3.1σ	WARN
00008	4.1	PPK_mean	ch40 (chip 2)	12700.75	3.8σ	WARN
00008	4.1	PPK_mean	ch33 (chip 2)	12729.00	3.4σ	WARN
00008	4.1	BL_mean	ch49 (chip 3)	8571.13	3.4σ	WARN
00008	4.1	PPK_mean	ch34 (chip 2)	12737.00	3.3σ	WARN
00008	4.1	PPK_mean	ch43 (chip 2)	12738.00	3.3σ	WARN
00008	4.1	PPK_mean	ch39 (chip 2)	12741.75	3.2σ	WARN
00008	4.1	PPK_mean	ch41 (chip 2)	12756.00	3.0σ	WARN
00012	3.4	BL_mean	ch104 (chip 6)	8353.00	3.2σ	WARN
00012	3.4	BL_mean	ch110 (chip 6)	8353.00	3.2σ	WARN
00012	3.4	NPK_mean	ch72 (chip 4)	4681.00	3.2σ	WARN
00013	3.3	FELVDDP	board	1783 mV	3.0σ	WARN
00013	3.4	PPK_mean	ch112 (chip 7)	12414.00	3.5σ	WARN
00013	4.1	PPK_mean	ch88 (chip 5)	12752.50	3.1σ	WARN
00013	4.3	FELVDDP	board	1656 mV	3.3σ	WARN
00017	3.1	rms	ch15 (chip 0)	12.84	3.6σ	WARN
00017	3.1	ped	ch32 (chip 2)	1240.36	3.1σ	WARN
00017	3.4	PPK_mean	ch96 (chip 6)	12375.00	4.0σ	WARN
00017	3.4	PPK_mean	ch110 (chip 6)	12448.00	3.2σ	WARN
00017	3.4	BL_mean	ch36 (chip 2)	8795.00	3.1σ	WARN
00017	3.4	NPK_mean	ch107 (chip 6)	4672.00	3.0σ	WARN
00017	4.1	PPK_mean	ch110 (chip 6)	12758.50	3.0σ	WARN
00019	3.1	rms	ch13 (chip 0)	12.56	3.0σ	WARN
00019	3.4	BL_mean	ch33 (chip 2)	8337.00	3.5σ	WARN
00019	3.4	NPK_mean	ch33 (chip 2)	4143.00	3.1σ	WARN
00021	3.1	ped	ch48 (chip 3)	1263.46	3.4σ	WARN

FEMB SN	Section	Parameter	Where	Value	Deviation	Level
00024	3.4	BL_mean	ch68 (chip 4)	8812.00	3.3σ	WARN
00024	3.4	BL_mean	ch57 (chip 3)	8349.00	3.3σ	WARN
00024	3.4	PPK_mean	ch68 (chip 4)	13039.00	3.1σ	WARN
00024	3.4	PPK_mean	ch76 (chip 4)	13037.00	3.1σ	WARN
00024	4.1	NPK_mean	ch40 (chip 2)	4624.50	3.2σ	WARN
00024	4.1	PPK_mean	ch68 (chip 4)	13211.75	3.1σ	WARN
00026	3.4	PPK_mean	ch44 (chip 2)	13073.00	3.5σ	WARN
00026	4.1	NPK_mean	ch39 (chip 2)	4163.25	3.3σ	WARN
00027	3.4	BL_mean	ch60 (chip 3)	8817.00	3.4σ	WARN
00027	3.4	PPK_mean	ch60 (chip 3)	13041.00	3.1σ	WARN
00027	3.4	BL_mean	ch68 (chip 4)	8364.00	3.1σ	WARN
00027	3.4	BL_mean	ch111 (chip 6)	8792.00	3.0σ	WARN
00027	4.1	PPK_mean	ch7 (chip 0)	13213.50	3.1σ	WARN
00031	2.1	Total_Power	board	7.915 W	4.1σ	WARN
00031	2.1	I_ColdADC	board	1.74 A	4.1σ	WARN
00031	3.1	ped	ch64 (chip 4)	1253.36	3.3σ	WARN
00031	3.2	I_ColdADC	board	1.74 A	4.2σ	WARN
00031	3.2	Total_Power	board	7.862 W	4.1σ	WARN
00031	3.4	NPK_mean	ch8 (chip 0)	4697.00	3.3σ	WARN
00031	3.4	NPK_mean	ch5 (chip 0)	4682.00	3.2σ	WARN
00031	3.4	BL_mean	ch5 (chip 0)	8794.00	3.1σ	WARN
00031	4.2	I_ColdADC	board	1.709 A	3.6σ	WARN
00031	4.2	Total_Power	board	8.463 W	3.1σ	WARN
00031	5	ADC_VREFP	chip1	1922.80	4.2σ	WARN
00031	5	ADC_VCMO	chip1	1213.70	3.1σ	WARN
00032	3.4	NPK_mean	ch48 (chip 3)	4739.00	3.8σ	WARN
00032	3.4	NPK_mean	ch56 (chip 3)	4730.00	3.7σ	WARN
00032	3.4	PPK_mean	ch63 (chip 3)	12449.00	3.2σ	WARN
00032	3.4	PPK_mean	ch58 (chip 3)	12458.00	3.1σ	WARN
00032	4.1	PPK_mean	ch58 (chip 3)	12696.50	3.9σ	WARN
00032	4.1	PPK_mean	ch55 (chip 3)	12718.50	3.5σ	WARN
00032	4.1	NPK_mean	ch56 (chip 3)	4637.50	3.4σ	WARN
00032	4.1	PPK_mean	ch52 (chip 3)	12743.75	3.2σ	WARN
00032	4.1	PPK_mean	ch48 (chip 3)	12755.00	3.1σ	WARN
00032	4.1	PPK_mean	ch49 (chip 3)	12755.50	3.0σ	WARN
00032	4.1	PPK_mean	ch63 (chip 3)	12757.50	3.0σ	WARN
00033	3.4	BL_mean	ch29 (chip 1)	8804.00	3.2σ	WARN
00034	3.1	ped	ch24 (chip 1)	1241.94	3.2σ	WARN
00034	3.1	ped	ch8 (chip 0)	1241.47	3.2σ	WARN

FEMB SN	Section	Parameter	Where	Value	Deviation	Level
00034	4.1	BL_mean	ch90 (chip 5)	8575.50	3.3σ	WARN
00034	4.1	BL_mean	ch79 (chip 4)	8928.45	3.0σ	WARN
00034	4.3	FERVDDP	board	1657 mV	3.5σ	WARN
00035	3.1	ped	ch88 (chip 5)	1232.32	3.1σ	WARN
00035	3.4	NPK_mean	ch88 (chip 5)	4669.00	3.0σ	WARN
00035	4.1	PPK_mean	ch68 (chip 4)	13231.50	3.4σ	WARN
00037	4.1	PPK_mean	ch94 (chip 5)	12753.50	3.1σ	WARN
00039	3.1	ped	ch109 (chip 6)	1289.33	3.6σ	WARN
00039	3.1	rms	ch5 (chip 0)	12.63	3.1σ	WARN
00039	3.4	NPK_mean	ch109 (chip 6)	4745.00	3.9σ	WARN
00039	3.4	BL_mean	ch109 (chip 6)	8816.00	3.4σ	WARN
00039	4.1	NPK_mean	ch112 (chip 7)	4607.75	3.0σ	WARN
00040	4.1	PPK_mean	ch118 (chip 7)	12726.00	3.5σ	WARN
00040	4.1	NPK_mean	ch49 (chip 3)	4158.25	3.4σ	WARN
00040	4.1	BL_mean	ch118 (chip 7)	8577.80	3.3σ	WARN
00040	4.1	BL_mean	ch21 (chip 1)	8933.56	3.1σ	WARN
00041	3.1	rms	ch72 (chip 4)	9.78	3.1σ	WARN
00044	3.1	ped	ch72 (chip 4)	1238.61	3.1σ	WARN
00045	3.1	ped	ch86 (chip 5)	578.45	3.2σ	WARN
00045	3.4	NPK_mean	ch40 (chip 2)	4669.00	3.0σ	WARN
00047	3.4	BL_mean	ch11 (chip 0)	8848.00	3.8σ	WARN
00047	3.4	PPK_mean	ch11 (chip 0)	13033.00	3.0σ	WARN
00047	4.1	BL_mean	ch11 (chip 0)	8977.40	3.9σ	WARN
00048	3.4	BL_mean	ch1 (chip 0)	8347.00	3.3σ	WARN
00048	3.4	BL_mean	ch6 (chip 0)	8363.00	3.1σ	WARN
00048	3.4	PPK_mean	ch116 (chip 7)	13036.00	3.1σ	WARN
00049	3.1	ped	ch120 (chip 7)	1247.25	3.2σ	WARN
00049	3.1	rms	ch127 (chip 7)	9.83	3.0σ	WARN
00051	2.1	I_ColdDATA	board	0.275 A	5.7σ	**ALERT**
00051	3.1	rms	ch36 (chip 2)	12.73	3.4σ	WARN
00051	3.2	I_ColdDATA	board	0.275 A	5.8σ	**ALERT**
00051	4.2	I_ColdDATA	board	0.275 A	5.7σ	**ALERT**
00051	4.3	CDVDDA	board	1179 mV	3.2σ	WARN
00054	3.1	rms	ch20 (chip 1)	12.57	3.0σ	WARN
00054	3.4	BL_mean	ch38 (chip 2)	8354.00	3.2σ	WARN
00054	4.1	BL_mean	ch54 (chip 3)	8586.76	3.1σ	WARN
00054	4.1	NPK_mean	ch66 (chip 4)	4609.75	3.0σ	WARN
00057	3.1	ped	ch56 (chip 3)	1305.49	3.8σ	WARN
00057	3.1	ped	ch40 (chip 2)	1269.79	3.4σ	WARN

FEMB SN	Section	Parameter	Where	Value	Deviation	Level
00057	3.1	`ped`	ch126 (chip 7)	576.84	3.2σ	WARN
00057	3.4	`NPK_mean`	ch120 (chip 7)	4689.00	3.2σ	WARN
00059	3.1	`rms`	ch13 (chip 0)	12.82	3.6σ	WARN
00059	3.1	`rms`	ch44 (chip 2)	12.63	3.1σ	WARN
00059	3.1	`rms`	ch20 (chip 1)	12.62	3.1σ	WARN
00059	3.1	`rms`	ch15 (chip 0)	12.60	3.1σ	WARN
00059	3.1	`rms`	ch12 (chip 0)	12.57	3.0σ	WARN
00061	3.1	`ped`	ch104 (chip 6)	1264.56	3.4σ	WARN
00061	3.1	`ped`	ch88 (chip 5)	1250.73	3.2σ	WARN
00061	3.1	`ped`	ch80 (chip 5)	1237.22	3.1σ	WARN
00061	5	`FE T`	chip1	882.30	3.2σ	WARN
00062	3.1	`rms`	ch88 (chip 5)	9.24	4.3σ	WARN
00062	3.4	`BL_mean`	ch38 (chip 2)	8349.00	3.3σ	WARN
00063	3.1	`rms`	ch113 (chip 7)	12.64	3.2σ	WARN
00063	3.4	`NPK_mean`	ch104 (chip 6)	4671.00	3.0σ	WARN
00063	3.4	`PPK_mean`	ch24 (chip 1)	12464.00	3.0σ	WARN

Appendix: Output Files

- `output_summary/summary_heatmap.png` – board × section worst level
- `output_summary/summary_boards.csv` – per-board level in each section and overall
- `secX_Y_*/Report_*.md` – detailed section reports with distribution plots

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FEMB QC Parameter Evaluation - Report Section 2.1: Initial Current Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Generated: 2026-10-16 16:37
Script: `eval_2_1_initial_power.py`

1. Scope

Section 2.1 of the per-board checkout report is the power-on reading of the four power-supply channels (BIAS, LArASIC, ColdDATA, ColdADC), taken before any configuration. It contains 17 numbers: V_set / V_meas / I_meas / P_meas for 4 rails, plus Total Power.

Category	Parameters	Treatment
Core	I_meas & V_meas of LArASIC / ColdDATA / ColdADC, Total Power (7)	Statistical judgement
Auxiliary	V_meas & I_meas of BIAS (2)	Shown for reference; I_BIAS ≈ 0 is better served by a fixed limit
Not evaluated	V_set (4)	Configuration setpoints, constant
Not evaluated	P_meas (4)	Redundant (P = V × I)

2. Judgement Method

For each parameter, mean and standard deviation (σ) are computed over all boards; each board is graded by its deviation |x − mean| / σ.

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
ALERT	5σ – 7σ	red
FAIL	> 7σ	dark red

3. Summary

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
I_LArASIC	core	0.4335 A	0.0035	0.4240	0.4390	6	44	0	0	0	–
I_ColdDATA	core	0.2330 A	0.0073	0.2230	0.2750	8	43	0	1	0	00051 (ALERT 5.7σ)
I_ColdADC	core	1.6041 A	0.0335	1.5560	1.7400	5	43	1	0	0	00031 (WARN 4.1σ)
V_LArASIC	core	2.9498 V	0.0045	2.9420	2.9550	9	44	0	0	0	–
V_ColdDATA	core	2.9735 V	0.0089	2.9540	2.9800	9	44	0	0	0	–
V_ColdADC	core	3.4220 V	0.0076	3.4090	3.4310	16	44	0	0	0	–
Total_Power	core	7.466 W	0.111	7.312	7.915	38	43	1	0	0	00031 (WARN 4.1σ)

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
V_BIAS	aux	4.9996 V	0.0005	4.9990	5.0000	2	44	0	0	0	–
I_BIAS	aux	0.0009 A	0.0020	0.0000	0.0060	3	44	0	0	0	–

Criteria (limits)

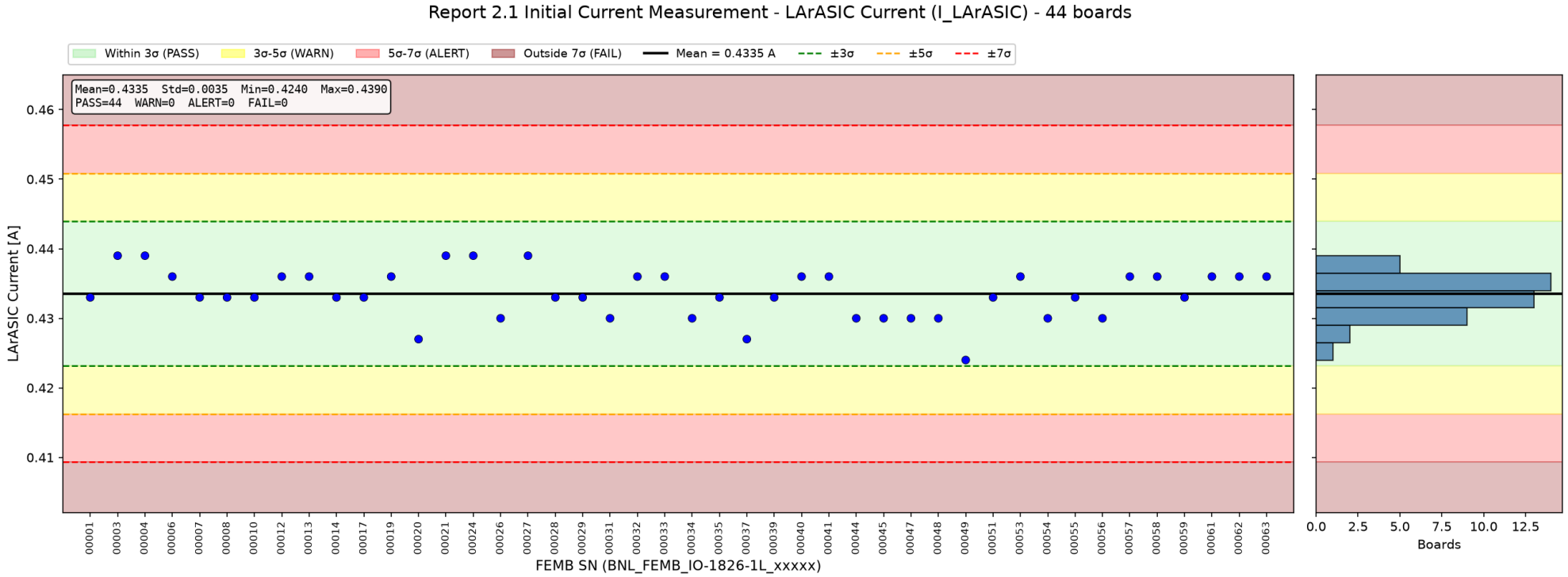
Parameter	Unit	−7σ	−5σ	−3σ	Mean	+3σ	+5σ	+7σ
I_LArASIC	A	0.4093	0.4162	0.4232	0.4335	0.4439	0.4508	0.4578
I_ColdDATA	A	0.1817	0.1964	0.2110	0.2330	0.2551	0.2697	0.2844
I_ColdADC	A	1.3698	1.4367	1.5037	1.6041	1.7046	1.7715	1.8385
V_LArASIC	V	2.9182	2.9272	2.9363	2.9498	2.9634	2.9724	2.9815
V_ColdDATA	V	2.9109	2.9288	2.9467	2.9735	3.0003	3.0181	3.0360
V_ColdADC	V	3.3691	3.3842	3.3994	3.4220	3.4447	3.4599	3.4750
Total_Power	W	6.692	6.913	7.134	7.466	7.797	8.018	8.239
V_BIAS	V	4.9961	4.9971	4.9981	4.9996	5.0011	5.0020	5.0030
I_BIAS	A	-0.0130	-0.0090	-0.0051	0.0009	0.0068	0.0108	0.0147

4. Core Parameters

4.1 LArASIC Current (`I_LArASIC`)

LArASIC rail current.
Mean = 0.4335 A, σ = 0.0035 A, range [0.4240, 0.4390], 6 distinct values over 44 boards.

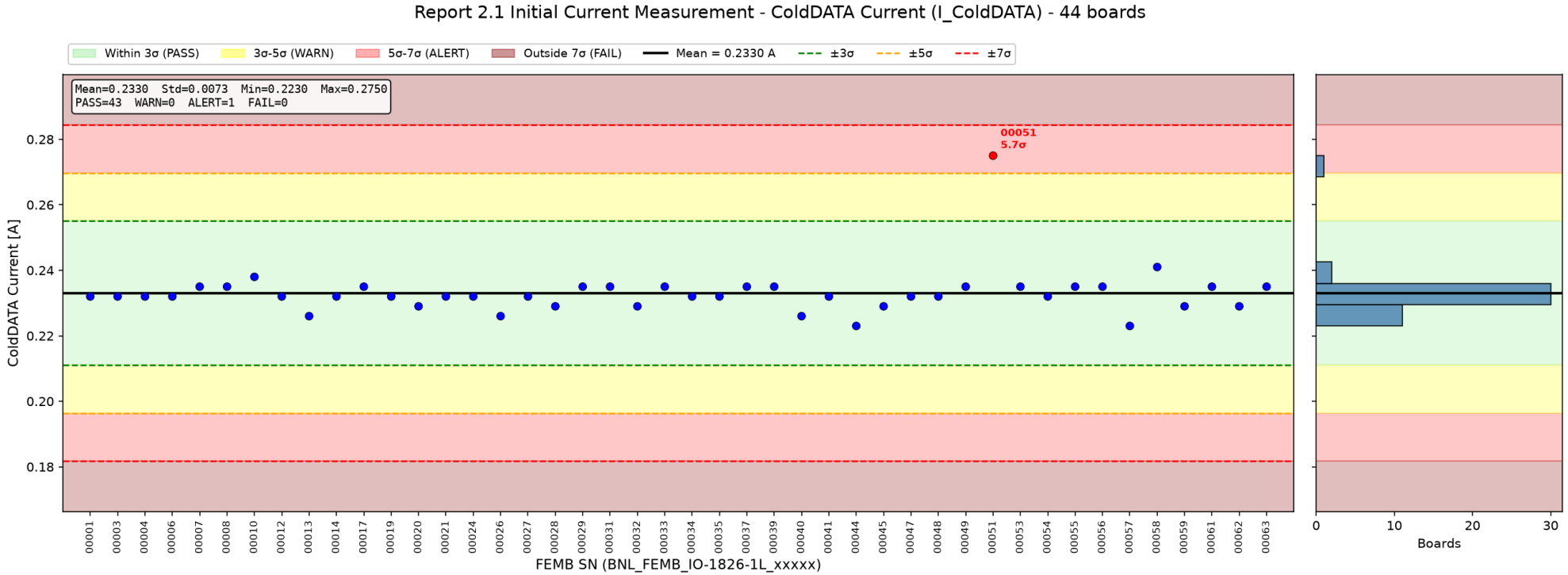
All boards within 3σ.



4.2 ColdDATA Current (`I_ColdDATA`)

ColdDATA rail current.
Mean = 0.2330 A, σ = 0.0073 A, range [0.2230, 0.2750], 8 distinct values over 44 boards.

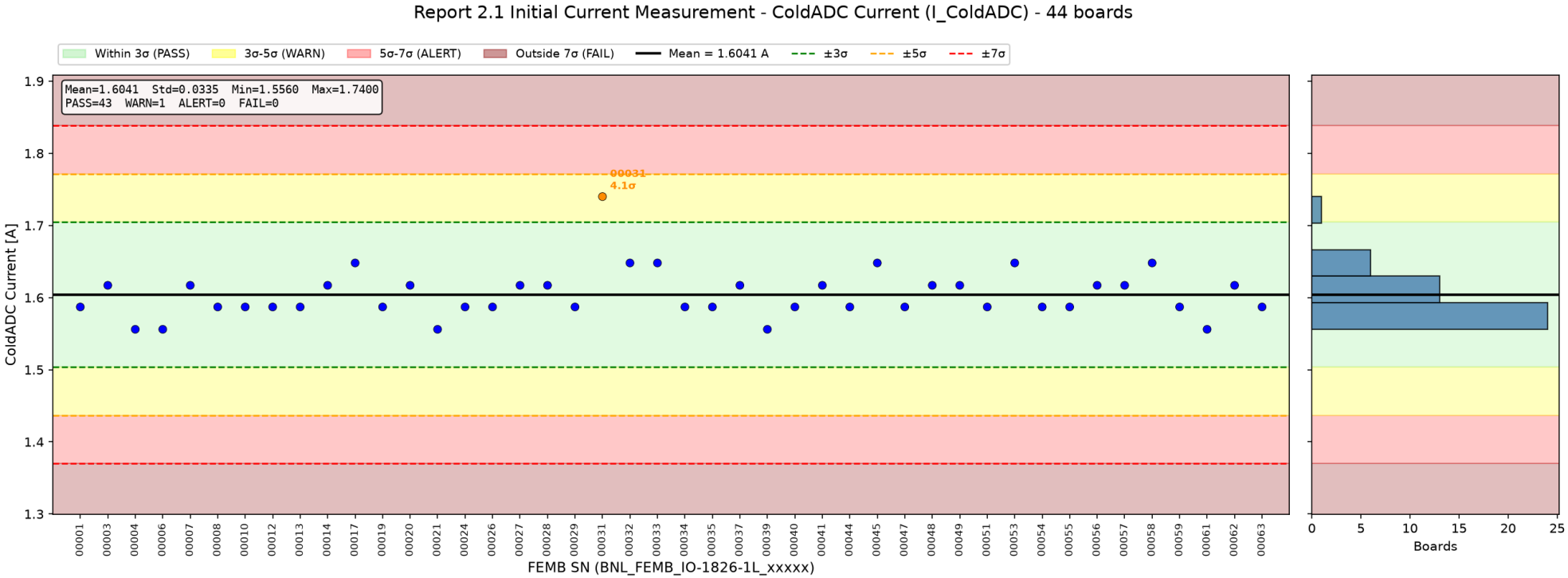
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00051	0.2750 A	5.7σ	**ALERT**



4.3 ColdADC Current (`I_ColdADC`)

ColdADC rail current.
Mean = 1.6041 A, σ = 0.0335 A, range [1.5560, 1.7400], 5 distinct values over 44 boards.

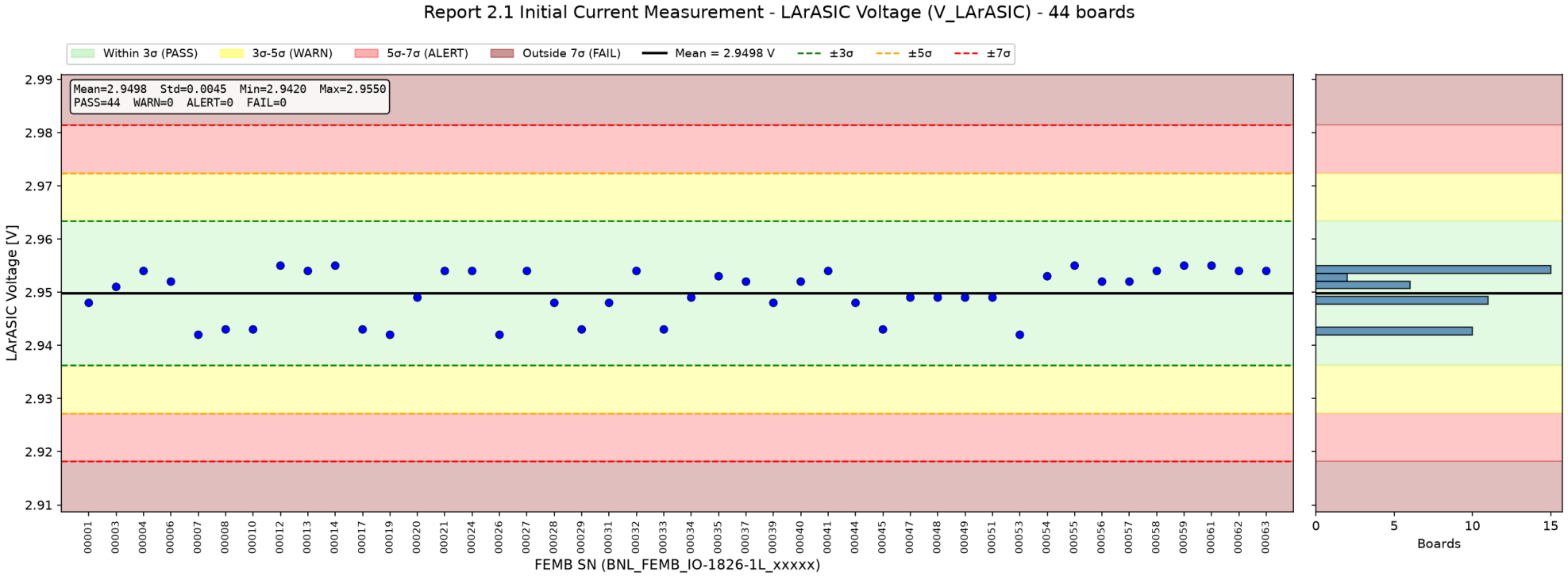
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	1.7400 A	4.1σ	**WARN**



4.4 LArASIC Voltage (`V_LArASIC`)

LArASIC rail voltage at the supply.
Mean = 2.9498 V, σ = 0.0045 V, range [2.9420, 2.9550], 9 distinct values over 44 boards.

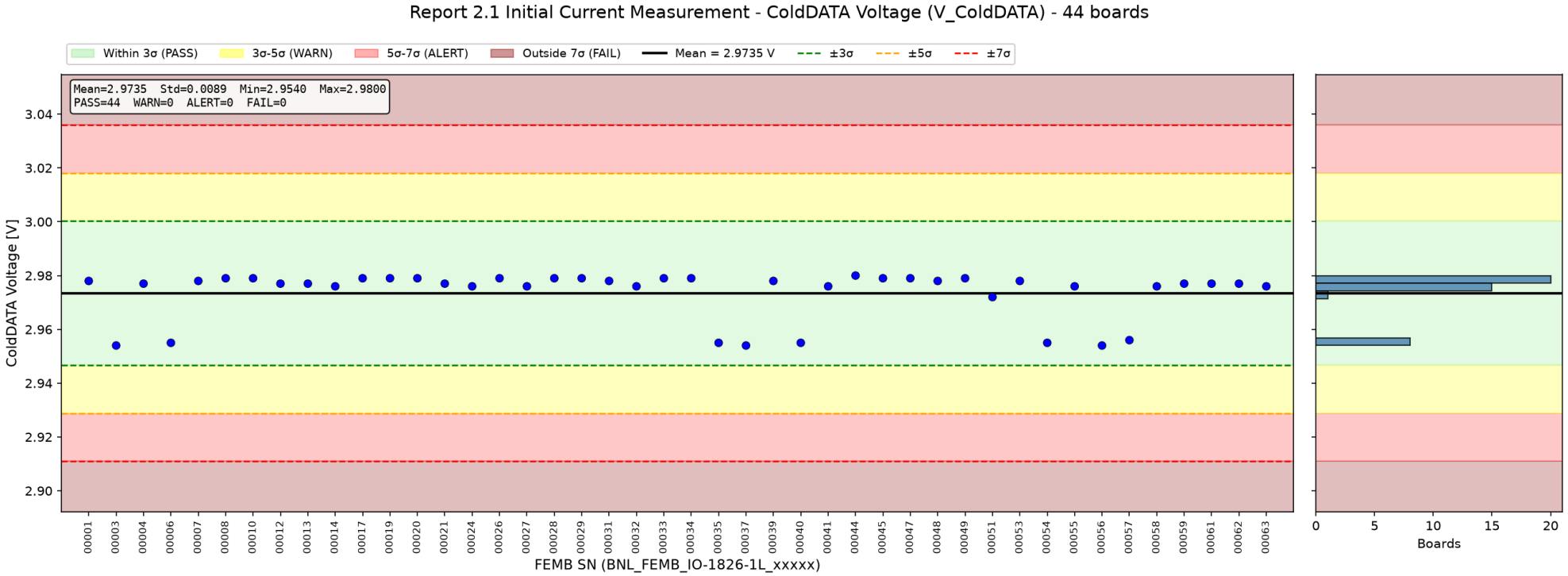
All boards within 3σ.



4.5 ColdDATA Voltage (`V_ColdDATA`)

ColdDATA rail voltage at the supply.
Mean = 2.9735 V, σ = 0.0089 V, range [2.9540, 2.9800], 9 distinct values over 44 boards.

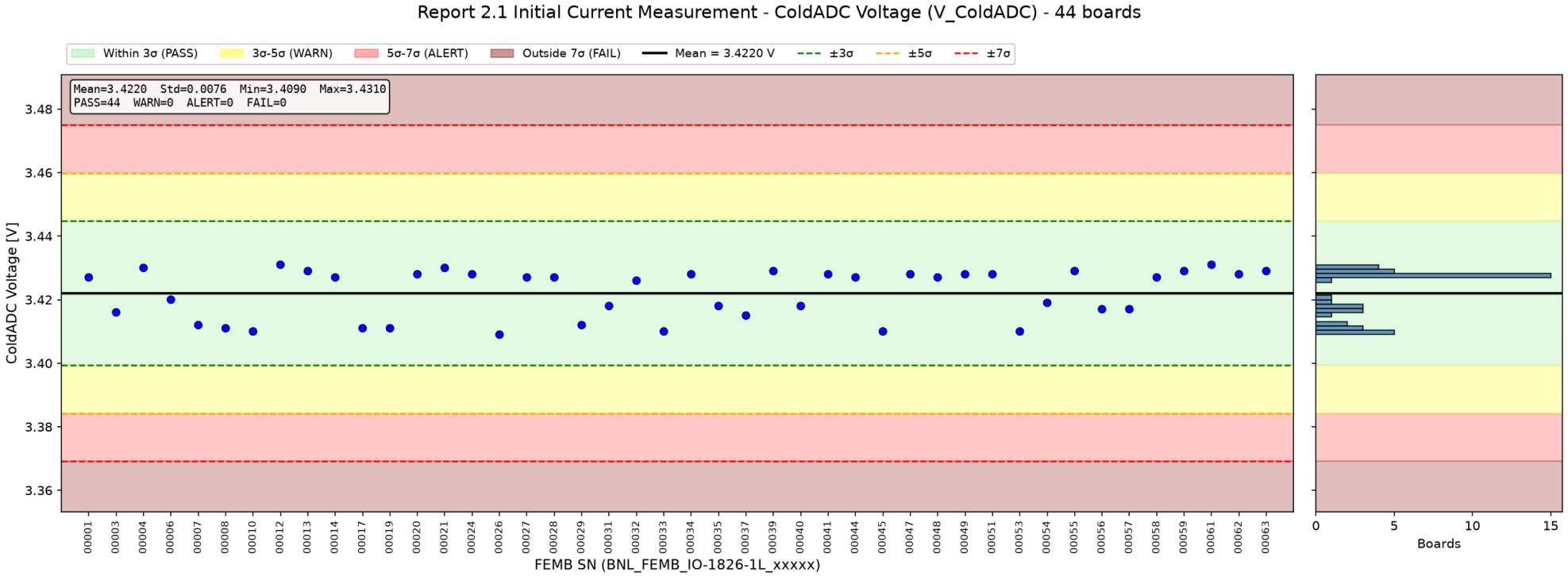
All boards within 3σ.



4.6 ColdADC Voltage (`V_ColdADC`)

ColdADC rail voltage at the supply.
Mean = 3.4220 V, σ = 0.0076 V, range [3.4090, 3.4310], 16 distinct values over 44 boards.

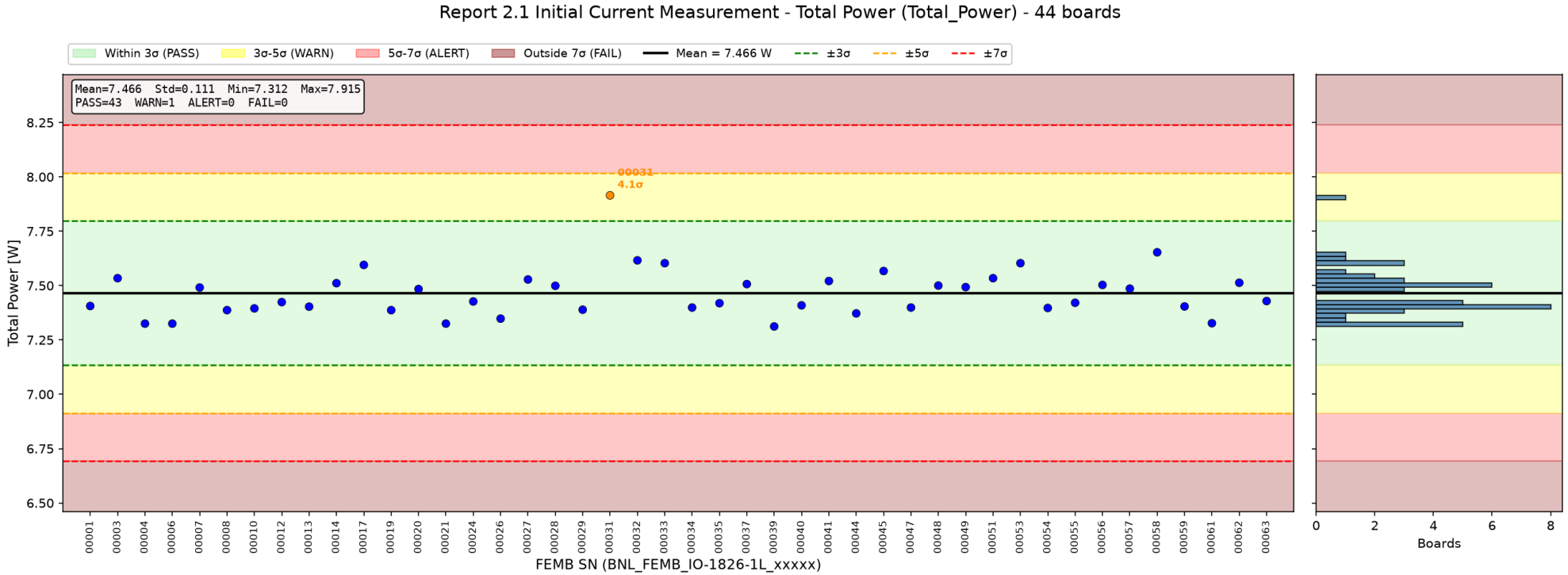
All boards within 3σ.



4.7 Total Power (`Total_Power`)

Sum of rail powers.
Mean = 7.466 W, σ = 0.111 W, range [7.312, 7.915], 38 distinct values over 44 boards.

FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	7.915 W	4.1σ	**WARN**

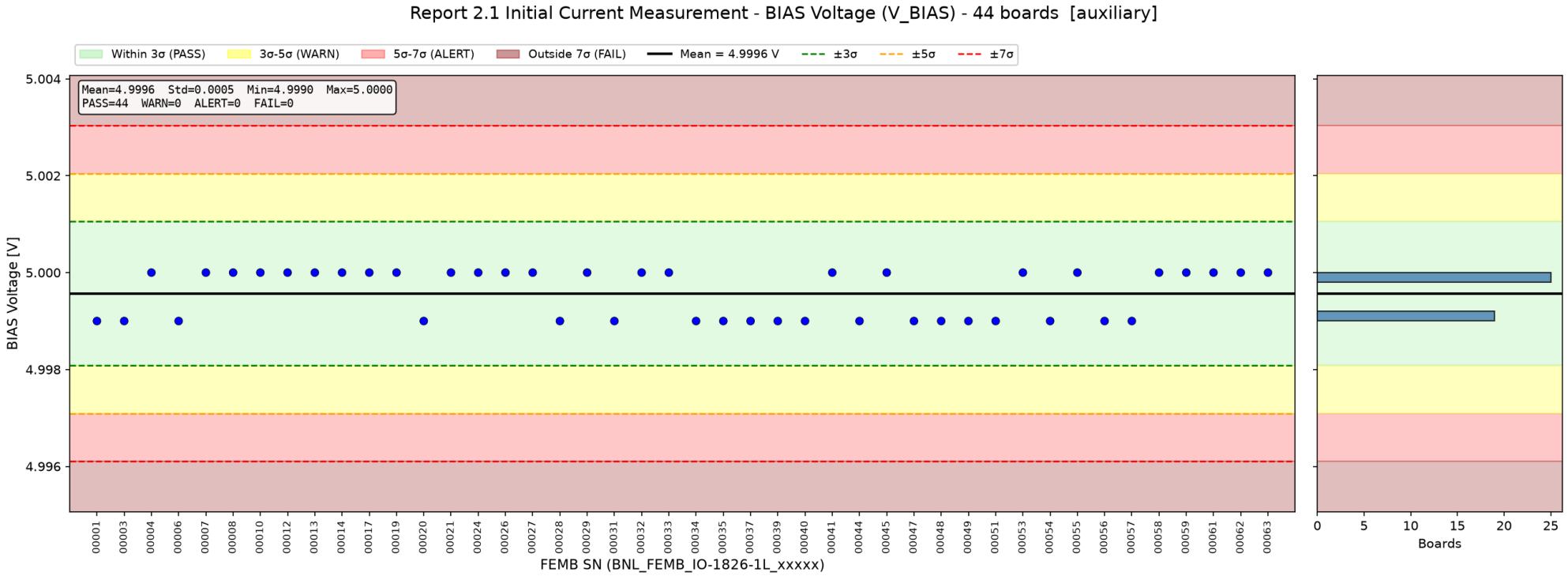


5. Auxiliary Parameters

5.1 BIAS Voltage (`V_BIAS`)

BIAS rail voltage.
Mean = 4.9996 V, σ = 0.0005 V, range [4.9990, 5.0000], 2 distinct values over 44 boards.

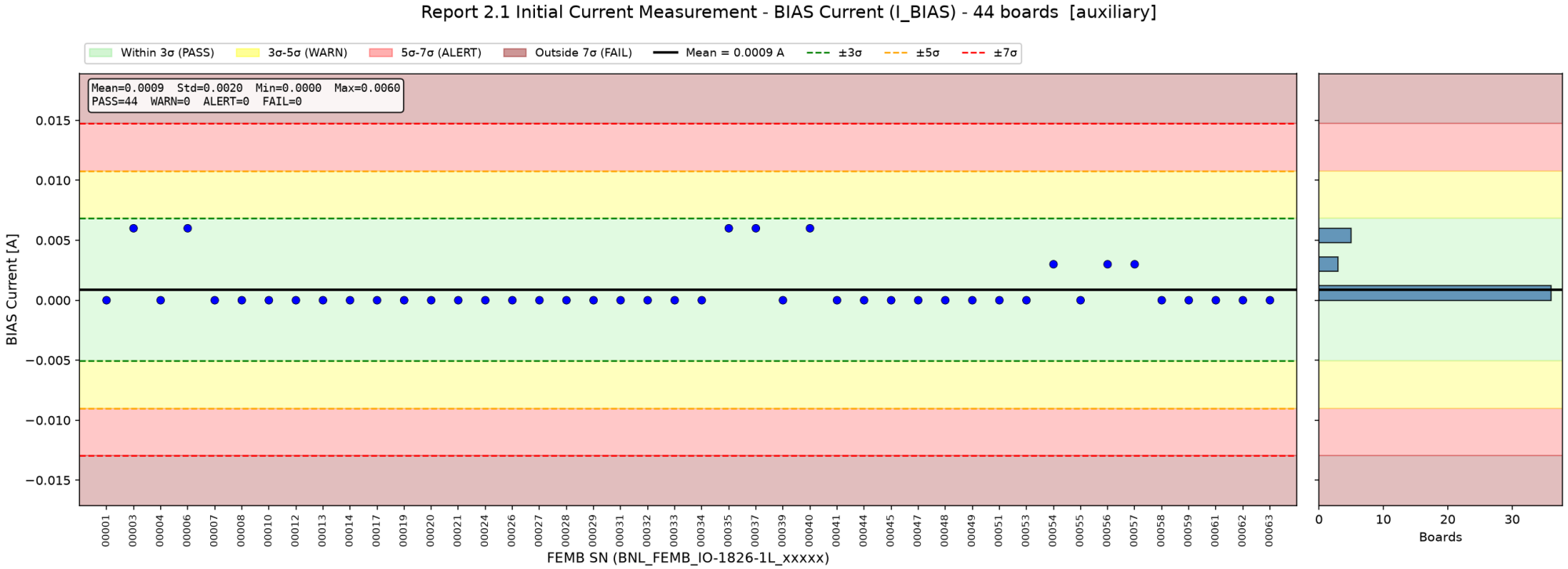
All boards within 3σ.



5.2 BIAS Current (`I_BIAS`)

BIAS rail current (≈ 0).
Mean = 0.0009 A, σ = 0.0020 A, range [0.0000, 0.0060], 3 distinct values over 44 boards.

All boards within 3σ.



6. Observations

- **BNL_FEMB_IO-1826-1L_00031**: I_ColdADC 4.1σ (WARN); Total_Power 4.1σ (WARN)
- **BNL_FEMB_IO-1826-1L_00051**: I_ColdDATA 5.7σ (ALERT)
- Readout resolution is coarse (≤ 10 distinct values) for: I_LArASIC, I_ColdDATA, I_ColdADC, V_LArASIC, V_ColdDATA. σ is therefore small and quantized; a minimum tolerance tied to instrument resolution should be considered before using these limits.
- Flagged boards are still included in the statistics and inflate σ; re-deriving limits without them would tighten the criteria.

Appendix A: Per-Board Judgement (core parameters)

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0.2σ	0.1σ	0.5σ	0.4σ	0.5σ	0.7σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	1.6σ	0.1σ	0.4σ	0.3σ	2.2σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	1.6σ	0.1σ	1.4σ	0.9σ	0.4σ	1.1σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0.7σ	0.1σ	1.4σ	0.5σ	2.1σ	0.3σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	0.2σ	0.3σ	0.4σ	1.7σ	0.5σ	1.3σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	0.2σ	0.3σ	0.5σ	1.5σ	0.6σ	1.5σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0.2σ	0.7σ	0.5σ	1.5σ	0.6σ	1.6σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0.7σ	0.1σ	0.5σ	1.1σ	0.4σ	1.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0.7σ	1.0σ	0.5σ	0.9σ	0.4σ	0.9σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0.2σ	0.1σ	0.4σ	1.1σ	0.3σ	0.7σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	0.2σ	0.3σ	1.3σ	1.5σ	0.6σ	1.5σ	1.2σ	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0.7σ	0.1σ	0.5σ	1.7σ	0.6σ	1.5σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	1.9σ	0.6σ	0.4σ	0.2σ	0.6σ	0.8σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	1.6σ	0.1σ	1.4σ	0.9σ	0.4σ	1.1σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	1.6σ	0.1σ	0.5σ	0.9σ	0.3σ	0.8σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	1.0σ	1.0σ	0.5σ	1.7σ	0.6σ	1.7σ	1.1σ	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	1.6σ	0.1σ	0.4σ	0.9σ	0.3σ	0.7σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0.2σ	0.6σ	0.4σ	0.4σ	0.6σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0.2σ	0.3σ	0.5σ	1.5σ	0.6σ	1.3σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	1.0σ	0.3σ	**4.1σ WARN**	0.4σ	0.5σ	0.5σ	**4.1σ WARN**	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0.7σ	0.6σ	1.3σ	0.9σ	0.3σ	0.5σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0.7σ	0.3σ	1.3σ	1.5σ	0.6σ	1.6σ	1.2σ	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	1.0σ	0.1σ	0.5σ	0.2σ	0.6σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0.2σ	0.1σ	0.5σ	0.7σ	2.1σ	0.5σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	1.9σ	0.3σ	0.4σ	0.5σ	2.2σ	0.9σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0.2σ	0.3σ	1.4σ	0.4σ	0.5σ	0.9σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0.7σ	1.0σ	0.5σ	0.5σ	2.1σ	0.5σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0.7σ	0.1σ	0.4σ	0.9σ	0.3σ	0.8σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	1.0σ	1.4σ	0.5σ	0.4σ	0.7σ	0.7σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	1.0σ	0.6σ	1.3σ	1.5σ	0.6σ	1.6σ	0.9σ	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	1.0σ	0.1σ	0.5σ	0.2σ	0.6σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	1.0σ	0.1σ	0.4σ	0.2σ	0.5σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	2.8σ	0.3σ	0.4σ	0.2σ	0.6σ	0.8σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0.2σ	**5.7σ ALERT**	0.5σ	0.2σ	0.2σ	0.8σ	0.6σ	ALERT
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0.7σ	0.3σ	1.3σ	1.7σ	0.5σ	1.6σ	1.2σ	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	1.0σ	0.1σ	0.5σ	0.7σ	2.1σ	0.4σ	0.6σ	PASS

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0.2σ	0.3σ	0.5σ	1.1σ	0.3σ	0.9σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	1.0σ	0.3σ	0.4σ	0.5σ	2.2σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0.7σ	1.4σ	0.4σ	0.5σ	2.0σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0.7σ	1.1σ	1.3σ	0.9σ	0.3σ	0.7σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0.2σ	0.6σ	0.5σ	1.1σ	0.4σ	0.9σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0.7σ	0.3σ	1.4σ	1.1σ	0.4σ	1.2σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0.7σ	0.6σ	0.4σ	0.9σ	0.4σ	0.8σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	0.7σ	0.3σ	0.5σ	0.9σ	0.3σ	0.9σ	0.3σ	PASS

Appendix B: Output Files

- `output/criteria_2_1.csv` – mean, σ and 3σ/5σ/7σ limits per parameter
- `output/judgement_2_1.csv` – value, deviation and level per board per parameter
- `output/plot_2_1_*.png` – distribution plots

← Back to summary

FEMB QC Parameter Evaluation - Report Section 3.1: SE OFF Noise Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Channels: 44 × 128 = 5632
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Configuration: single-ended, SE buffer OFF, 200 mV baseline, 14 mV/fC, 2 µs, no pulse
Generated: 2026-10-16 16:37
Script: `eval_3_1_noise.py`

1. Scope

Section 3.1 of the checkout report is the noise test: no calibration pulse is injected, and for each of the 128 channels the pedestal and RMS noise are evaluated. Pulse-response quantities (PPK / NPK / baseline) belong to section 3.4 and are not part of this section.

Parameter	Description
`rms` – RMS Noise	RMS of the baseline, ADC counts
`ped` – Pedestal	Mean baseline (DC offset), ADC counts

2. Judgement Method

For each parameter the mean and σ are computed over all 5632 channels pooled together; each channel is graded by |x − mean| / σ. A board takes the worst level among its channels.

Channel-level (noise / pulse) parameters use **5σ as the limit**: a channel beyond 5σ fails. (Board-level power and LDO parameters keep the wider 7σ limit.)

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
FAIL	> 5σ	red

For a Gaussian distribution, about 0.27 % of channels (≈ 15 of 5632) are expected beyond 3σ by chance, so isolated WARN channels are normal.

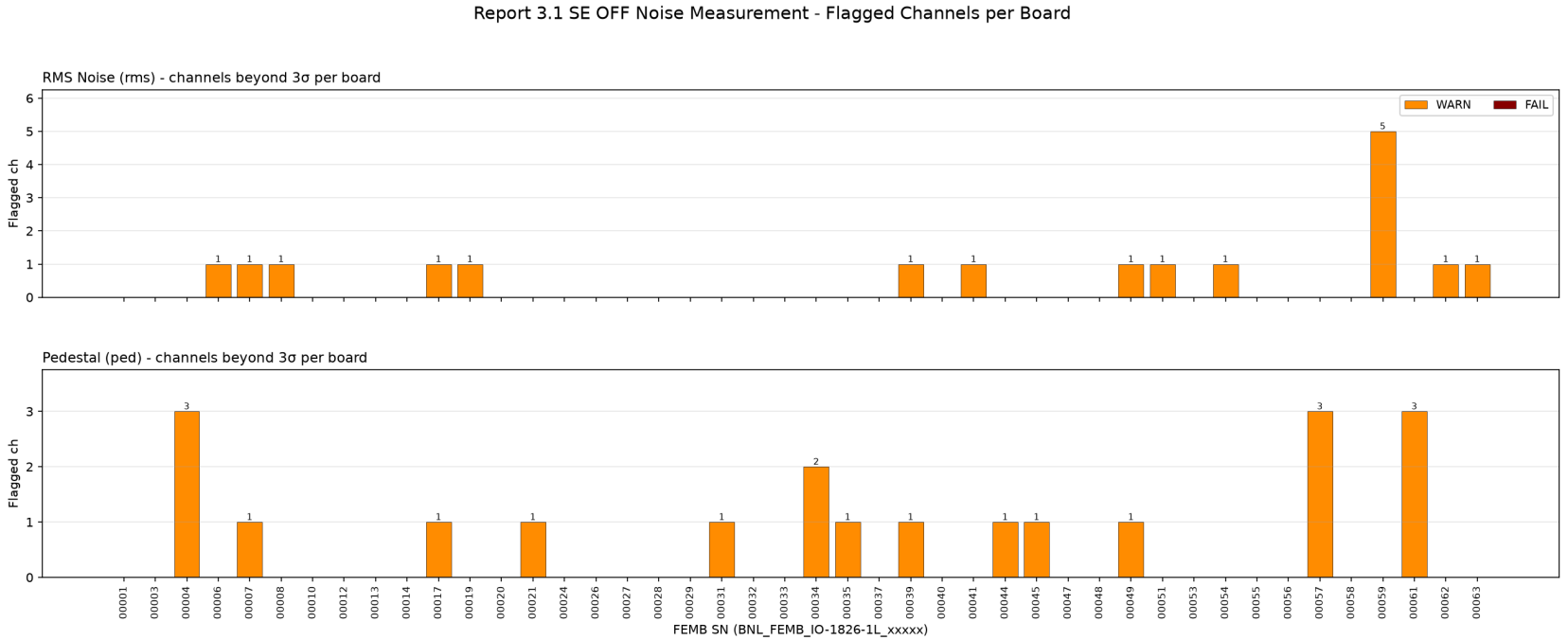
3. Summary

Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
`rms`	11.19	0.45	9.24	13.04	5615	17	0	–
`ped`	910.92	104.51	576.84	1305.49	5612	20	0	–

Criteria (limits, ADC counts)

Parameter	−5σ (FAIL)	−3σ	Mean	+3σ	+5σ (FAIL)
rms	8.92	9.83	11.19	12.56	13.47
ped	388.37	597.39	910.92	1224.45	1433.47

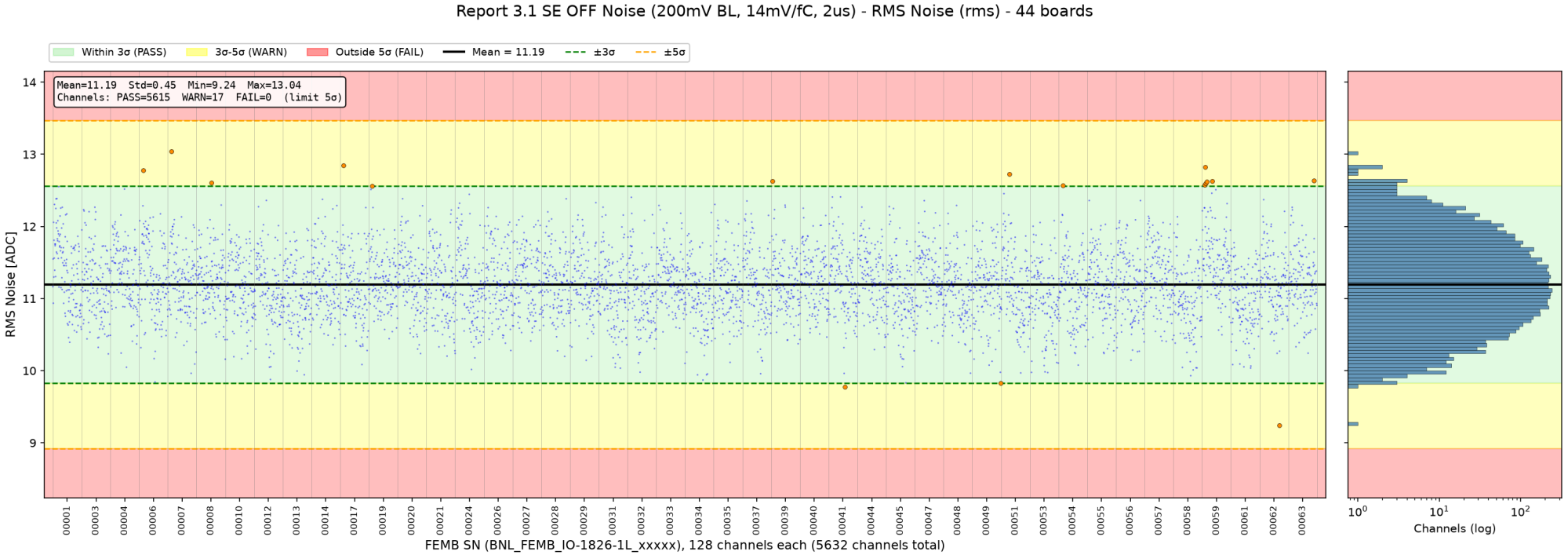
Flagged channels per board



4. Parameters

4.1 RMS Noise (`rms`)

Mean = 11.19, σ = 0.45, range [9.24, 13.04] ADC. Spread of per-board means: σ = 0.11; spread of per-channel means across boards: σ = 0.26.



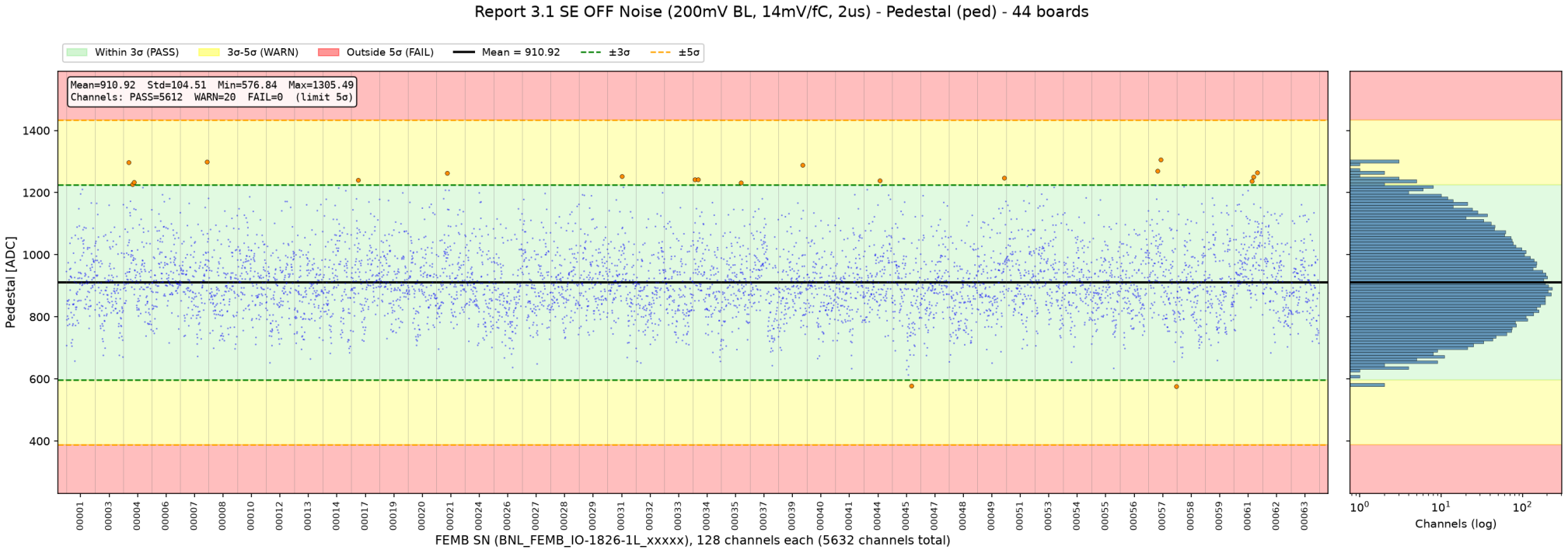
17 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00062	88	5	9.24	4.3σ	WARN
BNL_FEMB_IO-1826-1L_00007	15	0	13.04	4.1σ	WARN
BNL_FEMB_IO-1826-1L_00017	15	0	12.84	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00059	13	0	12.82	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00006	18	1	12.78	3.5σ	WARN
BNL_FEMB_IO-1826-1L_00051	36	2	12.73	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00063	113	7	12.64	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00059	44	2	12.63	3.2σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00039	5	0	12.63	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00059	20	1	12.62	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00041	72	4	9.78	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00008	66	4	12.61	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00059	15	0	12.60	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00059	12	0	12.57	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00054	20	1	12.57	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00049	127	7	9.83	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00019	13	0	12.56	3.0σ	WARN

4.2 Pedestal (ped)

Mean = 910.92, σ = 104.51, range [576.84, 1305.49] ADC. Spread of per-board means: σ = 23.59; spread of per-channel means across boards: σ = 65.47.



20 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00057	56	3	1305.49	3.8σ	WARN
BNL_FEMB_IO-1826-1L_00007	120	7	1298.34	3.7σ	WARN
BNL_FEMB_IO-1826-1L_00004	24	1	1297.25	3.7σ	WARN
BNL_FEMB_IO-1826-1L_00039	109	6	1289.33	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00057	40	2	1269.79	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00061	104	6	1264.56	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00021	48	3	1263.46	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00031	64	4	1253.36	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00061	88	5	1250.73	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00049	120	7	1247.25	3.2σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00057	126	7	576.84	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00045	86	5	578.45	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00034	24	1	1241.94	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00034	8	0	1241.47	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00017	32	2	1240.36	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00044	72	4	1238.61	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00061	80	5	1237.22	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00004	47	2	1233.97	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00035	88	5	1232.32	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00004	40	2	1226.81	3.0σ	WARN

5. Observations

- **rms**: no channel beyond 5σ.
- **ped**: no channel beyond 5σ.
- Thresholds are pooled over channels; `ped` varies channel-to-channel by design (σ ≈ 105 ADC), so its limits are wide and mainly catch dead / stuck channels.

Appendix A: Per-Board Judgement

Each cell: number of channels WARN / FAIL, and worst deviation.

FEMB SN	Test time	`rms`	`ped`	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0/0 (3.0σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0/0 (2.4σ)	0/0 (2.4σ)	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	0/0 (2.9σ)	3/0 (3.7σ)	WARN
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	1/0 (3.5σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	1/0 (4.1σ)	1/0 (3.7σ)	WARN
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	1/0 (3.1σ)	0/0 (2.2σ)	WARN
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0/0 (2.3σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0/0 (2.9σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0/0 (2.8σ)	0/0 (2.4σ)	PASS
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0/0 (2.6σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	1/0 (3.6σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	1/0 (3.0σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0/0 (2.6σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0/0 (2.3σ)	1/0 (3.4σ)	WARN
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	0/0 (2.8σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0/0 (2.3σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	0/0 (2.1σ)	0/0 (2.7σ)	PASS

FEMB SN	Test time	rms	ped	Worst
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0/0 (2.6σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0/0 (2.5σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0/0 (2.5σ)	1/0 (3.3σ)	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0/0 (2.7σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0/0 (2.4σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0/0 (2.9σ)	2/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0/0 (2.4σ)	1/0 (3.1σ)	WARN
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0/0 (2.7σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	1/0 (3.2σ)	1/0 (3.6σ)	WARN
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0/0 (2.3σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	1/0 (3.1σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	0/0 (2.4σ)	1/0 (3.1σ)	WARN
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	0/0 (3.0σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	0/0 (2.6σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0/0 (2.7σ)	0/0 (2.4σ)	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	1/0 (3.0σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	1/0 (3.4σ)	0/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0/0 (2.7σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	1/0 (3.0σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0/0 (2.6σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0/0 (2.6σ)	0/0 (2.0σ)	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0/0 (2.1σ)	3/0 (3.8σ)	WARN
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0/0 (2.8σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	5/0 (3.6σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0/0 (2.5σ)	3/0 (3.4σ)	WARN
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	1/0 (4.3σ)	0/0 (2.2σ)	WARN
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	1/0 (3.2σ)	0/0 (2.4σ)	WARN

Appendix B: Output Files

- `output/criteria_3_1.csv` – mean, σ and 3σ/5σ limits per parameter
- `output/judgement_3_1_channels.csv` – every channel beyond 3σ (board, channel, value, deviation, level)
- `output/judgement_3_1_boards.csv` – per-board counts and worst level
- `output/plot_3_1_*.png` – distribution plots and per-board summary

← Back to summary

FEMB QC Parameter Evaluation - Report Section 3.2: SE OFF Power Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Generated: 2026-10-16 16:37
Script: `eval_3_2_se_off_power.py`

1. Scope

Section 3.2 of the per-board checkout report is the power-supply reading taken after the FEMB is configured for the SE OFF noise measurement (single-ended, SE buffer OFF, 200 mV baseline, 14 mV/fC, 2 µs). It has the same 17 numbers as section 2.1: V_set / V_meas / I_meas / P_meas for the 4 rails (BIAS, LArASIC, ColdDATA, ColdADC), plus Total Power.

Category	Parameters	Treatment
Core	I_meas & V_meas of LArASIC / ColdDATA / ColdADC, Total Power (7)	Statistical judgement
Auxiliary	V_meas & I_meas of BIAS (2)	Shown for reference; I_BIAS ≈ 0 is better served by a fixed limit
Not evaluated	V_set (4)	Configuration setpoints, constant
Not evaluated	P_meas (4)	Redundant (P = V × I)

2. Judgement Method

For each parameter, mean and standard deviation (σ) are computed over all boards; each board is graded by its deviation |x − mean| / σ.

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
ALERT	5σ – 7σ	red
FAIL	> 7σ	dark red

3. Summary

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
I_LArASIC	core	0.4224 A	0.0028	0.4180	0.4270	4	44	0	0	0	–
I_ColdDATA	core	0.2320 A	0.0075	0.2230	0.2750	8	43	0	1	0	00051 (ALERT 5.7σ)
I_ColdADC	core	1.5958 A	0.0342	1.5260	1.7400	7	43	1	0	0	00031 (WARN 4.2σ)
V_LArASIC	core	2.9513 V	0.0046	2.9430	2.9570	11	44	0	0	0	–
V_ColdDATA	core	2.9733 V	0.0088	2.9540	2.9800	10	44	0	0	0	–
V_ColdADC	core	3.4217 V	0.0075	3.4080	3.4300	16	44	0	0	0	–
Total_Power	core	7.402 W	0.113	7.135	7.862	36	43	1	0	0	00031 (WARN 4.1σ)

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
V_BIAS	aux	4.9996 V	0.0005	4.9990	5.0000	2	44	0	0	0	–
I_BIAS	aux	0.0010 A	0.0022	0.0000	0.0060	3	44	0	0	0	–

Criteria (limits)

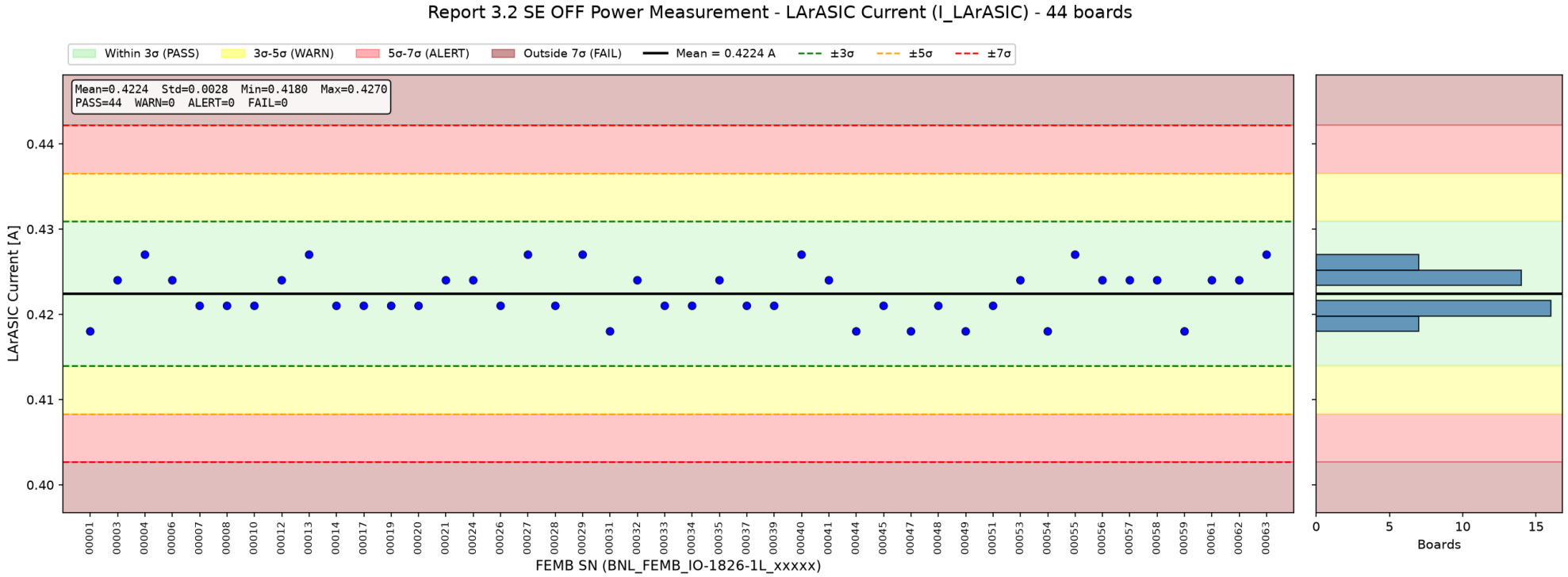
Parameter	Unit	−7σ	−5σ	−3σ	Mean	+3σ	+5σ	+7σ
I_LArASIC	A	0.4027	0.4083	0.4140	0.4224	0.4309	0.4365	0.4422
I_ColdDATA	A	0.1795	0.1945	0.2095	0.2320	0.2544	0.2694	0.2844
I_ColdADC	A	1.3566	1.4250	1.4933	1.5958	1.6983	1.7667	1.8350
V_LArASIC	V	2.9193	2.9284	2.9375	2.9513	2.9650	2.9741	2.9832
V_ColdDATA	V	2.9120	2.9295	2.9470	2.9733	2.9996	3.0171	3.0346
V_ColdADC	V	3.3696	3.3845	3.3994	3.4217	3.4441	3.4590	3.4739
Total_Power	W	6.609	6.836	7.062	7.402	7.742	7.968	8.195
V_BIAS	V	4.9961	4.9971	4.9981	4.9996	5.0011	5.0020	5.0030
I_BIAS	A	-0.0145	-0.0100	-0.0056	0.0010	0.0077	0.0121	0.0165

4. Core Parameters

4.1 LArASIC Current (`I_LArASIC`)

LArASIC rail current.
Mean = 0.4224 A, σ = 0.0028 A, range [0.4180, 0.4270], 4 distinct values over 44 boards.

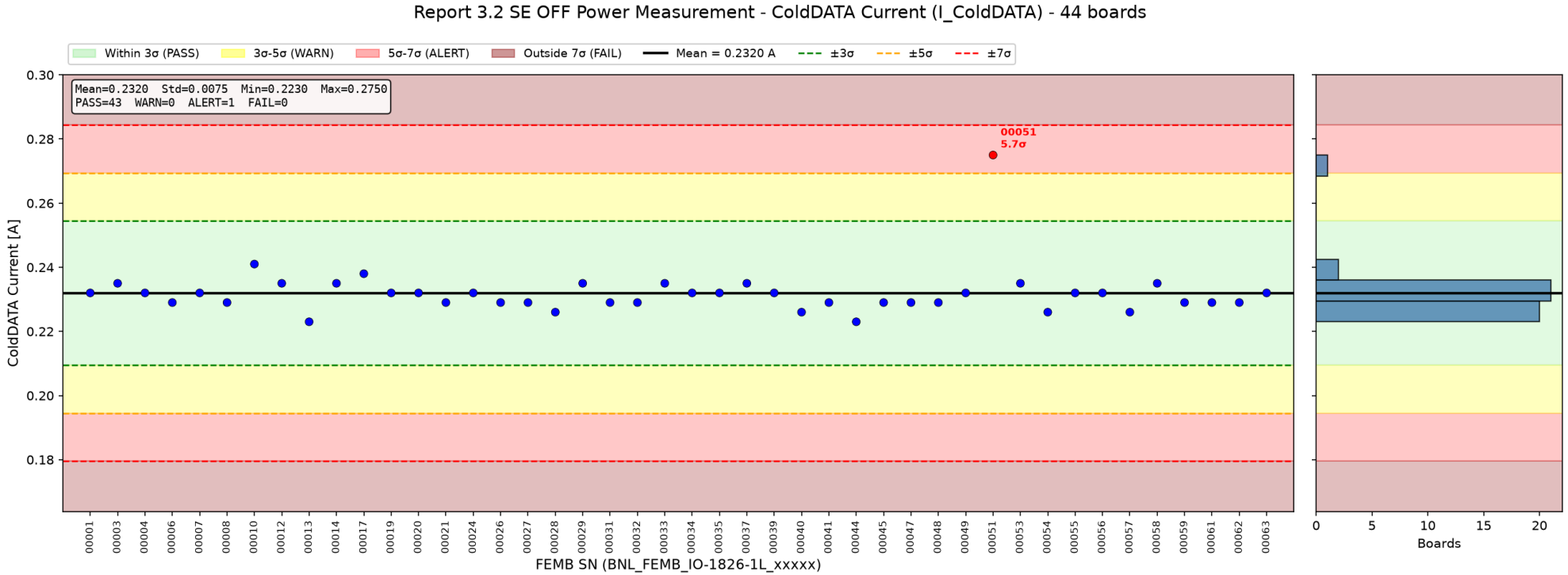
All boards within 3σ.



4.2 ColdDATA Current (`I_ColdDATA`)

ColdDATA rail current.
Mean = 0.2320 A, σ = 0.0075 A, range [0.2230, 0.2750], 8 distinct values over 44 boards.

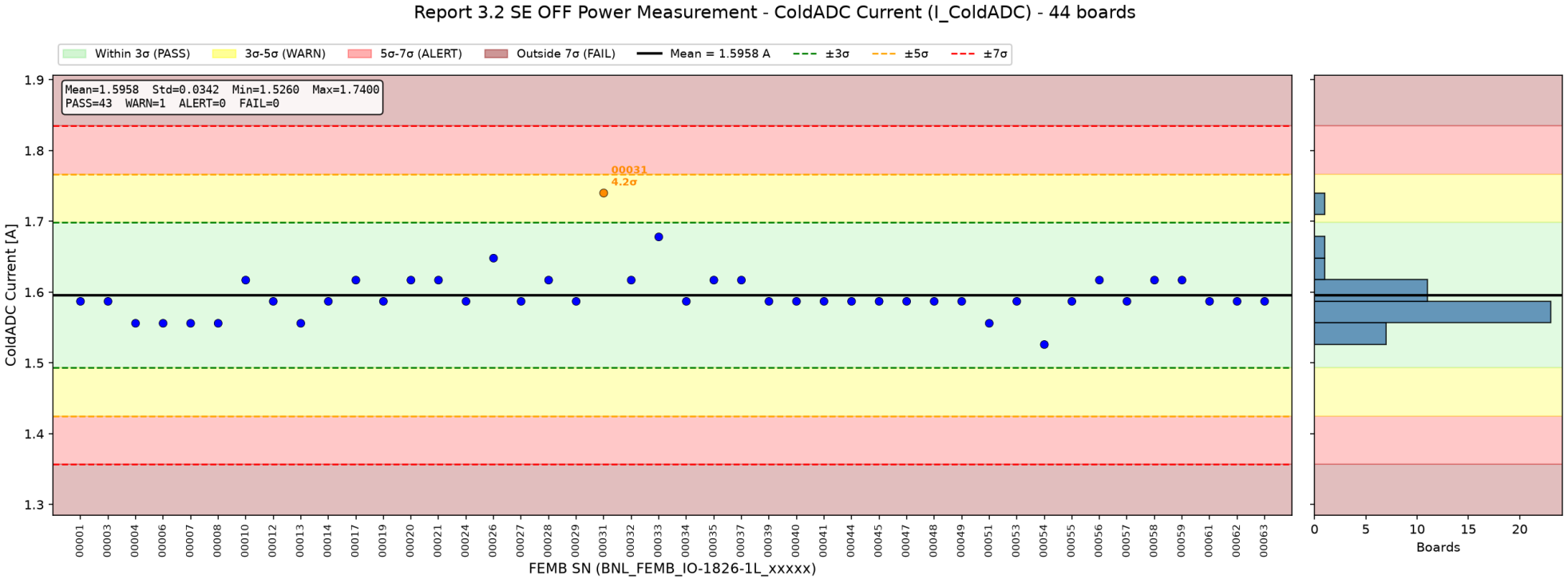
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00051	0.2750 A	5.7σ	**ALERT**



4.3 ColdADC Current (`I_ColdADC`)

ColdADC rail current.
Mean = 1.5958 A, σ = 0.0342 A, range [1.5260, 1.7400], 7 distinct values over 44 boards.

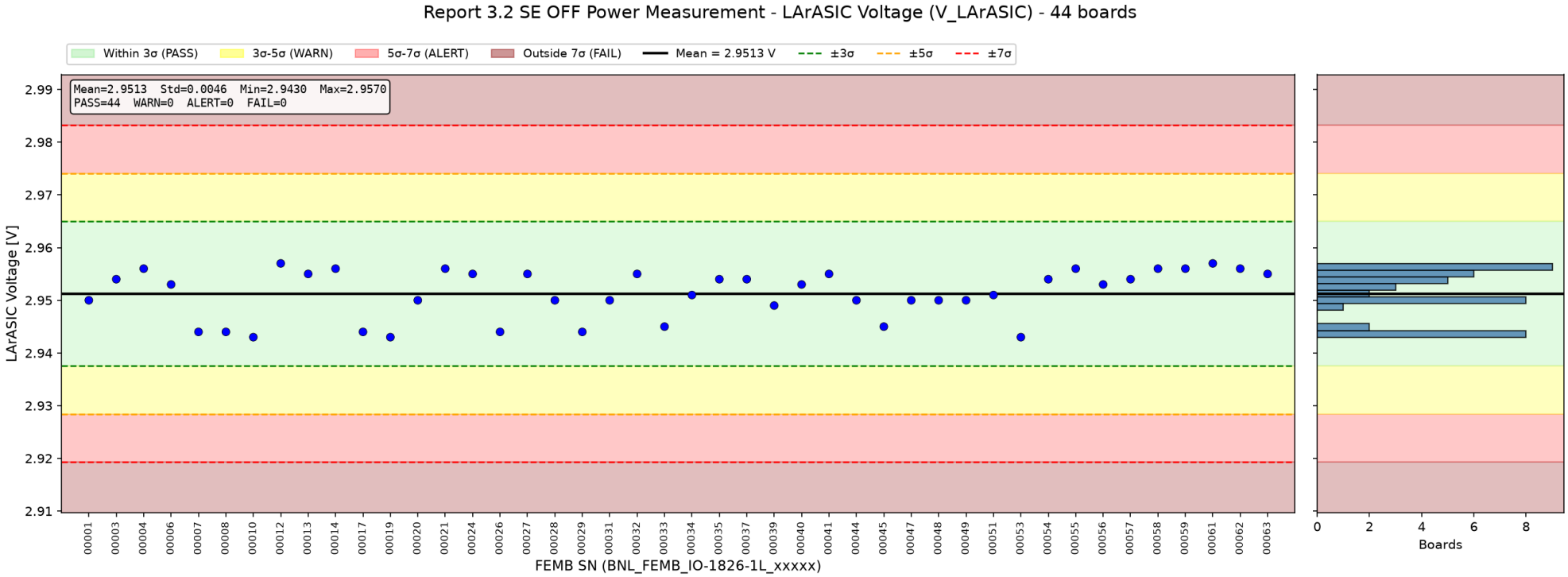
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	1.7400 A	4.2σ	**WARN**



4.4 LArASIC Voltage (`V_LArASIC`)

LArASIC rail voltage at the supply.
Mean = 2.9513 V, σ = 0.0046 V, range [2.9430, 2.9570], 11 distinct values over 44 boards.

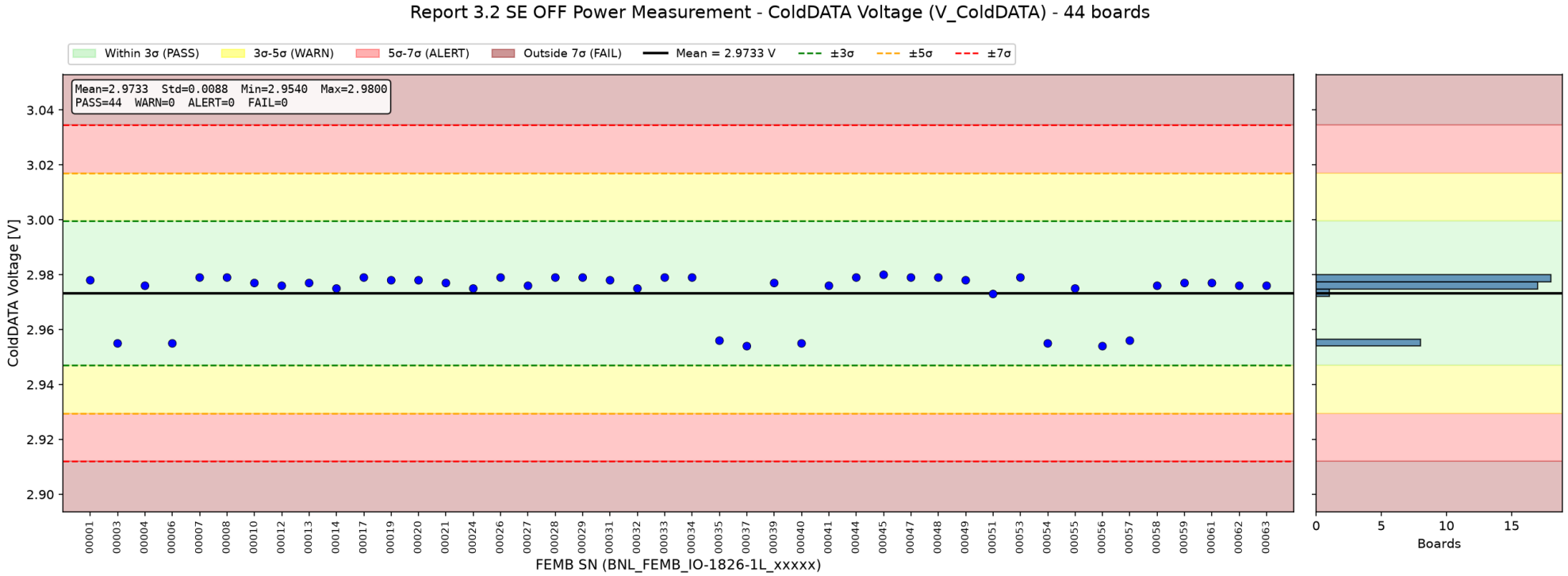
All boards within 3σ.



4.5 ColdDATA Voltage (`V_ColdDATA`)

ColdDATA rail voltage at the supply.
Mean = 2.9733 V, σ = 0.0088 V, range [2.9540, 2.9800], 10 distinct values over 44 boards.

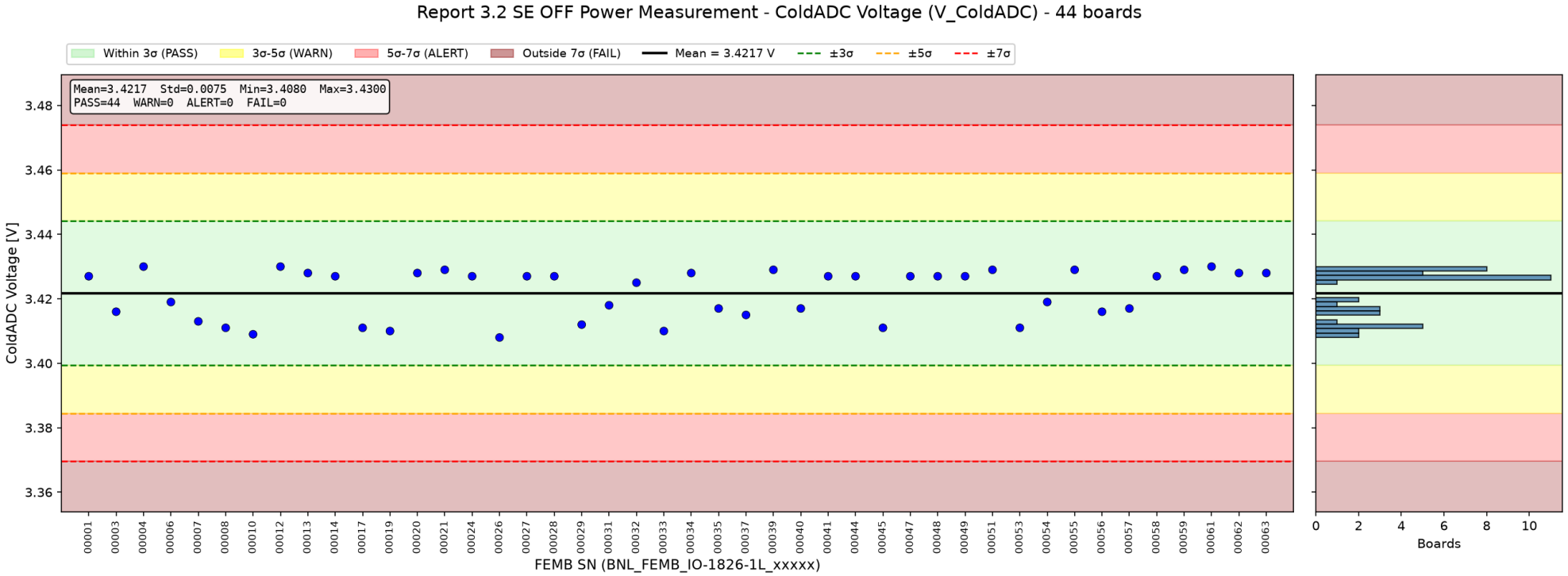
All boards within 3σ.



4.6 ColdADC Voltage (`V_ColdADC`)

ColdADC rail voltage at the supply.
Mean = 3.4217 V, σ = 0.0075 V, range [3.4080, 3.4300], 16 distinct values over 44 boards.

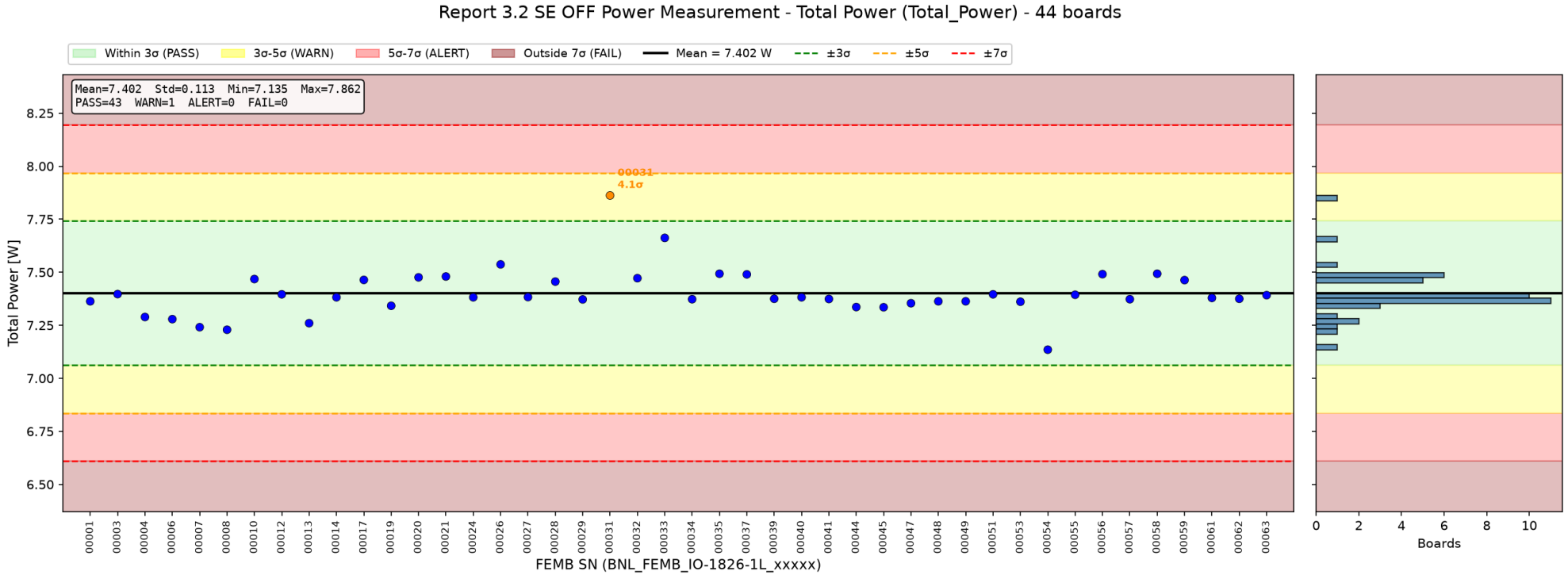
All boards within 3σ.



4.7 Total Power (`Total_Power`)

Sum of rail powers.
Mean = 7.402 W, σ = 0.113 W, range [7.135, 7.862], 36 distinct values over 44 boards.

FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	7.862 W	4.1σ	**WARN**

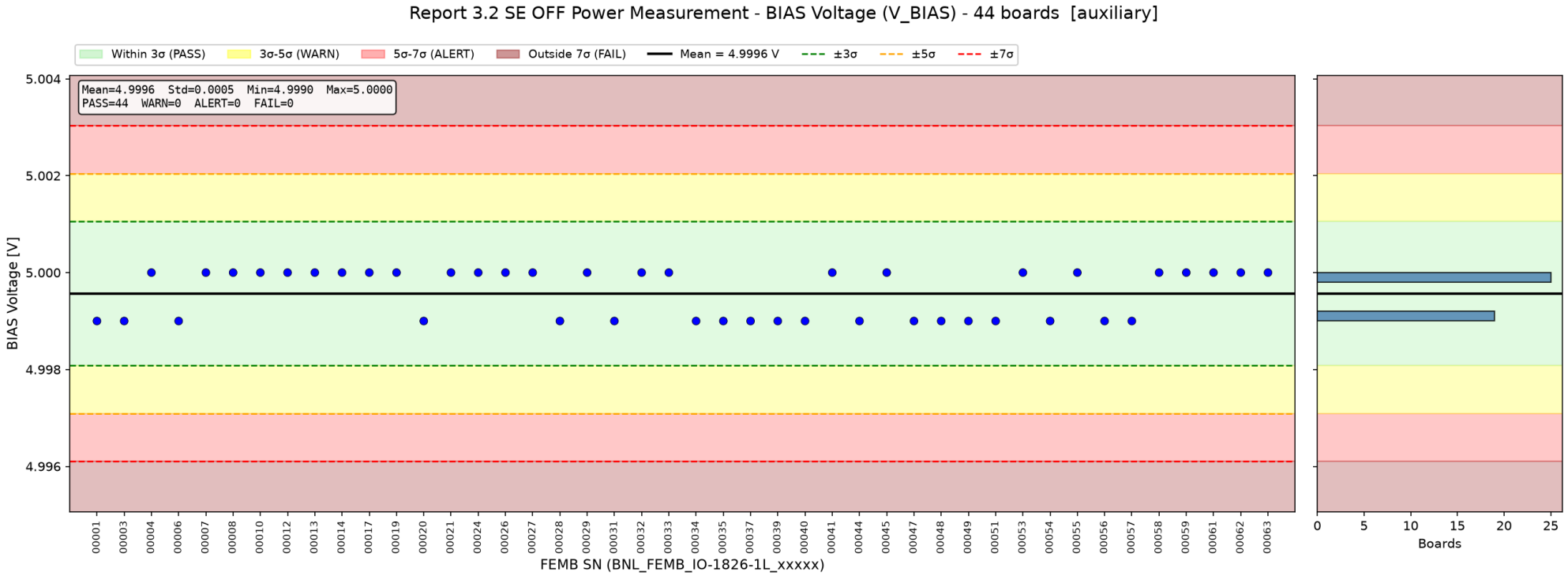


5. Auxiliary Parameters

5.1 BIAS Voltage (`V_BIAS`)

BIAS rail voltage.
Mean = 4.9996 V, σ = 0.0005 V, range [4.9990, 5.0000], 2 distinct values over 44 boards.

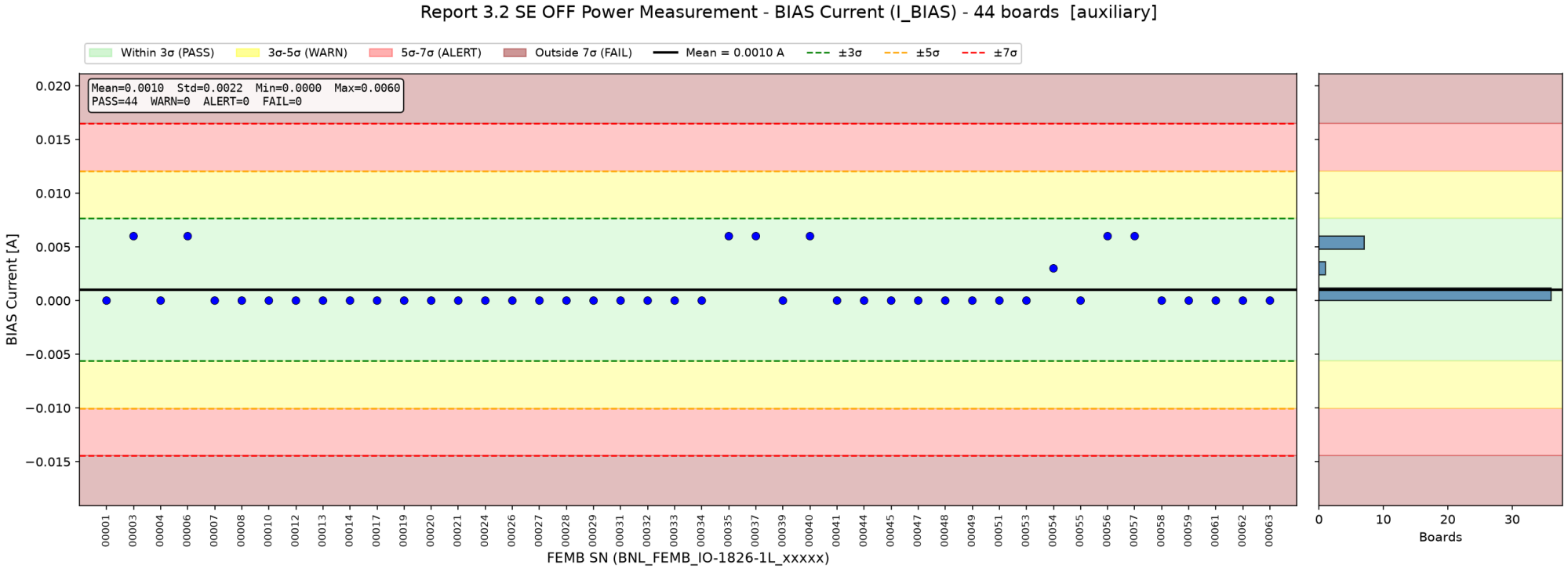
All boards within 3σ.



5.2 BIAS Current (`I_BIAS`)

BIAS rail current (≈ 0).
Mean = 0.0010 A, σ = 0.0022 A, range [0.0000, 0.0060], 3 distinct values over 44 boards.

All boards within 3σ.



6. Observations

- **BNL_FEMB_IO-1826-1L_00031**: I_ColdADC 4.2σ (WARN); Total_Power 4.1σ (WARN)
- **BNL_FEMB_IO-1826-1L_00051**: I_ColdDATA 5.7σ (ALERT)
- Readout resolution is coarse (≤ 10 distinct values) for: I_LArASIC, I_ColdDATA, I_ColdADC, V_ColdDATA. σ is therefore small and quantized; a minimum tolerance tied to instrument resolution should be considered before using these limits.
- Flagged boards are still included in the statistics and inflate σ; re-deriving limits without them would tighten the criteria.

Appendix A: Per-Board Judgement (core parameters)

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	1.6σ	0.0σ	0.3σ	0.3σ	0.5σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0.6σ	0.4σ	0.3σ	0.6σ	2.1σ	0.8σ	0.0σ	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	1.6σ	0.0σ	1.2σ	1.0σ	0.3σ	1.1σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0.6σ	0.4σ	1.2σ	0.4σ	2.1σ	0.4σ	1.1σ	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	0.5σ	0.0σ	1.2σ	1.6σ	0.7σ	1.2σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	0.5σ	0.4σ	1.2σ	1.6σ	0.7σ	1.4σ	1.5σ	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0.5σ	1.2σ	0.6σ	1.8σ	0.4σ	1.7σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0.6σ	0.4σ	0.3σ	1.3σ	0.3σ	1.1σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	1.6σ	1.2σ	1.2σ	0.8σ	0.4σ	0.8σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0.5σ	0.4σ	0.3σ	1.0σ	0.2σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	0.5σ	0.8σ	0.6σ	1.6σ	0.7σ	1.4σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0.5σ	0.0σ	0.3σ	1.8σ	0.5σ	1.6σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0.5σ	0.0σ	0.6σ	0.3σ	0.5σ	0.8σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0.6σ	0.4σ	0.6σ	1.0σ	0.4σ	1.0σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	0.6σ	0.0σ	0.3σ	0.8σ	0.2σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0.5σ	0.4σ	1.5σ	1.6σ	0.7σ	1.8σ	1.2σ	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	1.6σ	0.4σ	0.3σ	0.8σ	0.3σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0.5σ	0.8σ	0.6σ	0.3σ	0.7σ	0.7σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	1.6σ	0.4σ	0.3σ	1.6σ	0.7σ	1.3σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	1.6σ	0.4σ	**4.2σ WARN**	0.3σ	0.5σ	0.5σ	**4.1σ WARN**	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0.6σ	0.4σ	0.6σ	0.8σ	0.2σ	0.4σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0.5σ	0.4σ	2.4σ	1.4σ	0.7σ	1.6σ	2.3σ	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0.5σ	0.0σ	0.3σ	0.1σ	0.7σ	0.8σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0.6σ	0.0σ	0.6σ	0.6σ	2.0σ	0.6σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0.5σ	0.4σ	0.6σ	0.6σ	2.2σ	0.9σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0.5σ	0.0σ	0.3σ	0.5σ	0.4σ	1.0σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	1.6σ	0.8σ	0.3σ	0.4σ	2.1σ	0.6σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0.6σ	0.4σ	0.3σ	0.8σ	0.3σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	1.6σ	1.2σ	0.3σ	0.3σ	0.7σ	0.7σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	0.5σ	0.4σ	0.3σ	1.4σ	0.8σ	1.4σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	1.6σ	0.4σ	0.3σ	0.3σ	0.7σ	0.7σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0.5σ	0.4σ	0.3σ	0.3σ	0.7σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	1.6σ	0.0σ	0.3σ	0.3σ	0.5σ	0.7σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0.5σ	**5.7σ ALERT**	1.2σ	0.1σ	0.0σ	1.0σ	0.1σ	ALERT
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0.6σ	0.4σ	0.3σ	1.8σ	0.7σ	1.4σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	1.6σ	0.8σ	2.0σ	0.6σ	2.1σ	0.4σ	2.4σ	PASS

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	1.6σ	0.0σ	0.3σ	1.0σ	0.2σ	1.0σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0.6σ	0.0σ	0.6σ	0.4σ	2.2σ	0.8σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0.6σ	0.8σ	0.3σ	0.6σ	2.0σ	0.6σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0.6σ	0.4σ	0.6σ	1.0σ	0.3σ	0.7σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	1.6σ	0.4σ	0.6σ	1.0σ	0.4σ	1.0σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0.6σ	0.4σ	0.3σ	1.3σ	0.4σ	1.1σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0.6σ	0.4σ	0.3σ	1.0σ	0.3σ	0.8σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	1.6σ	0.0σ	0.3σ	0.8σ	0.3σ	0.8σ	0.1σ	PASS

Appendix B: Output Files

- `output/criteria_3_2.csv` – mean, σ and 3σ/5σ/7σ limits per parameter
- `output/judgement_3_2.csv` – value, deviation and level per board per parameter
- `output/plot_3_2_*.png` – distribution plots

← Back to summary

FEMB QC Parameter Evaluation - Report Section 3.3: SE OFF LDO Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Generated: 2026-10-16 16:37
Script: `eval_3_3_se_off_ldo.py`

1. Scope

Section 3.3 of the per-board checkout report lists the outputs of the 11 on-board LDO / monitor points, read in mV by the board monitor ADC while the FEMB is in the SE OFF configuration (same configuration as 3.1 / 3.2). One value per board per point; all 11 are evaluated, using the values as reported in the CSV.

Parameter	Description
`CDVDDA` – COLDATA VDDA	COLDATA analog supply
`CDVDDIO_HALF` – COLDATA VDDIO	COLDATA I/O supply
`ADCRVDDD1P2` – ADC-R VDDD 1.2V	Right-side ColdADC digital 1.2 V supply
`ADCLVDDD1P2` – ADC-L VDDD 1.2V	Left-side ColdADC digital 1.2 V supply
`FERVDDP` – FE-R VDDP	Right-side LArASIC supply
`FELVDDP` – FE-L VDDP	Left-side LArASIC supply
`ADCRP25V_1_HALF` – ADC-R 2.5V #1	Right-side ColdADC 2.5 V supply #1
`ADCLP25V_1_HALF` – ADC-L 2.5V #1	Left-side ColdADC 2.5 V supply #1
`ADCRP25V_HALF` – ADC-R 2.5V	Right-side ColdADC 2.5 V main supply
`ADCLP25V_HALF` – ADC-L 2.5V	Left-side ColdADC 2.5 V main supply
`GND` – Ground	Ground reference reading of the monitor ADC

2. Judgement Method

For each parameter, mean and standard deviation (σ) are computed over all boards; each board is graded by its deviation $|x - \text{mean}| / \sigma$.

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
ALERT	5σ – 7σ	red
FAIL	> 7σ	dark red

3. Summary

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
CDVDDA	core	1185.1 mV	2.1	1180.0	1189.0	9	44	0	0	0	–
CDVDDIO_HALF	core	2231.4 mV	5.8	2221.0	2241.0	20	44	0	0	0	–
ADCRVDDD1P2	core	1097.6 mV	1.9	1094.0	1101.0	8	44	0	0	0	–
ADCLVDDD1P2	core	1097.5 mV	1.9	1094.0	1102.0	9	44	0	0	0	–
FERVDDP	core	1793.4 mV	3.0	1785.0	1799.0	12	44	0	0	0	–
FELVDDP	core	1792.7 mV	3.2	1783.0	1799.0	14	43	1	0	0	00013 (WARN 3.1σ)
ADCRP25V_1_HALF	core	2252.5 mV	5.8	2243.0	2265.0	19	44	0	0	0	–
ADCLP25V_1_HALF	core	2250.4 mV	5.3	2240.0	2259.0	15	44	0	0	0	–
ADCRP25V_HALF	core	2251.7 mV	5.7	2237.0	2264.0	22	44	0	0	0	–
ADCLP25V_HALF	core	2252.1 mV	5.3	2240.0	2265.0	20	44	0	0	0	–
GND	core	29.5 mV	2.6	23.0	33.0	8	44	0	0	0	–

Criteria (limits)

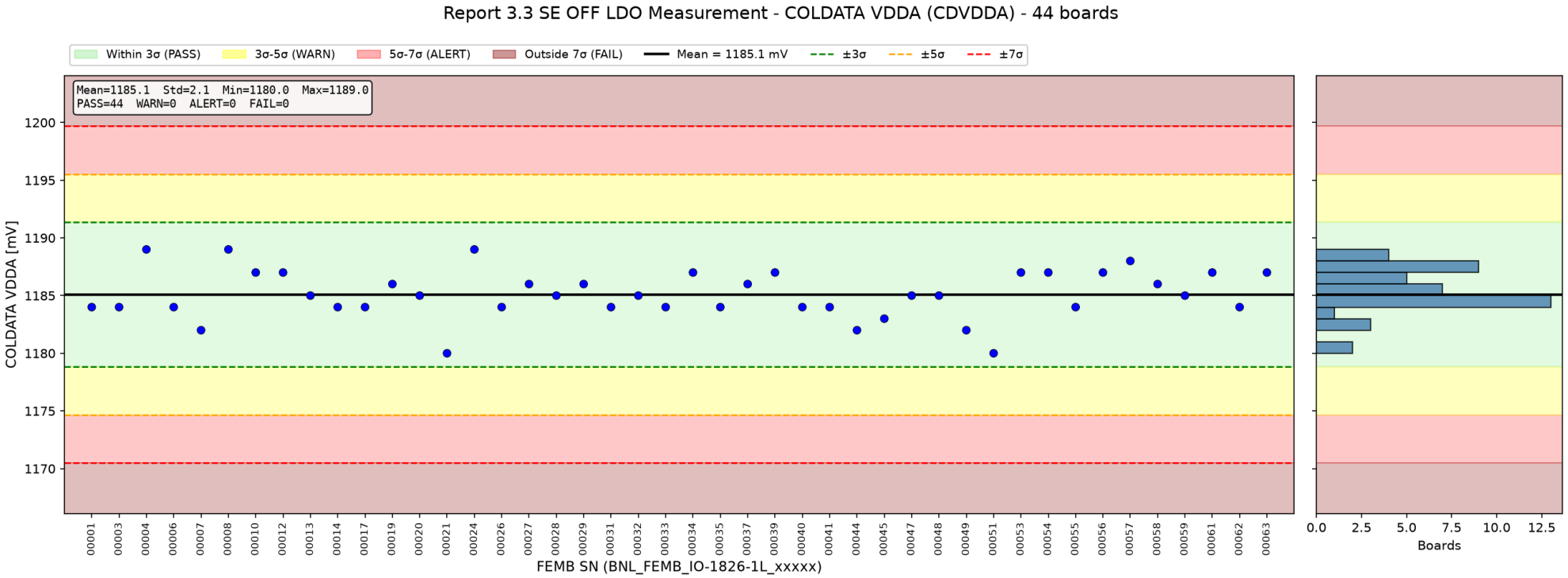
Parameter	Unit	−7σ	−5σ	−3σ	Mean	+3σ	+5σ	+7σ
CDVDDA	mV	1170.5	1174.7	1178.8	1185.1	1191.4	1195.5	1199.7
CDVDDIO_HALF	mV	2190.5	2202.2	2213.9	2231.4	2249.0	2260.6	2272.3
ADCRVDDD1P2	mV	1084.1	1088.0	1091.8	1097.6	1103.4	1107.2	1111.0
ADCLVDDD1P2	mV	1084.5	1088.2	1092.0	1097.5	1103.1	1106.9	1110.6
FERVDDP	mV	1772.4	1778.4	1784.4	1793.4	1802.3	1808.3	1814.3
FELVDDP	mV	1770.4	1776.8	1783.2	1792.7	1802.2	1808.6	1815.0
ADCRP25V_1_HALF	mV	2212.1	2223.6	2235.2	2252.5	2269.8	2281.3	2292.9
ADCLP25V_1_HALF	mV	2213.5	2224.0	2234.6	2250.4	2266.2	2276.7	2287.3
ADCRP25V_HALF	mV	2212.1	2223.4	2234.7	2251.7	2268.6	2279.9	2291.2
ADCLP25V_HALF	mV	2214.9	2225.5	2236.1	2252.1	2268.0	2278.6	2289.2
GND	mV	11.4	16.6	21.7	29.5	37.2	42.4	47.5

4. Parameters

4.1 COLDATA VDDA (`CDVDDA`)

COLDATA analog supply.
Mean = 1185.1 mV, σ = 2.1 mV, range [1180.0, 1189.0], 9 distinct values over 44 boards.

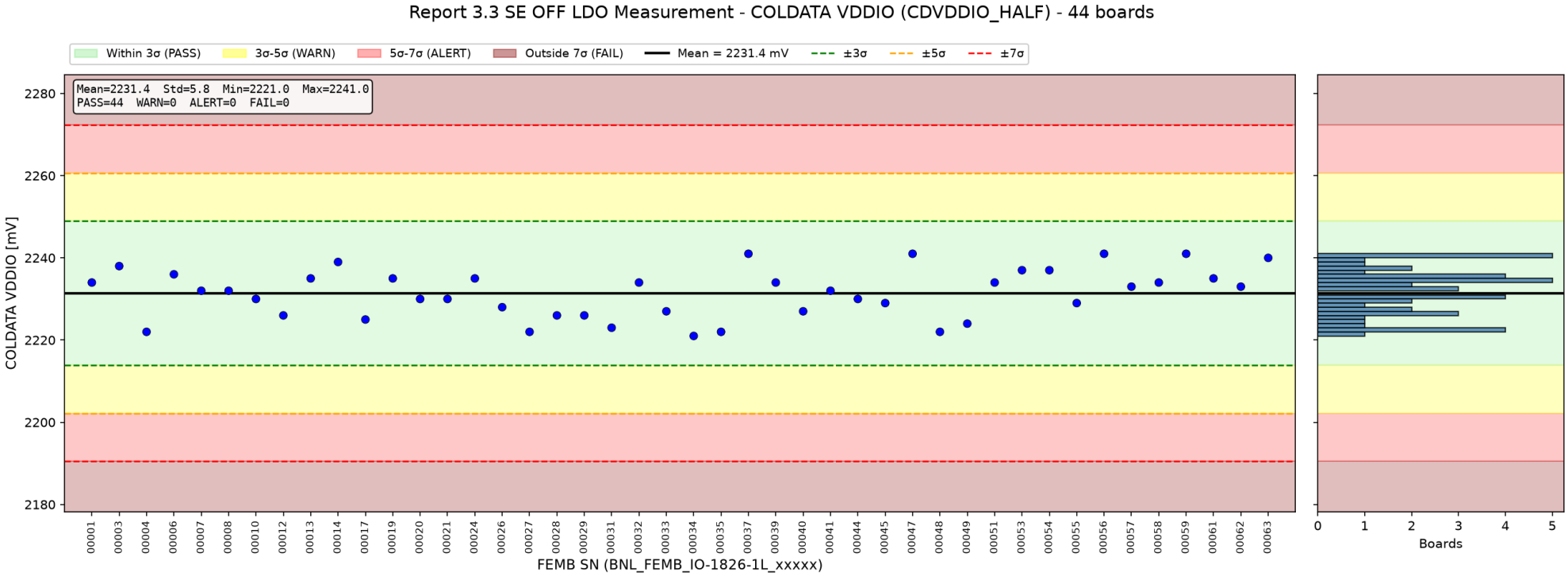
All boards within 3σ.



4.2 COLDATA VDDIO (`CDVDDIO_HALF`)

COLDATA I/O supply.
Mean = 2231.4 mV, σ = 5.8 mV, range [2221.0, 2241.0], 20 distinct values over 44 boards.

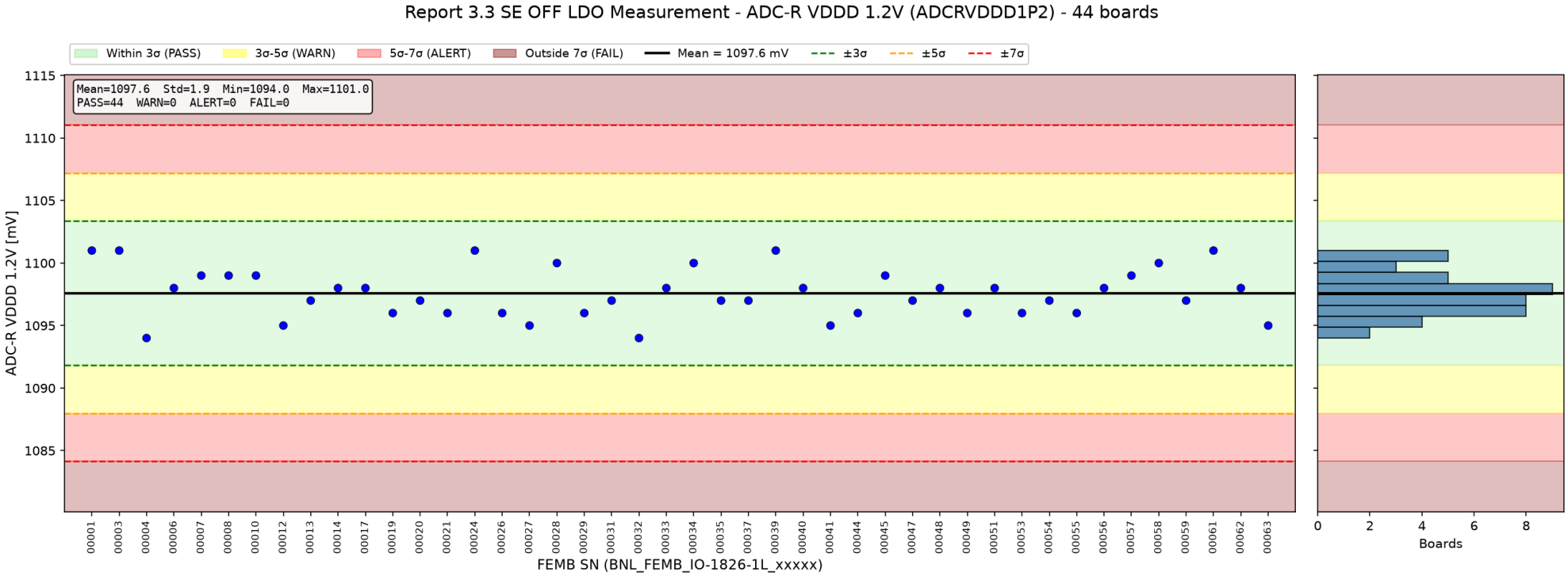
All boards within 3σ.



4.3 ADC-R VDDD 1.2V (`ADCRVDDD1P2`)

Right-side ColdADC digital 1.2 V supply.
Mean = 1097.6 mV, σ = 1.9 mV, range [1094.0, 1101.0], 8 distinct values over 44 boards.

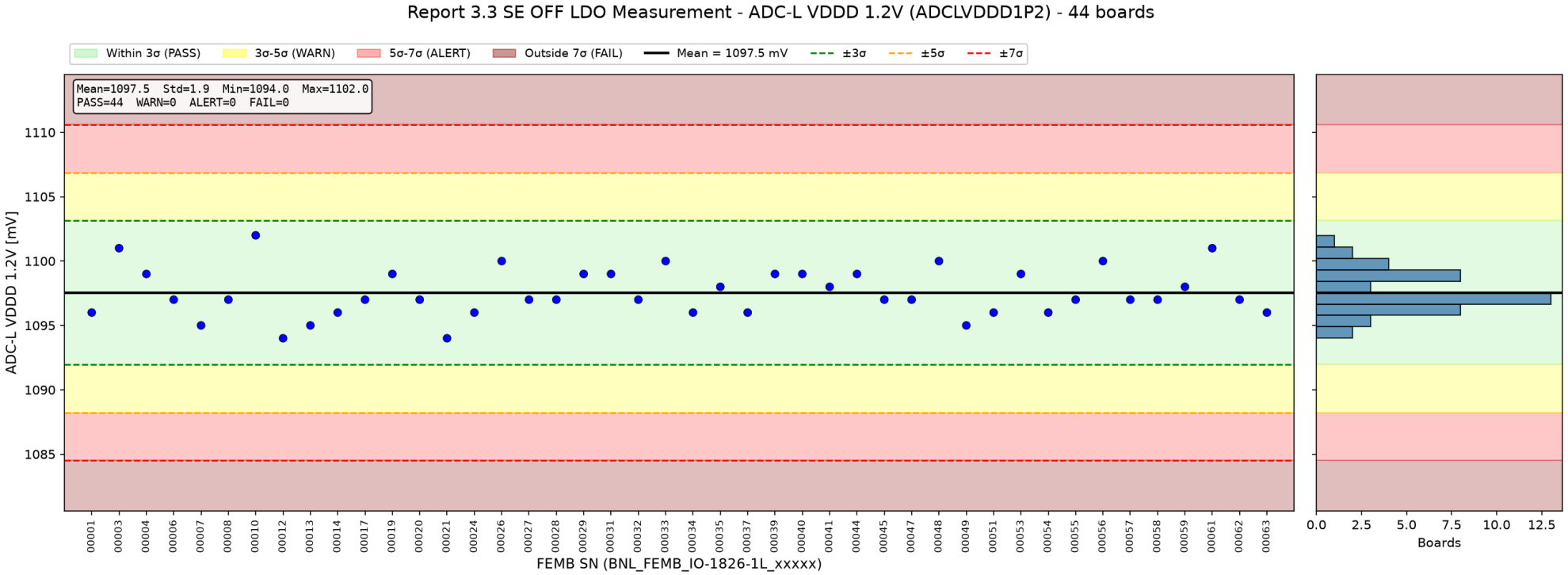
All boards within 3σ.



4.4 ADC-L VDDD 1.2V (`ADCLVDDD1P2`)

Left-side ColdADC digital 1.2 V supply.
Mean = 1097.5 mV, σ = 1.9 mV, range [1094.0, 1102.0], 9 distinct values over 44 boards.

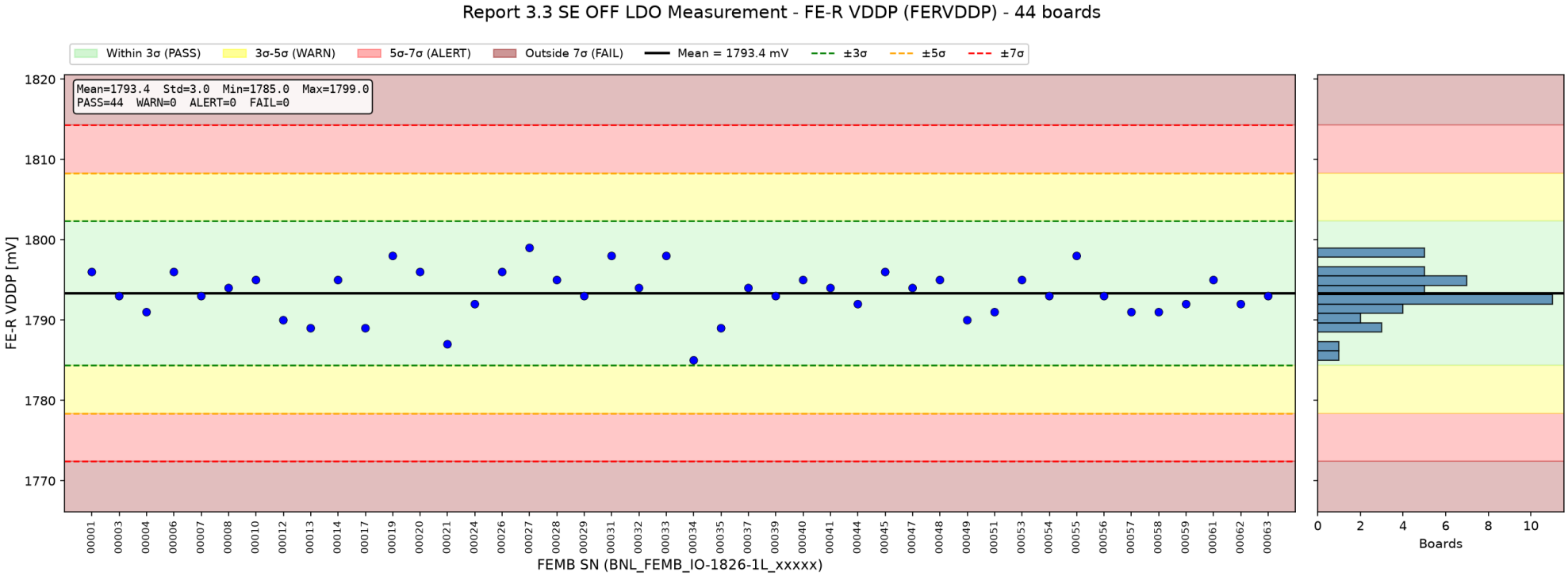
All boards within 3σ.



4.5 FE-R VDDP (`FERVDDP`)

Right-side LArASIC supply.
Mean = 1793.4 mV, σ = 3.0 mV, range [1785.0, 1799.0], 12 distinct values over 44 boards.

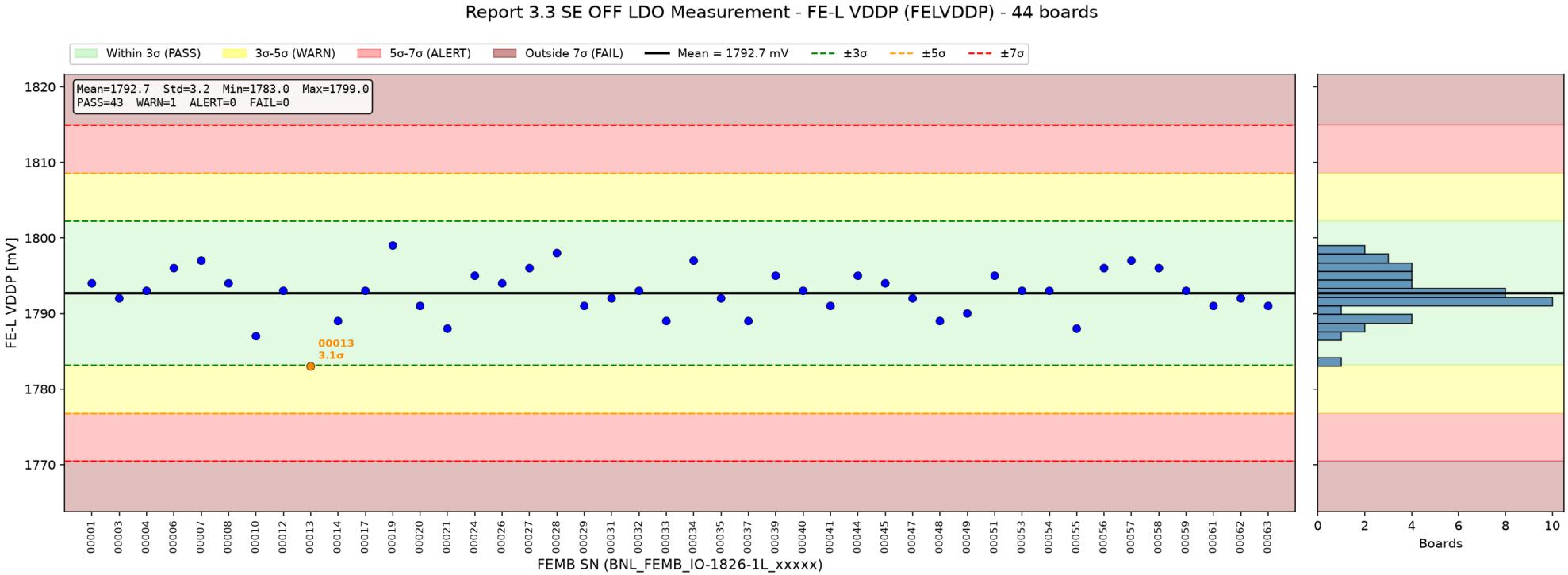
All boards within 3σ.



4.6 FE-L VDDP (`FELVDDP`)

Left-side LArASIC supply.
Mean = 1792.7 mV, σ = 3.2 mV, range [1783.0, 1799.0], 14 distinct values over 44 boards.

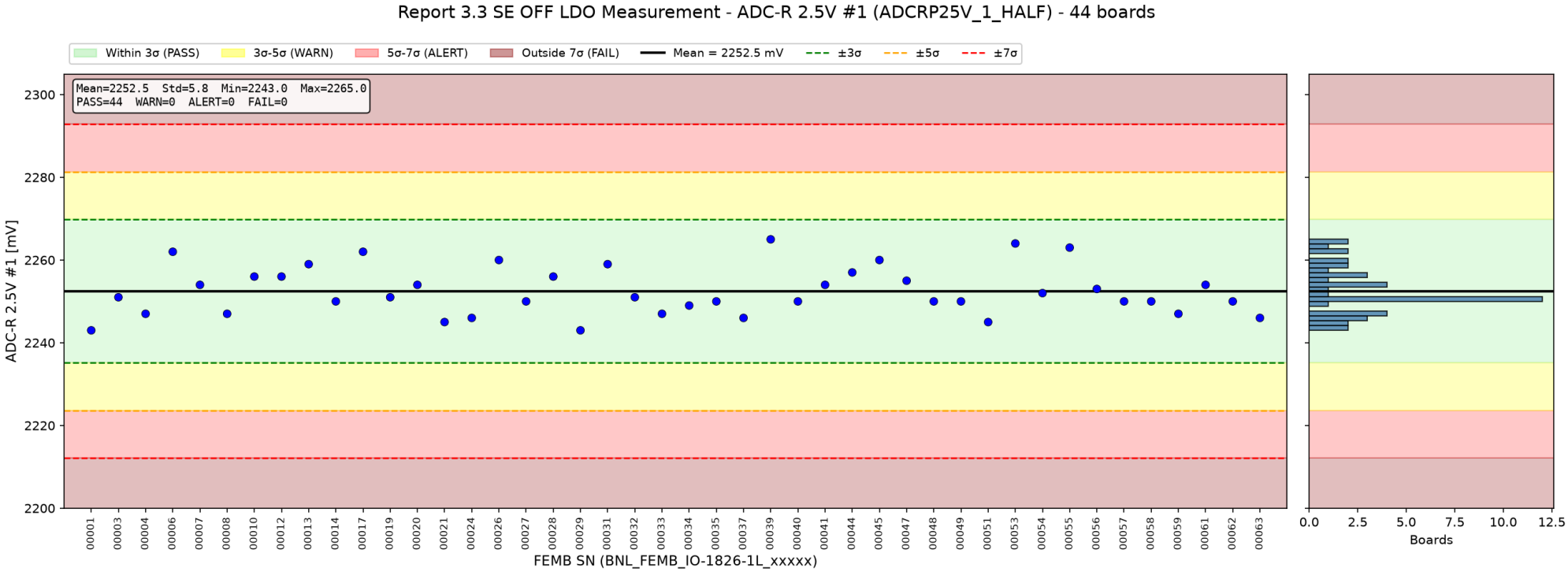
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00013	1783.0 mV	3.1σ	**WARN**



4.7 ADC-R 2.5V #1 (`ADCRP25V_1_HALF`)

Right-side ColdADC 2.5 V supply #1.
Mean = 2252.5 mV, σ = 5.8 mV, range [2243.0, 2265.0], 19 distinct values over 44 boards.

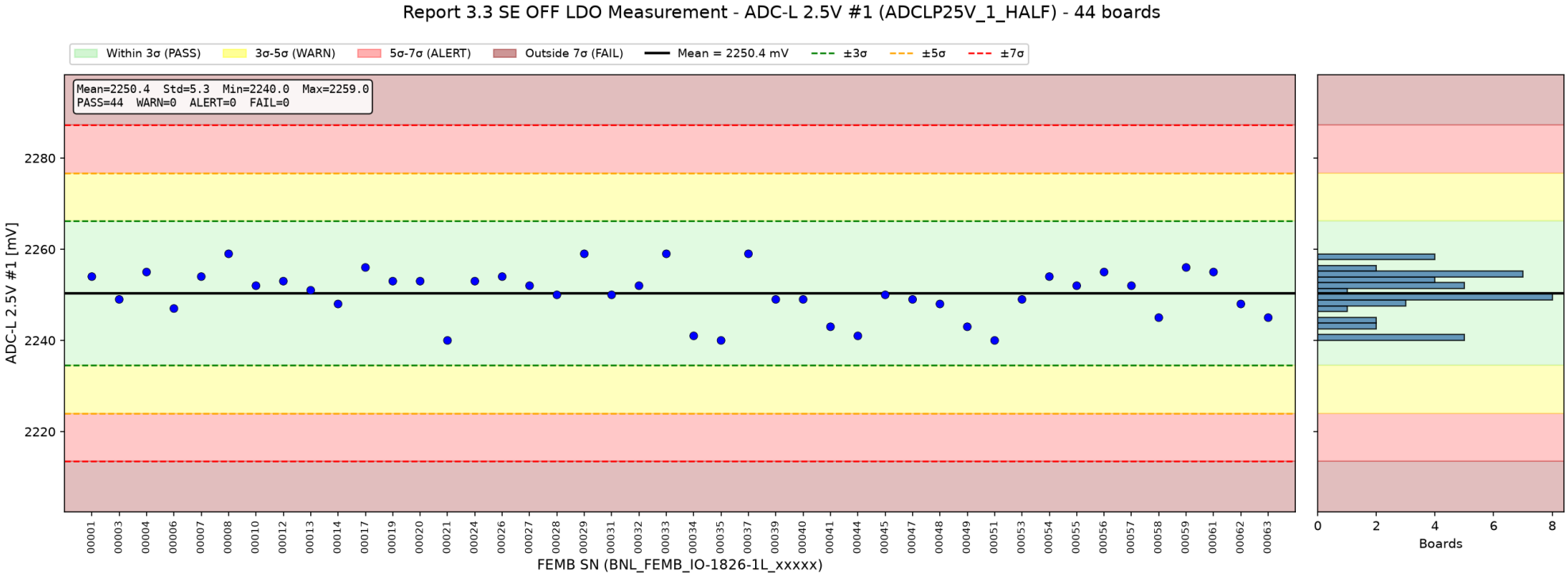
All boards within 3σ.



4.8 ADC-L 2.5V #1 (`ADCLP25V_1_HALF`)

Left-side ColdADC 2.5 V supply #1.
Mean = 2250.4 mV, σ = 5.3 mV, range [2240.0, 2259.0], 15 distinct values over 44 boards.

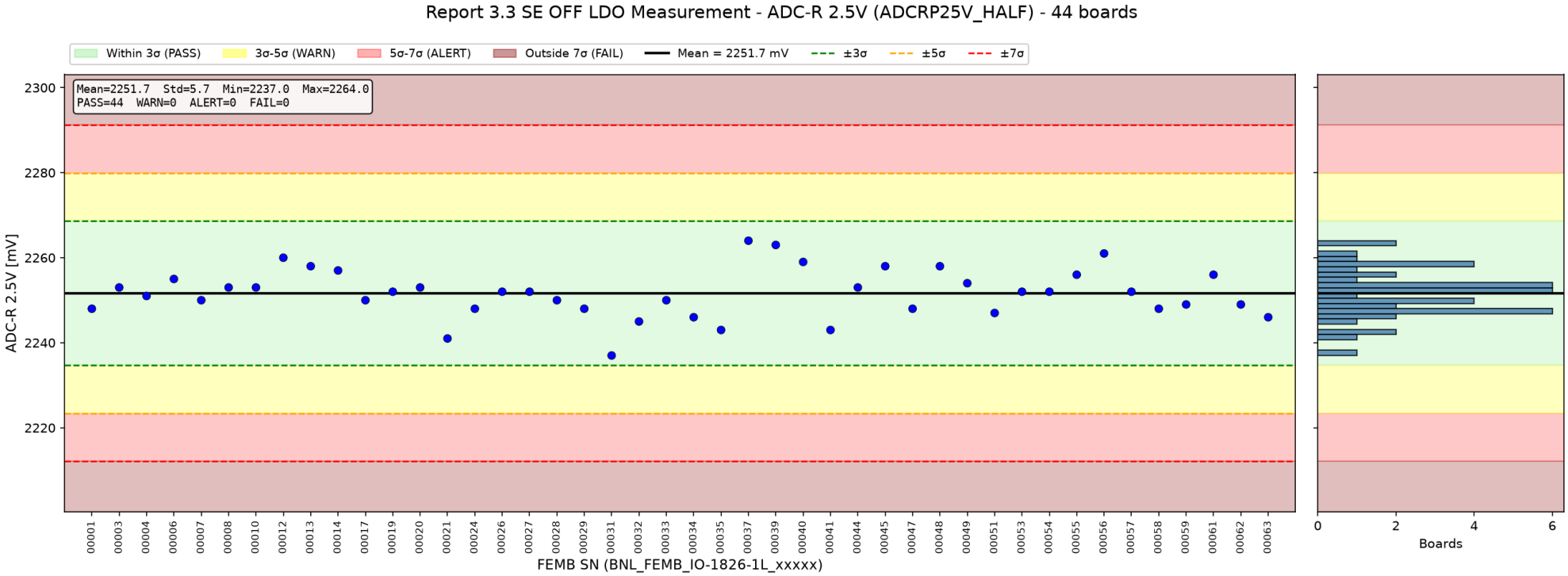
All boards within 3σ.



4.9 ADC-R 2.5V (`ADCRP25V_HALF`)

Right-side ColdADC 2.5 V main supply.
Mean = 2251.7 mV, σ = 5.7 mV, range [2237.0, 2264.0], 22 distinct values over 44 boards.

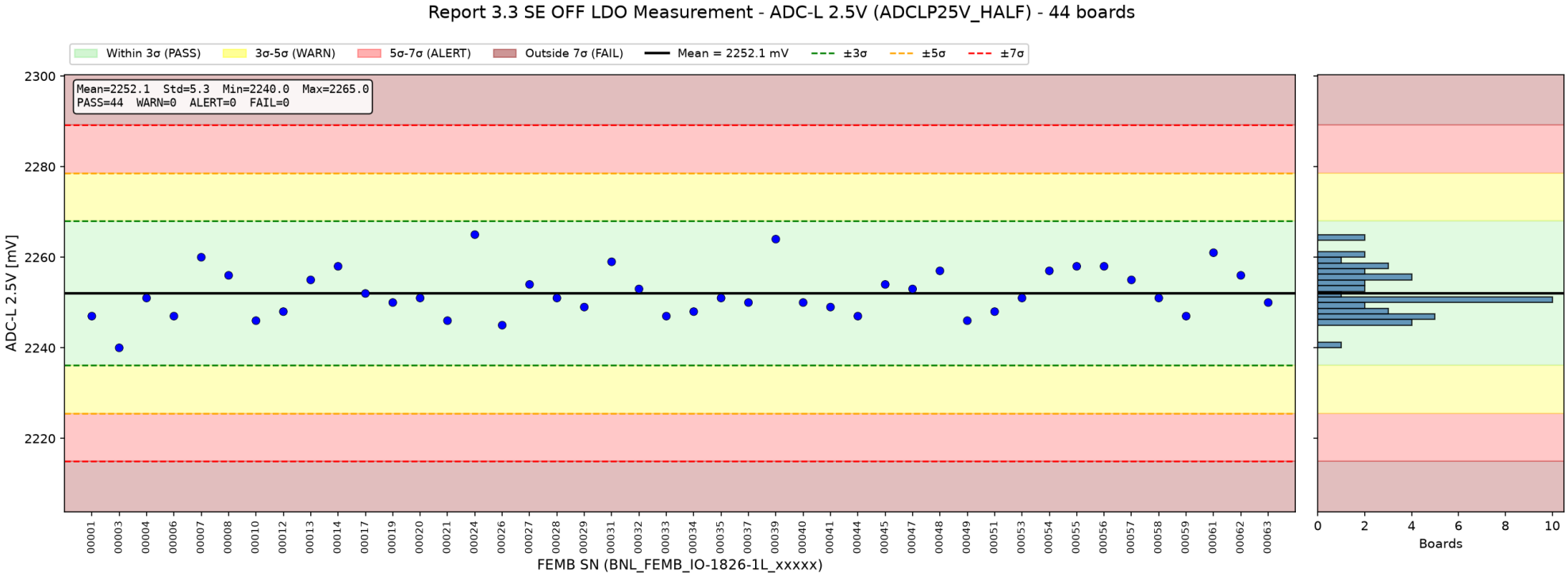
All boards within 3σ.



4.10 ADC-L 2.5V (`ADCLP25V_HALF`)

Left-side ColdADC 2.5 V main supply.
Mean = 2252.1 mV, σ = 5.3 mV, range [2240.0, 2265.0], 20 distinct values over 44 boards.

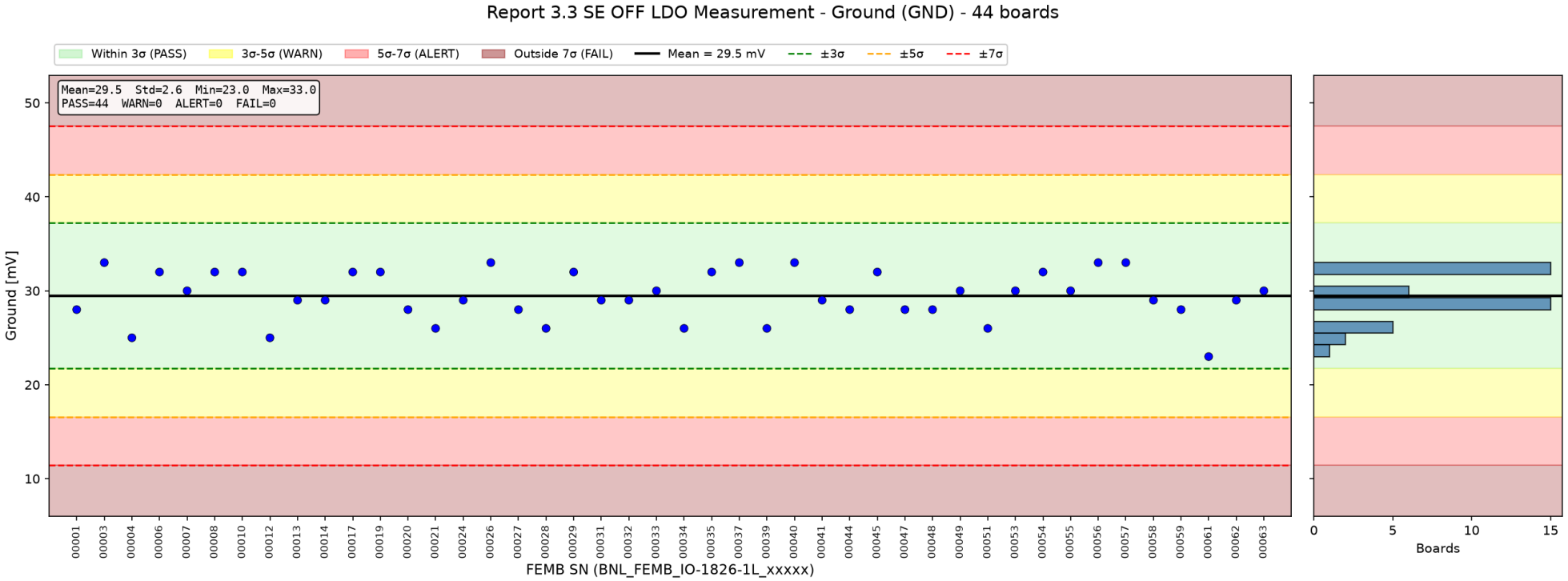
All boards within 3σ.



4.11 Ground (`GND`)

Ground reference reading of the monitor ADC.
Mean = 29.5 mV, σ = 2.6 mV, range [23.0, 33.0], 8 distinct values over 44 boards.

All boards within 3σ.



5. Observations

- **BNL_FEMB_IO-1826-1L_00013**: FELVDDP 3.1σ (WARN)
- Readout resolution is coarse (≤ 10 distinct values) for: CDVDDA, ADCRVDDD1P2, ADCLVDDD1P2, GND. σ is therefore small and quantized; a minimum tolerance tied to instrument resolution should be considered before using these limits.
- Flagged boards are still included in the statistics and inflate σ; re-deriving limits without them would tighten the criteria.

Appendix A: Per-Board Judgement

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0.5σ	0.4σ	1.8σ	0.8σ	0.9σ	0.4σ	1.6σ	0.7σ	0.6σ	1.0σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0.5σ	1.1σ	1.8σ	1.9σ	0.1σ	0.2σ	0.3σ	0.3σ	0.2σ	2.3σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	1.9σ	1.6σ	1.9σ	0.8σ	0.8σ	0.1σ	0.9σ	0.9σ	0.1σ	0.2σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0.5σ	0.8σ	0.2σ	0.3σ	0.9σ	1.0σ	1.7σ	0.6σ	0.6σ	1.0σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	1.5σ	0.1σ	0.7σ	1.4σ	0.1σ	1.4σ	0.3σ	0.7σ	0.3σ	1.5σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	1.9σ	0.1σ	0.7σ	0.3σ	0.2σ	0.4σ	0.9σ	1.6σ	0.2σ	0.7σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0.9σ	0.2σ	0.7σ	2.4σ	0.5σ	1.8σ	0.6σ	0.3σ	0.2σ	1.1σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0.9σ	0.9σ	1.3σ	1.9σ	1.1σ	0.1σ	0.6σ	0.5σ	1.5σ	0.8σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0.0σ	0.6σ	0.3σ	1.4σ	1.5σ	**3.1σ WARN**	1.1σ	0.1σ	1.1σ	0.6σ	0.2σ	WARN
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0.5σ	1.3σ	0.2σ	0.8σ	0.5σ	1.2σ	0.4σ	0.4σ	0.9σ	1.1σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	0.5σ	1.1σ	0.2σ	0.3σ	1.5σ	0.1σ	1.7σ	1.1σ	0.3σ	0.0σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0.4σ	0.6σ	0.8σ	0.8σ	1.5σ	2.0σ	0.3σ	0.5σ	0.1σ	0.4σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0.0σ	0.2σ	0.3σ	0.3σ	0.9σ	0.5σ	0.3σ	0.5σ	0.2σ	0.2σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	2.4σ	0.2σ	0.8σ	1.9σ	2.1σ	1.5σ	1.3σ	2.0σ	1.9σ	1.1σ	1.3σ	PASS

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	1.9σ	0.6σ	1.8σ	0.8σ	0.5σ	0.7σ	1.1σ	0.5σ	0.6σ	2.4σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0.5σ	0.6σ	0.8σ	1.3σ	0.9σ	0.4σ	1.3σ	0.7σ	0.1σ	1.3σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	0.4σ	1.6σ	1.3σ	0.3σ	1.9σ	1.0σ	0.4σ	0.3σ	0.1σ	0.4σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0.0σ	0.9σ	1.3σ	0.3σ	0.5σ	1.7σ	0.6σ	0.1σ	0.3σ	0.2σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0.4σ	0.9σ	0.8σ	0.8σ	0.1σ	0.5σ	1.6σ	1.6σ	0.6σ	0.6σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0.5σ	1.4σ	0.3σ	0.8σ	1.5σ	0.2σ	1.1σ	0.1σ	2.6σ	1.3σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0.0σ	0.4σ	1.9σ	0.3σ	0.2σ	0.1σ	0.3σ	0.3σ	1.2σ	0.2σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0.5σ	0.8σ	0.2σ	1.3σ	1.5σ	1.2σ	0.9σ	1.6σ	0.3σ	1.0σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0.9σ	1.8σ	1.3σ	0.8σ	2.8σ	1.4σ	0.6σ	1.8σ	1.0σ	0.8σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0.5σ	1.6σ	0.3σ	0.2σ	1.5σ	0.2σ	0.4σ	2.0σ	1.5σ	0.2σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0.4σ	1.6σ	0.3σ	0.8σ	0.2σ	1.2σ	1.1σ	1.6σ	2.2σ	0.4σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0.9σ	0.4σ	1.8σ	0.8σ	0.1σ	0.7σ	2.2σ	0.3σ	2.0σ	2.2σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0.5σ	0.8σ	0.2σ	0.8σ	0.5σ	0.1σ	0.4σ	0.3σ	1.3σ	0.4σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0.5σ	0.1σ	1.3σ	0.2σ	0.2σ	0.5σ	0.3σ	1.4σ	1.5σ	0.6σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	1.5σ	0.2σ	0.8σ	0.8σ	0.5σ	0.7σ	0.8σ	1.8σ	0.2σ	1.0σ	0.6σ	PASS

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	1.0σ	0.4σ	0.7σ	0.3σ	0.9σ	0.4σ	1.3σ	0.1σ	1.1σ	0.4σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	0.0σ	1.6σ	0.3σ	0.3σ	0.2σ	0.2σ	0.4σ	0.3σ	0.6σ	0.2σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0.0σ	1.6σ	0.2σ	1.3σ	0.5σ	1.2σ	0.4σ	0.4σ	1.1σ	0.9σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	1.5σ	1.3σ	0.8σ	1.4σ	1.1σ	0.9σ	0.4σ	1.4σ	0.4σ	1.1σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	2.4σ	0.4σ	0.2σ	0.8σ	0.8σ	0.7σ	1.3σ	2.0σ	0.8σ	0.8σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0.9σ	1.0σ	0.8σ	0.8σ	0.5σ	0.1σ	2.0σ	0.3σ	0.1σ	0.2σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	0.9σ	1.0σ	0.3σ	0.8σ	0.1σ	0.1σ	0.1σ	0.7σ	0.1σ	0.9σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0.5σ	0.4σ	0.8σ	0.3σ	1.5σ	1.5σ	1.8σ	0.3σ	0.8σ	1.1σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0.9σ	1.6σ	0.2σ	1.3σ	0.1σ	1.0σ	0.1σ	0.9σ	1.7σ	1.1σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	1.4σ	0.3σ	0.7σ	0.3σ	0.8σ	1.4σ	0.4σ	0.3σ	0.1σ	0.6σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0.4σ	0.4σ	1.3σ	0.3σ	0.8σ	1.0σ	0.4σ	1.0σ	0.6σ	0.2σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0.0σ	1.6σ	0.3σ	0.2σ	0.5σ	0.1σ	0.9σ	1.1σ	0.5σ	1.0σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0.9σ	0.6σ	1.8σ	1.9σ	0.5σ	0.5σ	0.3σ	0.9σ	0.8σ	1.7σ	2.5σ	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0.5σ	0.3σ	0.2σ	0.3σ	0.5σ	0.2σ	0.4σ	0.4σ	0.5σ	0.7σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	0.9σ	1.5σ	1.3σ	0.8σ	0.1σ	0.5σ	1.1σ	1.0σ	1.0σ	0.4σ	0.2σ	PASS

Appendix B: Output Files

- `output/criteria_3_3.csv` – mean, σ and 3σ/5σ/7σ limits per parameter
- `output/judgement_3_3.csv` – value, deviation and level per board per parameter
- `output/plot_3_3_*.png` – distribution plots

← Back to summary

FEMB QC Parameter Evaluation - Report Section 3.4: SE OFF Pulse Response

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Channels: 44 × 128 = 5632
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Configuration: single-ended, SE buffer OFF, 900 mV baseline, 14 mV/fC, 2 µs, calibration pulse DAC 0x10
Generated: 2026-10-16 16:37
Script: `eval_3_4_se_off_pulse.py`

1. Scope

Section 3.4 of the checkout report is the pulse-response test in the SE OFF configuration: the on-chip calibration pulse (DAC 0x10) is injected into every channel and, for each of the 128 channels, the positive peak, the baseline and the negative peak of the pulse are recorded (`se_pulse_ana()`, `report_log07csv`).

The CSV also stores `PPK_max/min`, `NPK_max/min` and `BL_max/min`; for every board they are identical to the corresponding `_mean` array (one value per channel), so only the `_mean` arrays are evaluated.

Parameter	Description
`PPK_mean` – Positive Peak	Positive peak of the averaged calibration pulse, ADC counts
`BL_mean` – Baseline	Baseline level of the pulse waveform, ADC counts
`NPK_mean` – Negative Peak	Negative peak (undershoot minimum) of the pulse waveform, ADC counts

2. Judgement Method

For each parameter the mean and σ are computed over all 5632 channels pooled together; each channel is graded by |x − mean| / σ. A board takes the worst level among its channels.

Channel-level (noise / pulse) parameters use **5σ as the limit**: a channel beyond 5σ fails. (Board-level power and LDO parameters keep the wider 7σ limit.)

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
FAIL	> 5σ	red

For a Gaussian distribution, about 0.27 % of channels (≈ 15 of 5632) are expected beyond 3σ by chance, so isolated WARN channels are normal.

3. Summary

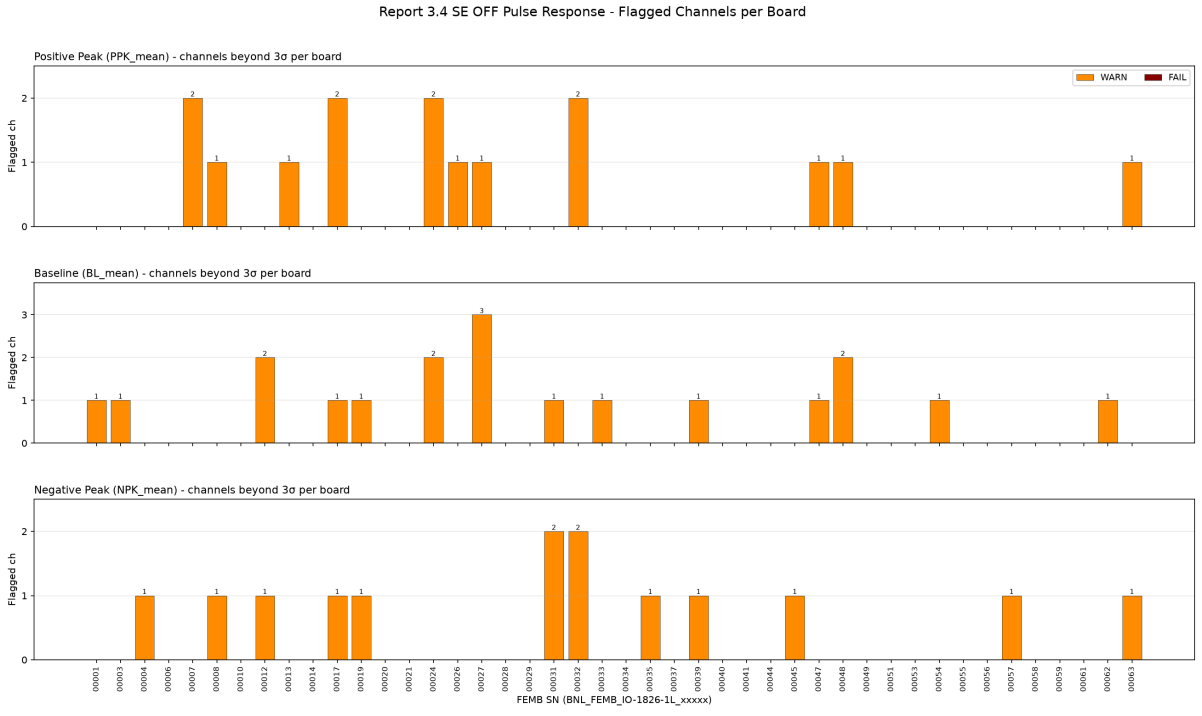
Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
`PPK_mean`	12747.59	94.23	12375.00	13073.00	5617	15	0	–
`BL_mean`	8579.36	70.03	8330.00	8848.00	5613	19	0	–

Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
NPK_mean	4410.82	85.63	4143.00	4745.00	5618	14	0	–

Criteria (limits, ADC counts)

Parameter	−5σ (FAIL)	−3σ	Mean	+3σ	+5σ (FAIL)
PPK_mean	12276.44	12464.90	12747.59	13030.28	13218.73
BL_mean	8229.19	8369.26	8579.36	8789.46	8929.53
NPK_mean	3982.66	4153.93	4410.82	4667.71	4838.97

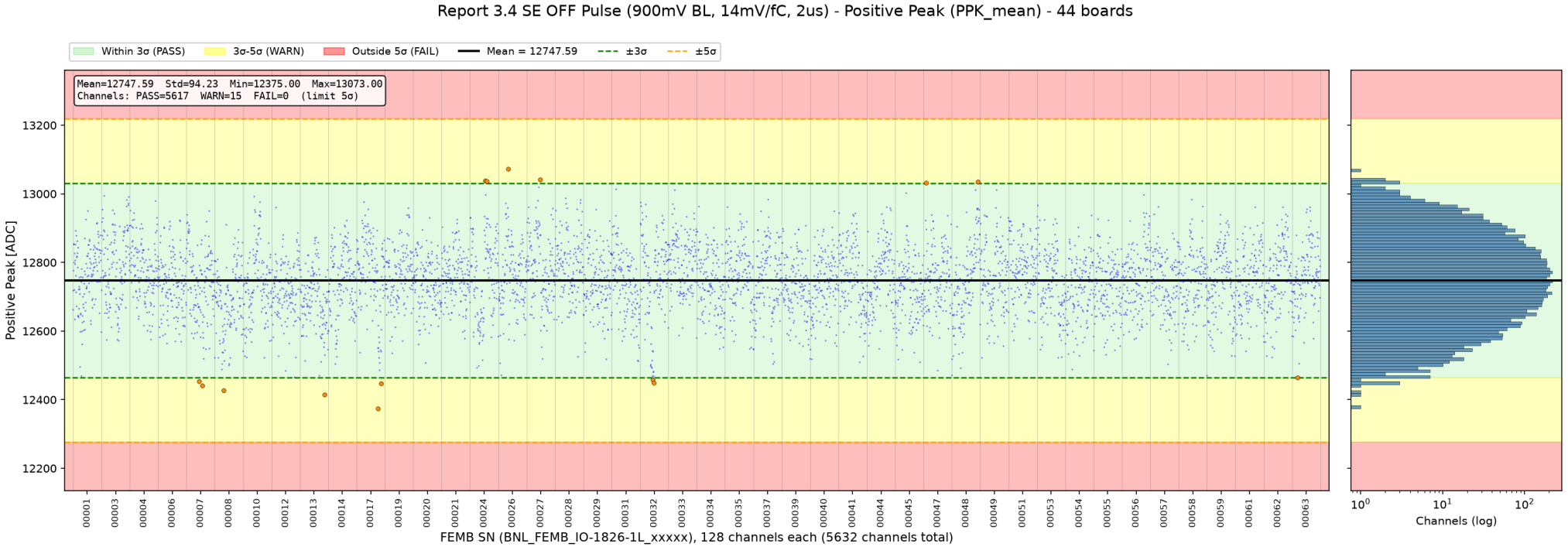
Flagged channels per board



4. Parameters

4.1 Positive Peak (`PPK_mean`)

Mean = 12747.59, σ = 94.23, range [12375.00, 13073.00] ADC. Spread of per-board means: σ = 22.56; spread of per-channel means across boards: σ = 45.81.



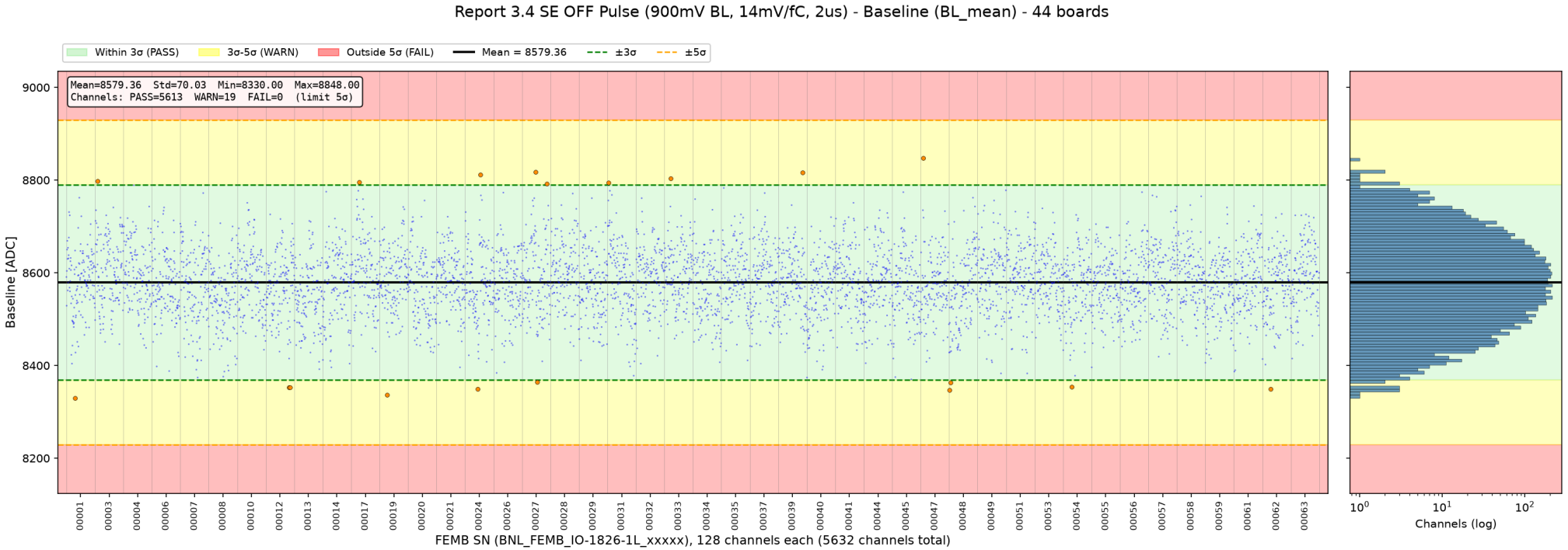
15 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00017	96	6	12375.00	4.0σ	WARN
BNL_FEMB_IO-1826-1L_00013	112	7	12414.00	3.5σ	WARN
BNL_FEMB_IO-1826-1L_00026	44	2	13073.00	3.5σ	WARN
BNL_FEMB_IO-1826-1L_00008	40	2	12427.00	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00007	72	4	12441.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00017	110	6	12448.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00032	63	3	12449.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00007	56	3	12453.00	3.1σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00027	60	3	13041.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00024	68	4	13039.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00032	58	3	12458.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00024	76	4	13037.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00048	116	7	13036.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00047	11	0	13033.00	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00063	24	1	12464.00	3.0σ	WARN

4.2 Baseline (`BL_mean`)

Mean = 8579.36, σ = 70.03, range [8330.00, 8848.00] ADC. Spread of per-board means: σ = 13.60; spread of per-channel means across boards: σ = 28.11.



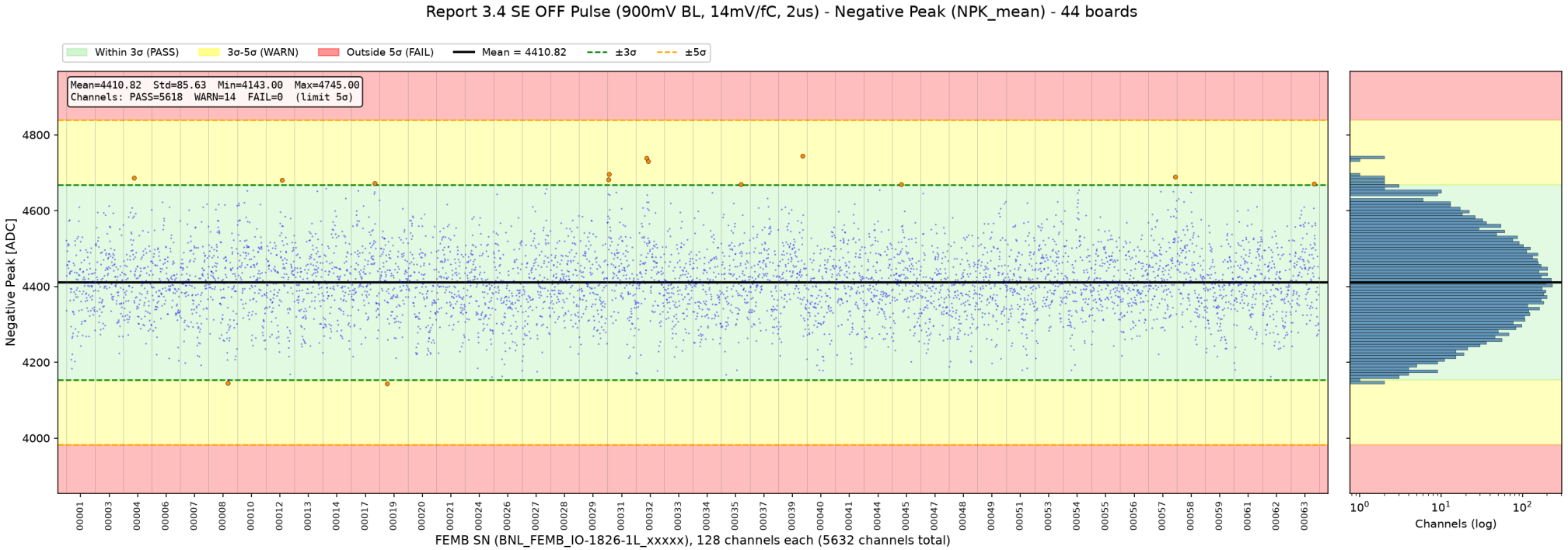
19 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00047	11	0	8848.00	3.8σ	WARN
BNL_FEMB_IO-1826-1L_00001	38	2	8330.00	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00019	33	2	8337.00	3.5σ	WARN
BNL_FEMB_IO-1826-1L_00027	60	3	8817.00	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00039	109	6	8816.00	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00024	68	4	8812.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00048	1	0	8347.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00024	57	3	8349.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00062	38	2	8349.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00012	104	6	8353.00	3.2σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00012	110	6	8353.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00054	38	2	8354.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00033	29	1	8804.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00003	13	0	8798.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00048	6	0	8363.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00017	36	2	8795.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00027	68	4	8364.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00031	5	0	8794.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00027	111	6	8792.00	3.0σ	WARN

4.3 Negative Peak (`NPK_mean`)

Mean = 4410.82, σ = 85.63, range [4143.00, 4745.00] ADC. Spread of per-board means: σ = 16.70; spread of per-channel means across boards: σ = 44.88.



14 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00039	109	6	4745.00	3.9σ	WARN
BNL_FEMB_IO-1826-1L_00032	48	3	4739.00	3.8σ	WARN
BNL_FEMB_IO-1826-1L_00032	56	3	4730.00	3.7σ	WARN
BNL_FEMB_IO-1826-1L_00031	8	0	4697.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00057	120	7	4689.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00004	47	2	4686.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00031	5	0	4682.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00012	72	4	4681.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00019	33	2	4143.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00008	86	5	4145.00	3.1σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00017	107	6	4672.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00063	104	6	4671.00	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00035	88	5	4669.00	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00045	40	2	4669.00	3.0σ	WARN

5. Observations

- **PPK_mean**: no channel beyond 5σ.
- **BL_mean**: no channel beyond 5σ.
- **NPK_mean**: no channel beyond 5σ.

Appendix A: Per-Board Judgement

Each cell: number of channels WARN / FAIL, and worst deviation.

FEMB SN	Test time	PPK_mean	BL_mean	NPK_mean	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0/0 (3.0σ)	1/0 (3.6σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0/0 (2.6σ)	1/0 (3.1σ)	0/0 (2.3σ)	WARN
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	0/0 (2.5σ)	0/0 (3.0σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0/0 (1.9σ)	0/0 (2.7σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	2/0 (3.3σ)	0/0 (2.9σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	1/0 (3.4σ)	0/0 (2.9σ)	1/0 (3.1σ)	WARN
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0/0 (2.6σ)	0/0 (2.5σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0/0 (3.0σ)	2/0 (3.2σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	1/0 (3.5σ)	0/0 (2.7σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0/0 (2.6σ)	0/0 (2.8σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	2/0 (4.0σ)	1/0 (3.1σ)	1/0 (3.1σ)	WARN
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0/0 (2.3σ)	1/0 (3.5σ)	1/0 (3.1σ)	WARN
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0/0 (2.6σ)	0/0 (2.5σ)	0/0 (2.2σ)	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0/0 (1.9σ)	0/0 (2.6σ)	0/0 (2.4σ)	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	2/0 (3.1σ)	2/0 (3.3σ)	0/0 (2.9σ)	WARN
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	1/0 (3.5σ)	0/0 (2.8σ)	0/0 (2.9σ)	WARN
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	1/0 (3.1σ)	3/0 (3.4σ)	0/0 (2.9σ)	WARN
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0/0 (2.5σ)	0/0 (2.7σ)	0/0 (2.2σ)	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0/0 (2.4σ)	0/0 (2.6σ)	0/0 (2.0σ)	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0/0 (2.8σ)	1/0 (3.1σ)	2/0 (3.3σ)	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	2/0 (3.2σ)	0/0 (2.6σ)	2/0 (3.8σ)	WARN
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0/0 (2.8σ)	1/0 (3.2σ)	0/0 (2.4σ)	WARN
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0/0 (2.3σ)	0/0 (2.6σ)	0/0 (2.8σ)	PASS

FEMB SN	Test time	PPK_mean	BL_mean	NPK_mean	Worst
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0/0 (2.3σ)	0/0 (2.9σ)	1/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0/0 (2.6σ)	0/0 (2.8σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0/0 (2.0σ)	1/0 (3.4σ)	1/0 (3.9σ)	WARN
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0/0 (2.4σ)	0/0 (2.9σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0/0 (2.5σ)	0/0 (2.5σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	0/0 (2.3σ)	0/0 (2.5σ)	0/0 (3.0σ)	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	0/0 (2.7σ)	0/0 (3.0σ)	1/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	1/0 (3.0σ)	1/0 (3.8σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	1/0 (3.1σ)	2/0 (3.3σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	0/0 (2.8σ)	0/0 (2.8σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0/0 (1.9σ)	0/0 (2.8σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0/0 (2.4σ)	0/0 (2.3σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	0/0 (2.8σ)	1/0 (3.2σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0/0 (2.3σ)	0/0 (2.6σ)	0/0 (2.0σ)	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0/0 (2.6σ)	0/0 (2.8σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0/0 (2.6σ)	0/0 (2.7σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0/0 (2.5σ)	0/0 (2.7σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0/0 (2.9σ)	0/0 (2.5σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0/0 (2.3σ)	0/0 (2.7σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0/0 (2.4σ)	1/0 (3.3σ)	0/0 (2.9σ)	WARN
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	1/0 (3.0σ)	0/0 (2.3σ)	1/0 (3.0σ)	WARN

Appendix B: Output Files

- `output/criteria_3_4.csv` – mean, σ and 3σ/5σ limits per parameter
- `output/judgement_3_4_channels.csv` – every channel beyond 3σ (board, channel, value, deviation, level)
- `output/judgement_3_4_boards.csv` – per-board counts and worst level
- `output/plot_3_4_*.png` – distribution plots and per-board summary

← Back to summary

FEMB QC Parameter Evaluation - Report Section 4.1: DIFF Pulse Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Channels: 44 × 128 = 5632
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Configuration: differential ColdADC input (SDD on), 900 mV baseline, 14 mV/fC, 2 µs, calibration pulse DAC 0x10
Generated: 2026-10-16 16:37
Script: `eval_4_1_diff_pulse.py`

1. Scope

Section 4.1 of the checkout report repeats the pulse-response test of section 3.4 with the ColdADC in differential mode (FE differential driver SDD on): the calibration pulse (DAC 0x10) is injected into every channel and the positive peak, baseline and negative peak are measured.

Data source. The checkout CSV stores only the 4.1 True/False result. The per-channel values are read from `Pulse_Raw_DIFF_900mVBL_14_0mVfC_2_0us_0x10.bin` in each board's report folder, the pickle `[PPK, NPK, BL]` written by `QC_tools.GetPeaks()` during the checkout. (Cross-check: the same file type for section 3.4 agrees with the 3.4 CSV values to < 1 ADC; the CSV values are rounded.)

Parameter	Description
`PPK_mean` – Positive Peak	Positive peak of the averaged calibration pulse, ADC counts
`BL_mean` – Baseline	Baseline level of the pulse waveform, ADC counts
`NPK_mean` – Negative Peak	Negative peak (undershoot minimum) of the pulse waveform, ADC counts

2. Judgement Method

For each parameter the mean and σ are computed over all 5632 channels pooled together; each channel is graded by |x − mean| / σ. A board takes the worst level among its channels.

Channel-level (noise / pulse) parameters use **5σ as the limit**: a channel beyond 5σ fails. (Board-level power and LDO parameters keep the wider 7σ limit.)

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
FAIL	> 5σ	red

For a Gaussian distribution, about 0.27 % of channels (≈ 15 of 5632) are expected beyond 3σ by chance, so isolated WARN channels are normal.

3. Summary

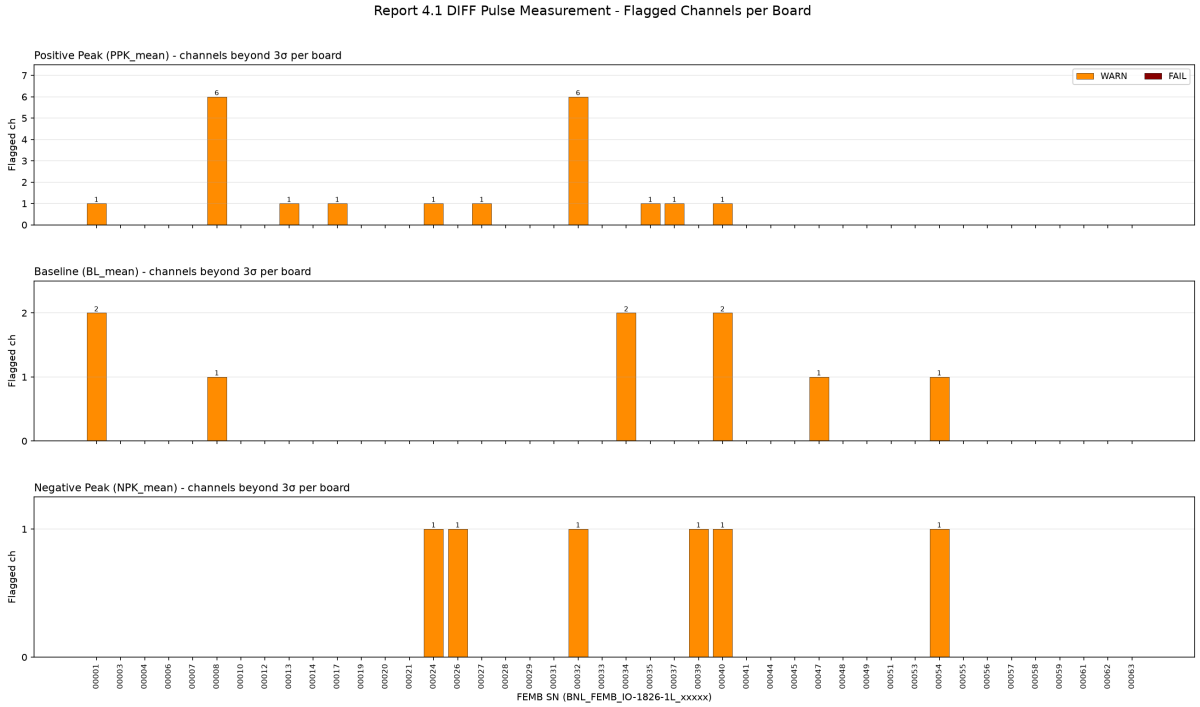
Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
`PPK_mean`	12981.12	73.96	12696.50	13231.50	5612	20	0	–

Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
BL_mean	8759.93	55.69	8559.92	8977.40	5623	9	0	–
NPK_mean	4397.00	70.25	4158.25	4637.50	5626	6	0	–

Criteria (limits, ADC counts)

Parameter	−5σ (FAIL)	−3σ	Mean	+3σ	+5σ (FAIL)
PPK_mean	12611.30	12759.23	12981.12	13203.01	13350.94
BL_mean	8481.49	8592.86	8759.93	8926.99	9038.37
NPK_mean	4045.76	4186.25	4397.00	4607.75	4748.24

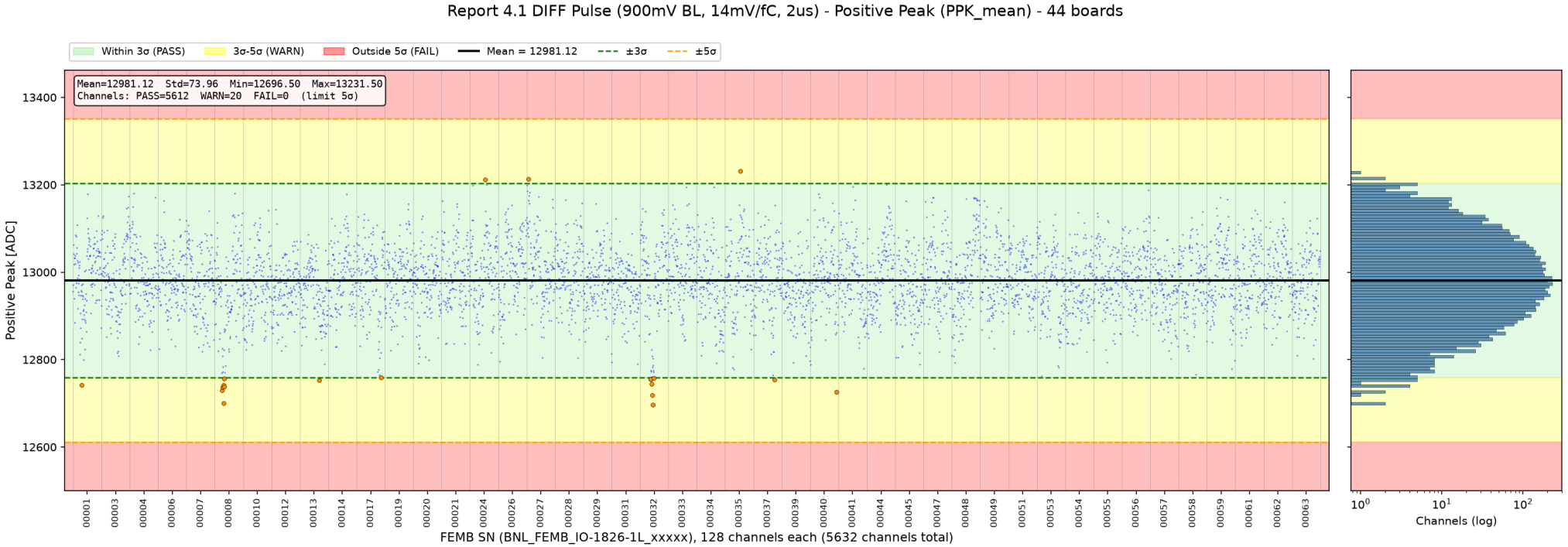
Flagged channels per board



4. Parameters

4.1 Positive Peak (`PPK_mean`)

Mean = 12981.12, σ = 73.96, range [12696.50, 13231.50] ADC. Spread of per-board means: σ = 15.44; spread of per-channel means across boards: σ = 22.16.



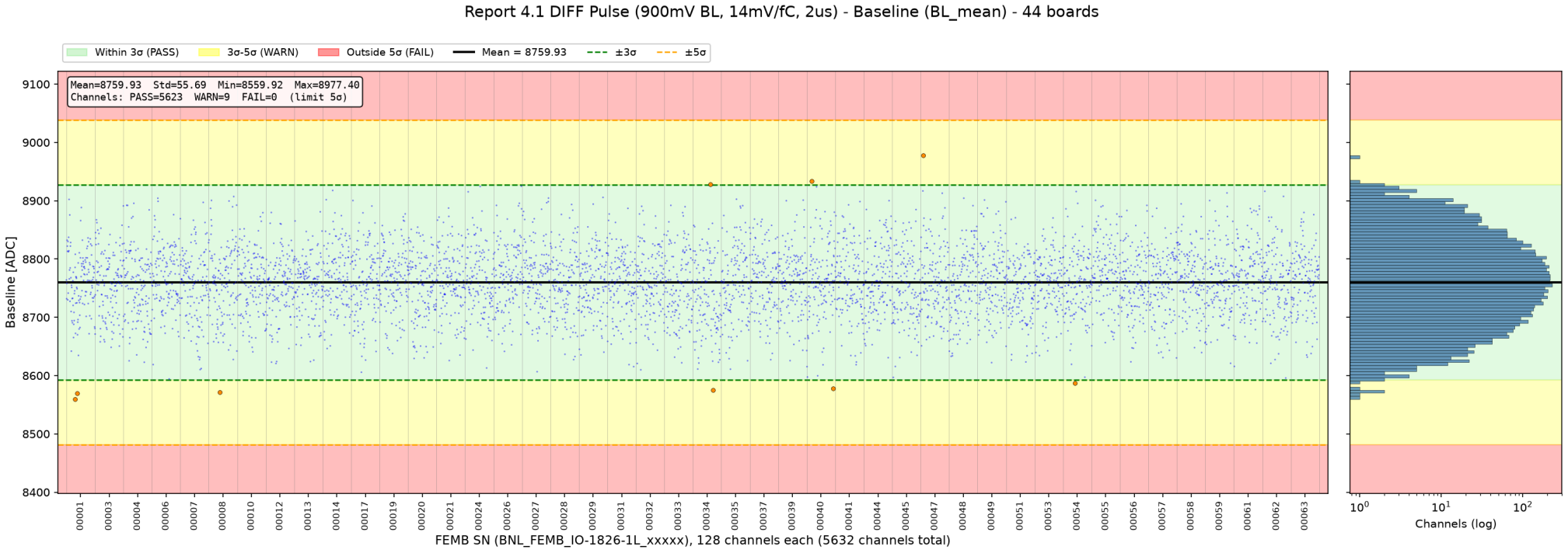
20 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00032	58	3	12696.50	3.8σ	WARN
BNL_FEMB_IO-1826-1L_00008	40	2	12700.75	3.8σ	WARN
BNL_FEMB_IO-1826-1L_00032	55	3	12718.50	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00040	118	7	12726.00	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00008	33	2	12729.00	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00035	68	4	13231.50	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00008	34	2	12737.00	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00008	43	2	12738.00	3.3σ	WARN

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00008	39	2	12741.75	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00001	38	2	12742.00	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00032	52	3	12743.75	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00027	7	0	13213.50	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00024	68	4	13211.75	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00013	88	5	12752.50	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00037	94	5	12753.50	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00032	48	3	12755.00	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00032	49	3	12755.50	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00008	41	2	12756.00	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00032	63	3	12757.50	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00017	110	6	12758.50	3.0σ	WARN

4.2 Baseline (`BL_mean`)

Mean = 8759.93, σ = 55.69, range [8559.92, 8977.40] ADC. Spread of per-board means: σ = 7.36; spread of per-channel means across boards: σ = 17.82.

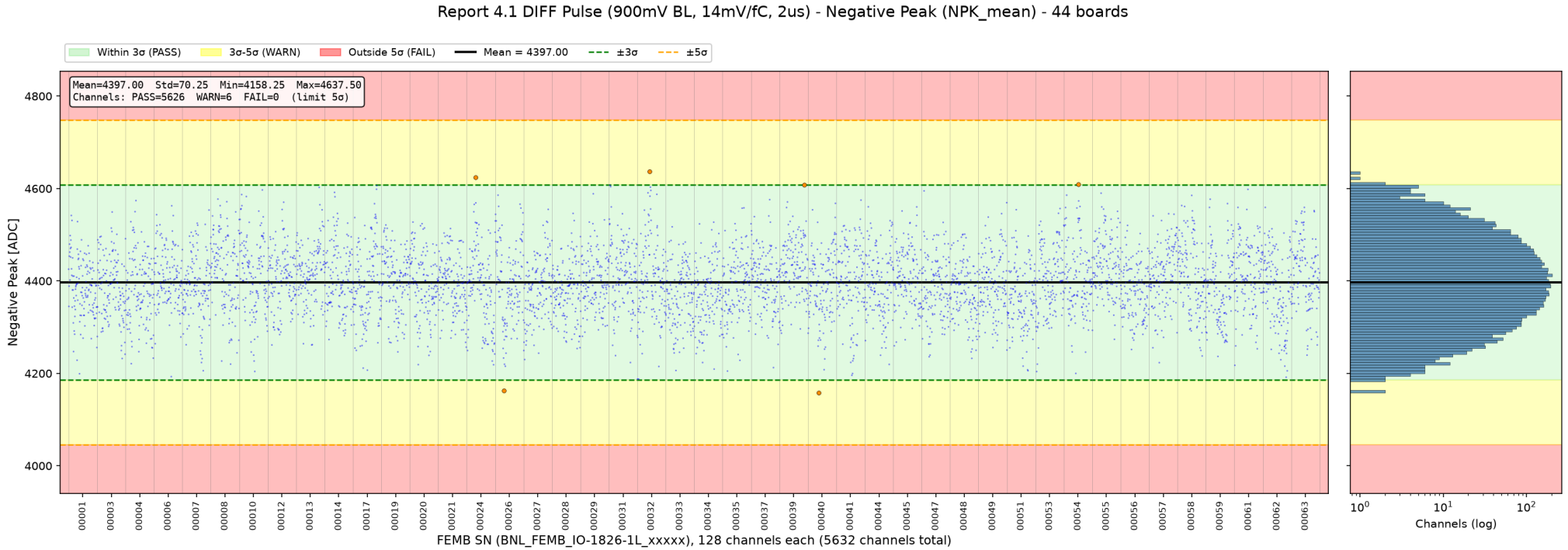


9 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00047	11	0	8977.40	3.9σ	WARN
BNL_FEMB_IO-1826-1L_00001	38	2	8559.92	3.6σ	WARN
BNL_FEMB_IO-1826-1L_00001	49	3	8569.29	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00008	49	3	8571.13	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00034	90	5	8575.50	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00040	118	7	8577.80	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00040	21	1	8933.56	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00054	54	3	8586.76	3.1σ	WARN
BNL_FEMB_IO-1826-1L_00034	79	4	8928.45	3.0σ	WARN

4.3 Negative Peak (`NPK_mean`)

Mean = 4397.00, σ = 70.25, range [4158.25, 4637.50] ADC. Spread of per-board means: σ = 16.47; spread of per-channel means across boards: σ = 23.71.



6 channel(s) beyond 3σ:

FEMB SN	Channel	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00032	56	3	4637.50	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00040	49	3	4158.25	3.4σ	WARN
BNL_FEMB_IO-1826-1L_00026	39	2	4163.25	3.3σ	WARN
BNL_FEMB_IO-1826-1L_00024	40	2	4624.50	3.2σ	WARN
BNL_FEMB_IO-1826-1L_00054	66	4	4609.75	3.0σ	WARN
BNL_FEMB_IO-1826-1L_00039	112	7	4607.75	3.0σ	WARN

5. Observations

- **PPK_mean**: no channel beyond 5σ.
- **BL_mean**: no channel beyond 5σ.

- **NPK_mean**: no channel beyond 5σ.
- **PPK_mean**: BNL_FEMB_IO-1826-1L_00008 chip 2 has 6 flagged channels (ch33, ch34, ch39, ch40, ch41, ch43) – clustered on one chip, suggests a chip-level offset rather than random channels.
- **PPK_mean**: BNL_FEMB_IO-1826-1L_00032 chip 3 has 6 flagged channels (ch48, ch49, ch52, ch55, ch58, ch63) – clustered on one chip, suggests a chip-level offset rather than random channels.

Appendix A: Per-Board Judgement

Each cell: number of channels WARN / FAIL, and worst deviation.

FEMB SN	Test time	PPK_mean	BL_mean	NPK_mean	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	1/0 (3.2σ)	2/0 (3.6σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0/0 (2.6σ)	0/0 (2.0σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	0/0 (2.7σ)	0/0 (2.5σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0/0 (2.5σ)	0/0 (3.0σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	0/0 (2.2σ)	0/0 (2.7σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	6/0 (3.8σ)	1/0 (3.4σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0/0 (2.2σ)	0/0 (2.1σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0/0 (2.2σ)	0/0 (2.5σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	1/0 (3.1σ)	0/0 (2.4σ)	0/0 (2.9σ)	WARN
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0/0 (2.7σ)	0/0 (2.8σ)	0/0 (2.8σ)	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	1/0 (3.0σ)	0/0 (2.8σ)	0/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0/0 (2.4σ)	0/0 (2.2σ)	0/0 (2.1σ)	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0/0 (2.6σ)	0/0 (2.3σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0/0 (2.5σ)	0/0 (2.4σ)	0/0 (2.0σ)	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	1/0 (3.1σ)	0/0 (3.0σ)	1/0 (3.2σ)	WARN
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0/0 (2.5σ)	0/0 (3.0σ)	1/0 (3.3σ)	WARN
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	1/0 (3.1σ)	0/0 (2.8σ)	0/0 (2.4σ)	WARN
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0/0 (2.5σ)	0/0 (2.4σ)	0/0 (2.1σ)	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0/0 (2.4σ)	0/0 (3.0σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0/0 (2.6σ)	0/0 (2.6σ)	0/0 (3.0σ)	PASS
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	6/0 (3.8σ)	0/0 (2.7σ)	1/0 (3.4σ)	WARN
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0/0 (2.9σ)	0/0 (2.5σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0/0 (2.9σ)	2/0 (3.3σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	1/0 (3.4σ)	0/0 (2.7σ)	0/0 (2.7σ)	WARN
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	1/0 (3.1σ)	0/0 (2.4σ)	0/0 (2.6σ)	WARN
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0/0 (2.5σ)	0/0 (2.5σ)	1/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	1/0 (3.4σ)	2/0 (3.3σ)	1/0 (3.4σ)	WARN
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0/0 (2.9σ)	0/0 (2.5σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	0/0 (3.0σ)	0/0 (2.5σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	0/0 (2.6σ)	0/0 (2.9σ)	0/0 (2.3σ)	PASS

FEMB SN	Test time	PPK_mean	BL_mean	NPK_mean	Worst
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	0/0 (3.0σ)	1/0 (3.9σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0/0 (2.6σ)	0/0 (2.6σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	0/0 (2.5σ)	0/0 (2.5σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0/0 (1.7σ)	0/0 (2.5σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0/0 (3.0σ)	0/0 (2.3σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	0/0 (2.2σ)	1/0 (3.1σ)	1/0 (3.0σ)	WARN
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0/0 (2.5σ)	0/0 (2.5σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0/0 (2.8σ)	0/0 (2.4σ)	0/0 (2.1σ)	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0/0 (2.4σ)	0/0 (2.3σ)	0/0 (2.5σ)	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0/0 (2.9σ)	0/0 (2.4σ)	0/0 (2.7σ)	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0/0 (2.7σ)	0/0 (2.8σ)	0/0 (2.6σ)	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0/0 (2.2σ)	0/0 (2.6σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0/0 (2.4σ)	0/0 (2.9σ)	0/0 (2.9σ)	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	0/0 (2.4σ)	0/0 (2.3σ)	0/0 (2.3σ)	PASS

Appendix B: Output Files

- `output/criteria_4_1.csv` – mean, σ and 3σ/5σ limits per parameter
- `output/judgement_4_1_channels.csv` – every channel beyond 3σ (board, channel, value, deviation, level)
- `output/judgement_4_1_boards.csv` – per-board counts and worst level
- `output/plot_4_1_*.png` – distribution plots and per-board summary

← Back to summary

FEMB QC Parameter Evaluation - Report Section 4.2: DIFF Power Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Generated: 2026-10-16 16:37
Script: `eval_4_2_diff_power.py`

1. Scope

Section 4.2 of the per-board checkout report is the power-supply reading taken while the FEMB is in the DIFF configuration of section 4.1 (ColdADC differential input, FE differential driver on, 900 mV baseline, 14 mV/fC, 2 µs, calibration pulse on). It has the same 17 numbers as sections 2.1 and 3.2: V_set / V_meas / I_meas / P_meas for the 4 rails (BIAS, LArASIC, ColdDATA, ColdADC), plus Total Power. In the CSV it is the third power block.

Category	Parameters	Treatment
Core	I_meas & V_meas of LArASIC / ColdDATA / ColdADC, Total Power (7)	Statistical judgement
Auxiliary	V_meas & I_meas of BIAS (2)	Shown for reference; I_BIAS ≈ 0 is better served by a fixed limit
Not evaluated	V_set (4)	Configuration setpoints, constant
Not evaluated	P_meas (4)	Redundant (P = V × I)

2. Judgement Method

For each parameter, mean and standard deviation (σ) are computed over all boards; each board is graded by its deviation |x − mean| / σ.

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
ALERT	5σ – 7σ	red
FAIL	> 7σ	dark red

3. Summary

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
I_LArASIC	core	0.6736 A	0.0049	0.6650	0.6840	7	44	0	0	0	–
I_ColdDATA	core	0.2340 A	0.0072	0.2230	0.2750	8	43	0	1	0	00051 (ALERT 5.7σ)
I_ColdADC	core	1.6006 A	0.0301	1.5260	1.7090	6	43	1	0	0	00031 (WARN 3.6σ)
V_LArASIC	core	2.9079 V	0.0045	2.9000	2.9140	12	44	0	0	0	–
V_ColdDATA	core	2.9713 V	0.0088	2.9520	2.9780	10	44	0	0	0	–
V_ColdADC	core	3.4201 V	0.0076	3.4060	3.4290	14	44	0	0	0	–

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
Total_Power	core	8.134 W	0.106	7.898	8.463	44	43	1	0	0	00031 (WARN 3.1σ)
V_BIAS	aux	4.9996 V	0.0005	4.9990	5.0000	2	44	0	0	0	–
I_BIAS	aux	0.0011 A	0.0023	0.0000	0.0060	2	44	0	0	0	–

Criteria (limits)

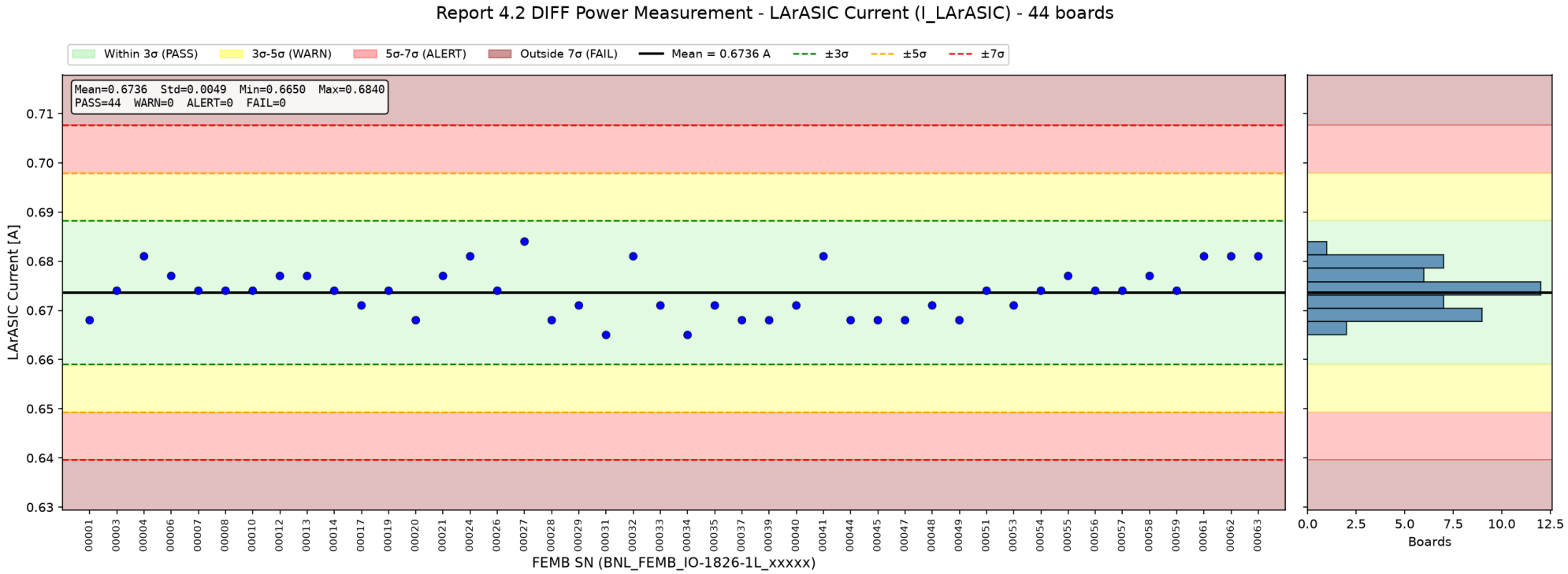
Parameter	Unit	−7σ	−5σ	−3σ	Mean	+3σ	+5σ	+7σ
I_LArASIC	A	0.6396	0.6493	0.6590	0.6736	0.6882	0.6979	0.7077
I_ColdDATA	A	0.1833	0.1978	0.2123	0.2340	0.2557	0.2702	0.2847
I_ColdADC	A	1.3900	1.4502	1.5103	1.6006	1.6908	1.7510	1.8112
V_LArASIC	V	2.8765	2.8855	2.8944	2.9079	2.9214	2.9303	2.9393
V_ColdDATA	V	2.9098	2.9274	2.9449	2.9713	2.9976	3.0152	3.0328
V_ColdADC	V	3.3667	3.3820	3.3972	3.4201	3.4429	3.4582	3.4734
Total_Power	W	7.391	7.603	7.815	8.134	8.452	8.664	8.877
V_BIAS	V	4.9961	4.9971	4.9981	4.9996	5.0011	5.0020	5.0030
I_BIAS	A	-0.0151	-0.0105	-0.0059	0.0011	0.0080	0.0127	0.0173

4. Core Parameters

4.1 LArASIC Current (`I_LArASIC`)

LArASIC rail current.
Mean = 0.6736 A, σ = 0.0049 A, range [0.6650, 0.6840], 7 distinct values over 44 boards.

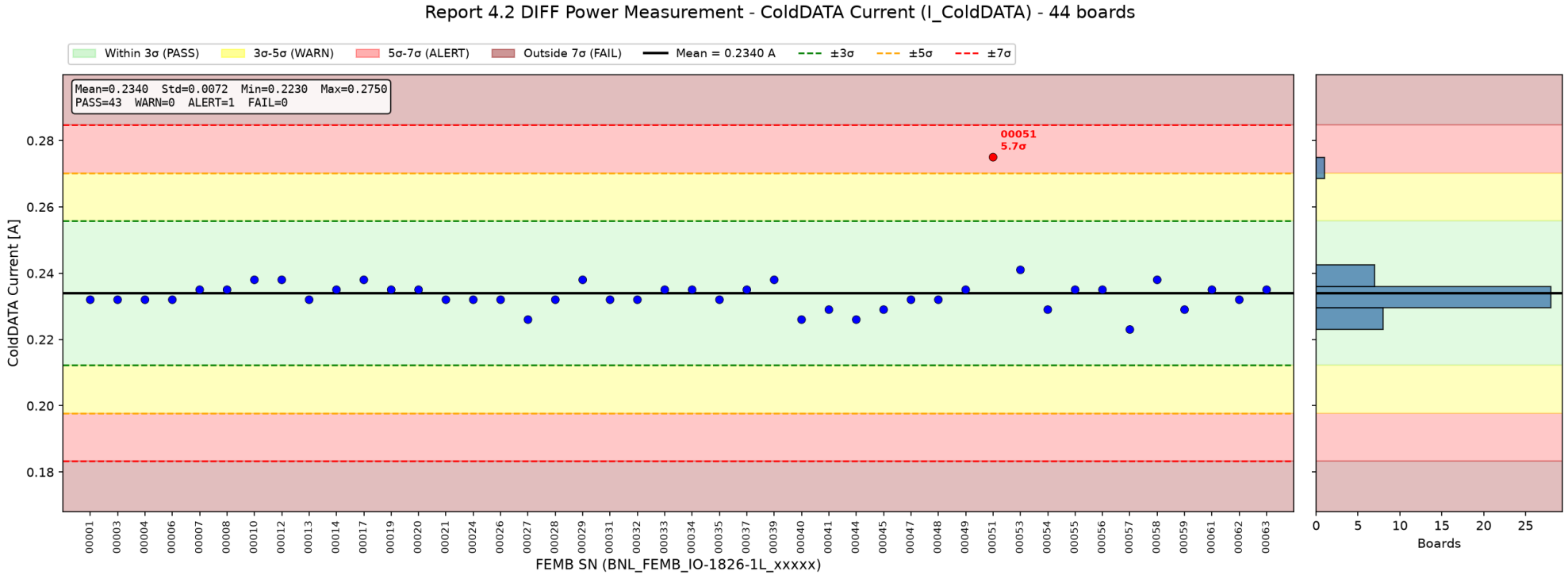
All boards within 3σ.



4.2 ColdDATA Current (`I_ColdDATA`)

ColdDATA rail current.
Mean = 0.2340 A, σ = 0.0072 A, range [0.2230, 0.2750], 8 distinct values over 44 boards.

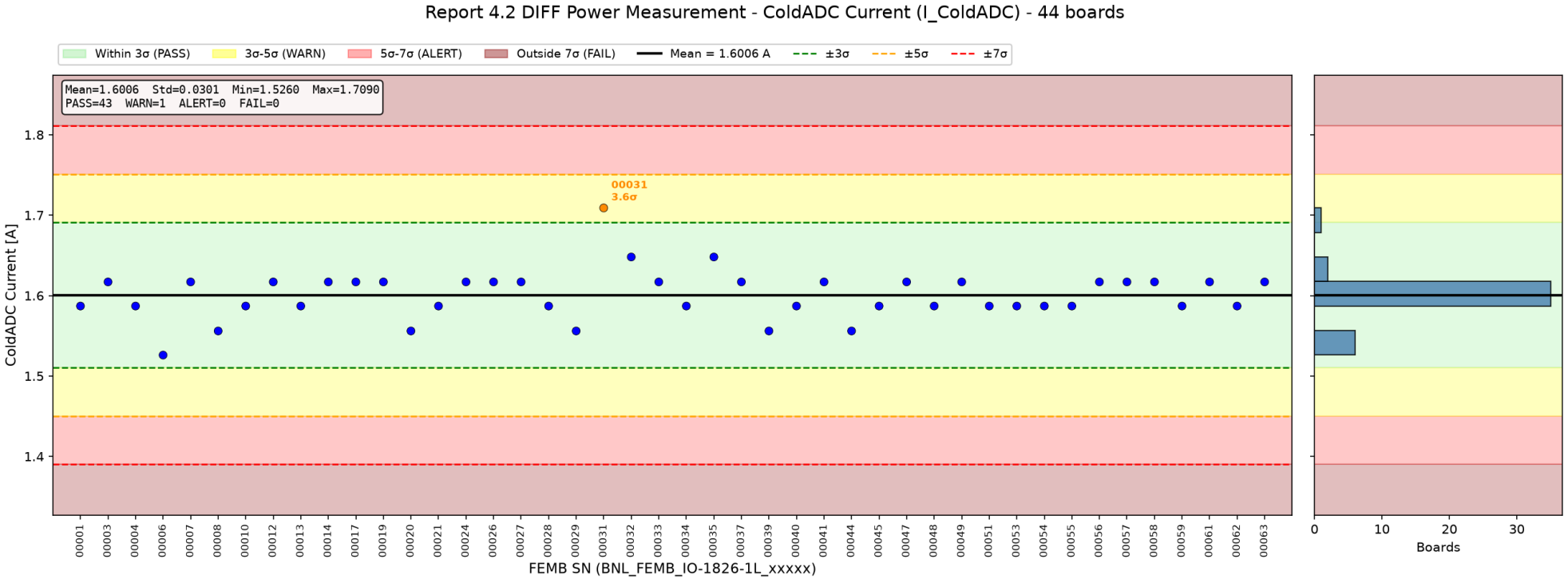
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00051	0.2750 A	5.7σ	**ALERT**



4.3 ColdADC Current (`I_ColdADC`)

ColdADC rail current.
Mean = 1.6006 A, σ = 0.0301 A, range [1.5260, 1.7090], 6 distinct values over 44 boards.

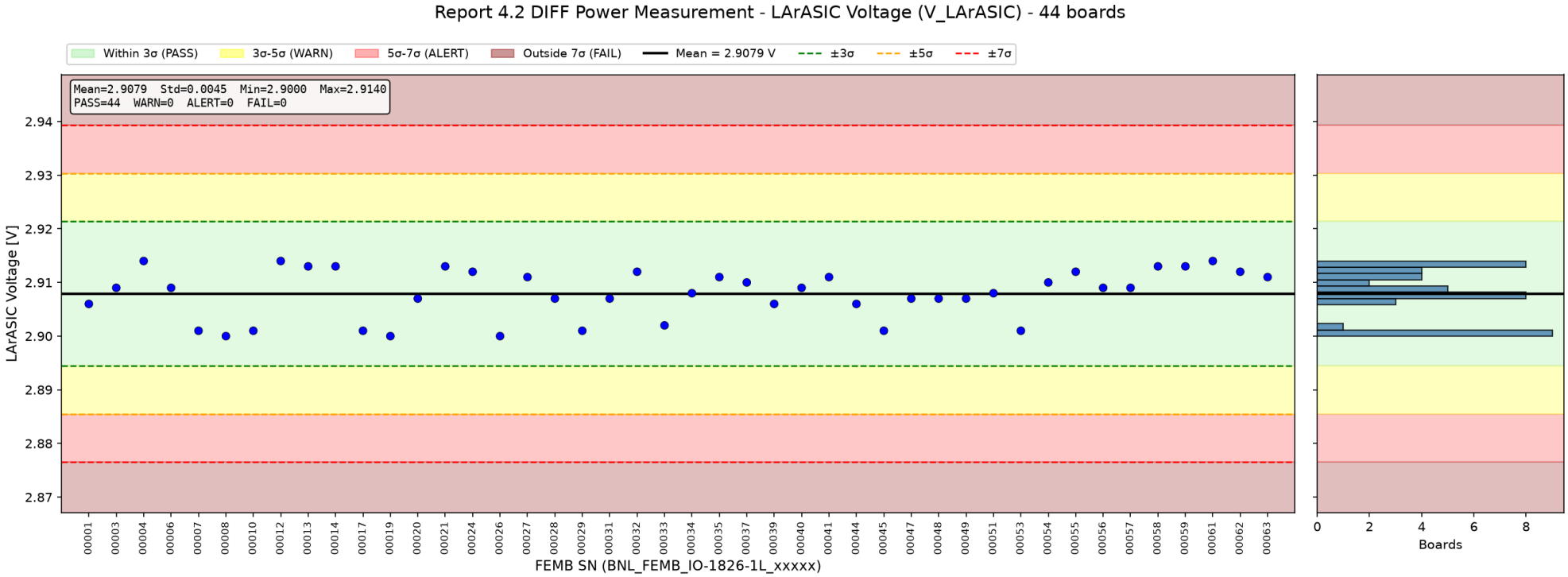
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	1.7090 A	3.6σ	**WARN**



4.4 LArASIC Voltage (`V_LArASIC`)

LArASIC rail voltage at the supply.
Mean = 2.9079 V, σ = 0.0045 V, range [2.9000, 2.9140], 12 distinct values over 44 boards.

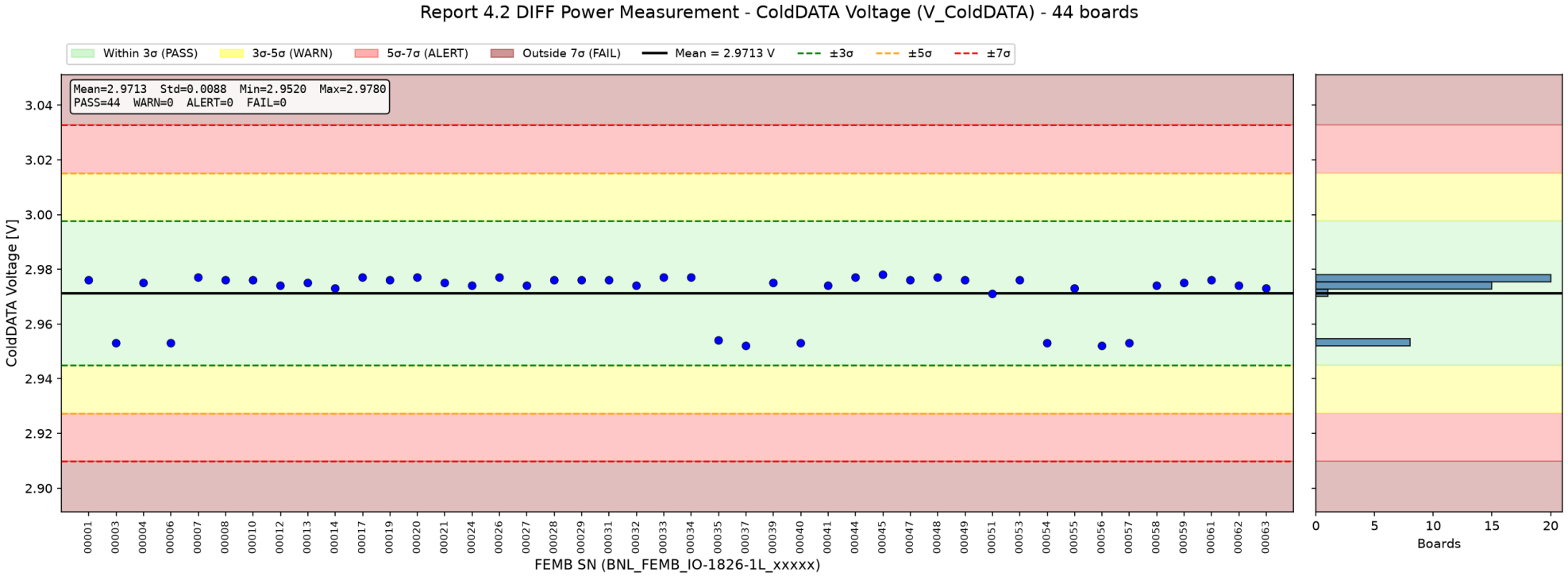
All boards within 3σ.



4.5 ColdDATA Voltage (`V_ColdDATA`)

ColdDATA rail voltage at the supply.
Mean = 2.9713 V, σ = 0.0088 V, range [2.9520, 2.9780], 10 distinct values over 44 boards.

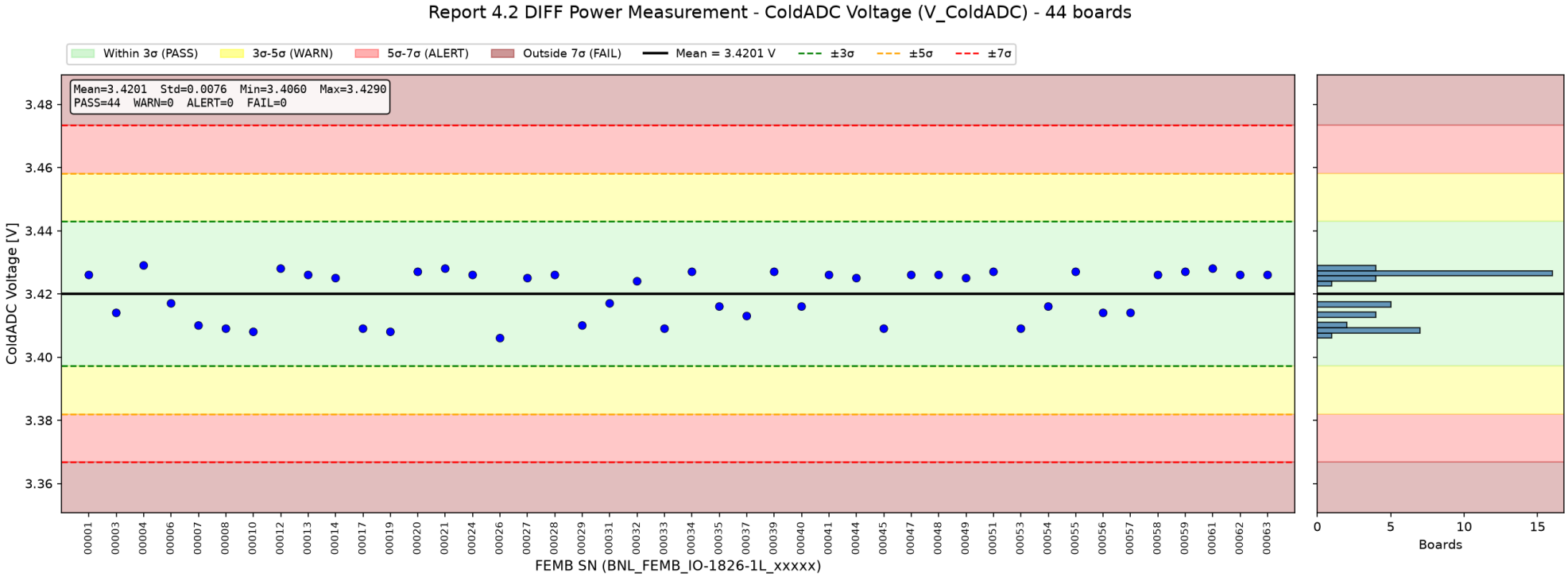
All boards within 3σ.



4.6 ColdADC Voltage (`V_ColdADC`)

ColdADC rail voltage at the supply.
Mean = 3.4201 V, σ = 0.0076 V, range [3.4060, 3.4290], 14 distinct values over 44 boards.

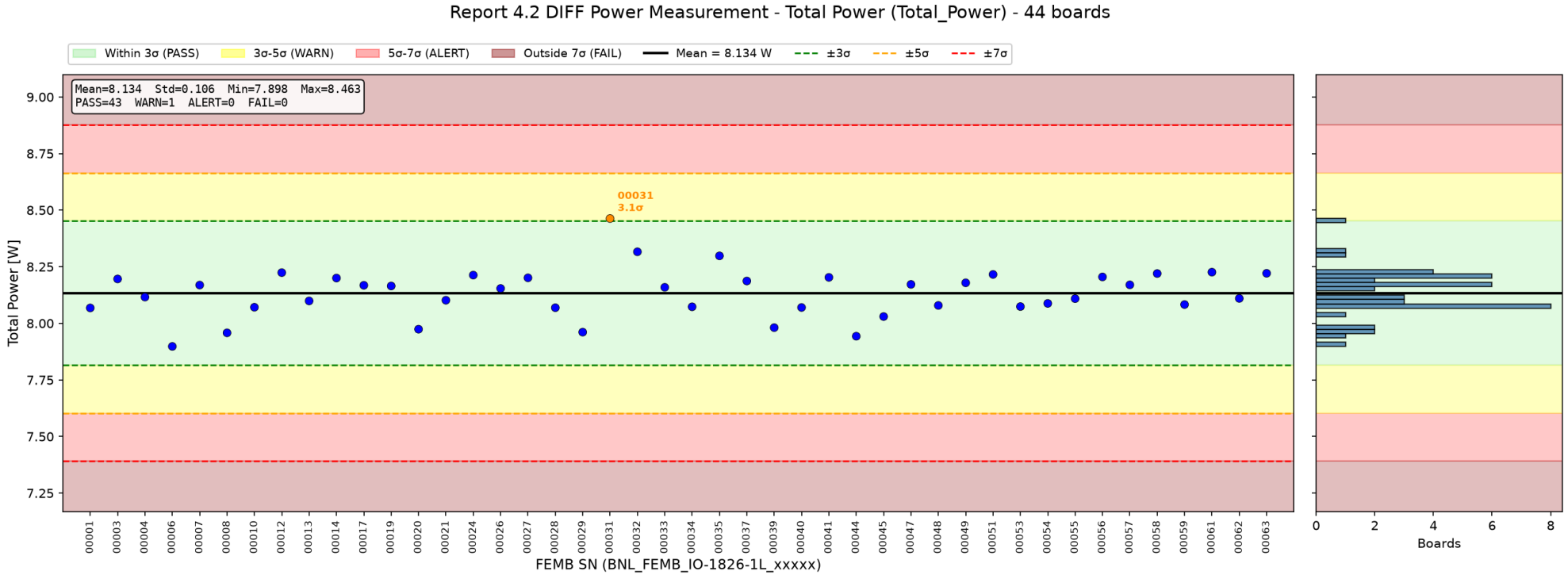
All boards within 3σ.



4.7 Total Power (`Total_Power`)

Sum of rail powers.
Mean = 8.134 W, σ = 0.106 W, range [7.898, 8.463], 44 distinct values over 44 boards.

FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	8.463 W	3.1σ	**WARN**

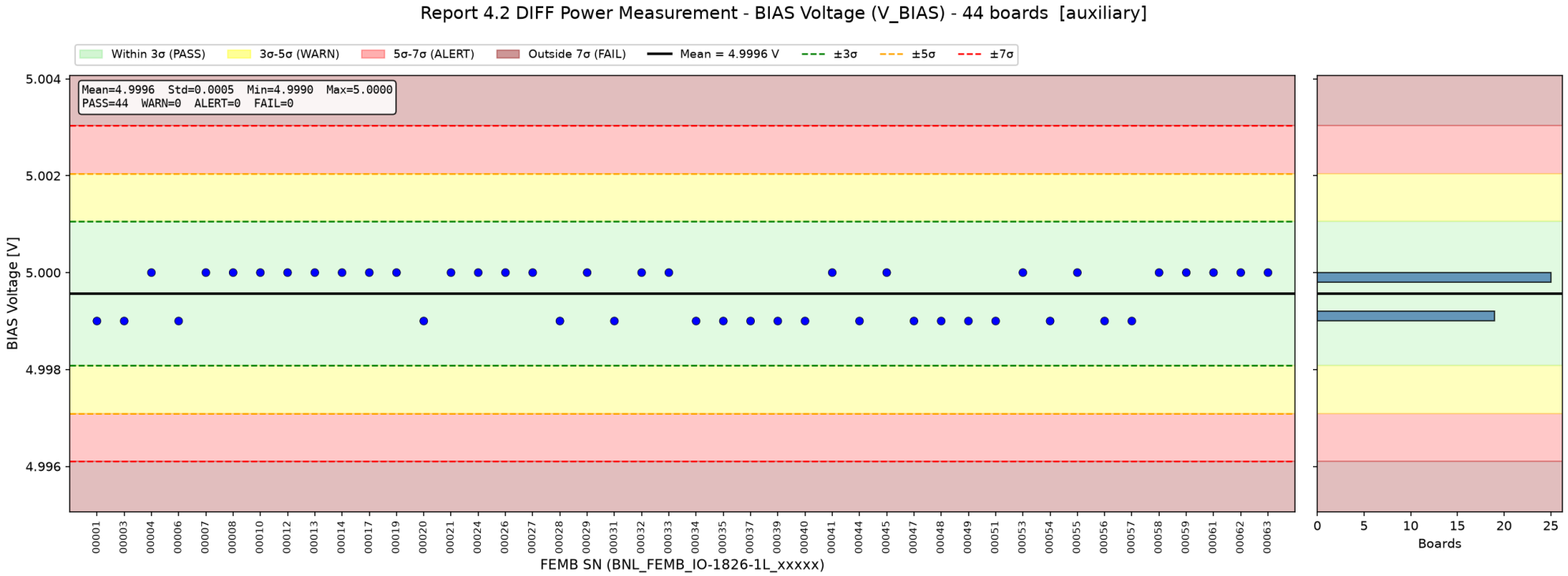


5. Auxiliary Parameters

5.1 BIAS Voltage (`V_BIAS`)

BIAS rail voltage.
Mean = 4.9996 V, σ = 0.0005 V, range [4.9990, 5.0000], 2 distinct values over 44 boards.

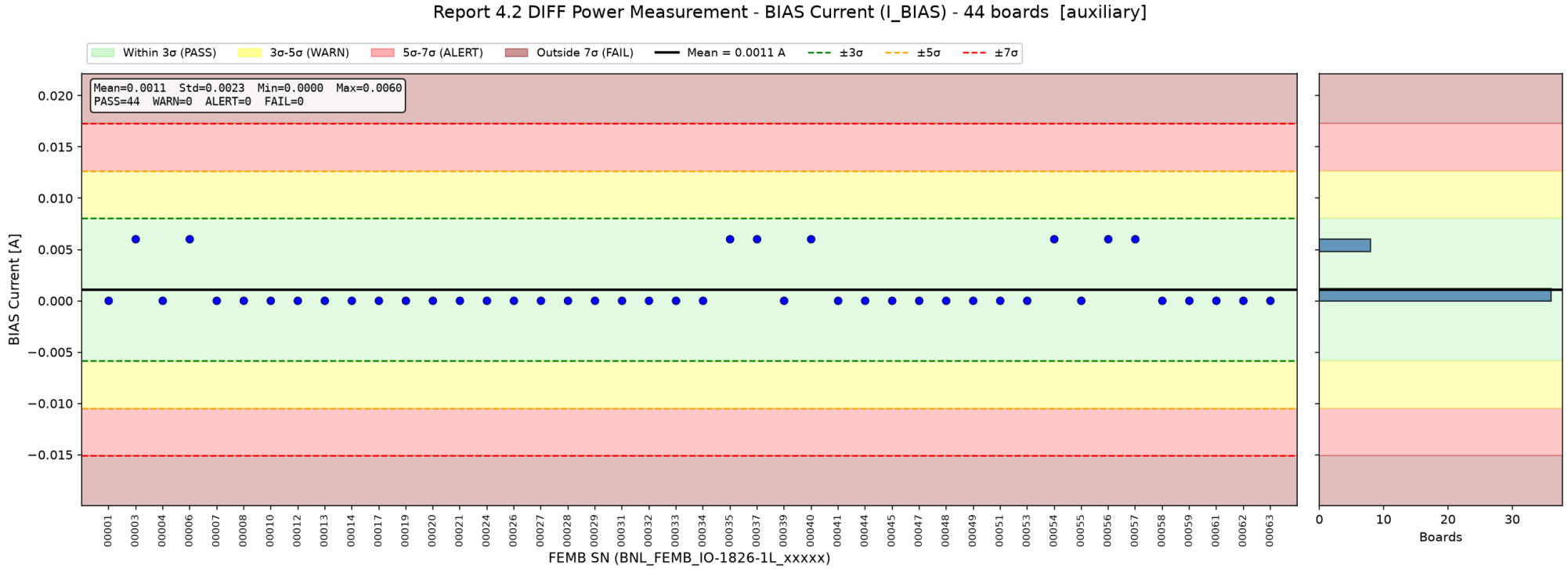
All boards within 3σ.



5.2 BIAS Current (`I_BIAS`)

BIAS rail current (≈ 0).
Mean = 0.0011 A, σ = 0.0023 A, range [0.0000, 0.0060], 2 distinct values over 44 boards.

All boards within 3σ.



6. Observations

- **BNL_FEMB_IO-1826-1L_00031**: I_ColdADC 3.6σ (WARN); Total_Power 3.1σ (WARN)
- **BNL_FEMB_IO-1826-1L_00051**: I_ColdDATA 5.7σ (ALERT)
- Readout resolution is coarse (≤ 10 distinct values) for: I_LArASIC, I_ColdDATA, I_ColdADC, V_ColdDATA. σ is therefore small and quantized; a minimum tolerance tied to instrument resolution should be considered before using these limits.
- Flagged boards are still included in the statistics and inflate σ; re-deriving limits without them would tighten the criteria.

Appendix A: Per-Board Judgement (core parameters)

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	1.2σ	0.3σ	0.5σ	0.4σ	0.5σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0.1σ	0.3σ	0.5σ	0.2σ	2.1σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	1.5σ	0.3σ	0.5σ	1.4σ	0.4σ	1.2σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0.7σ	0.3σ	2.5σ	0.2σ	2.1σ	0.4σ	2.2σ	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	0.1σ	0.1σ	0.5σ	1.5σ	0.7σ	1.3σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	0.1σ	0.1σ	1.5σ	1.8σ	0.5σ	1.5σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0.1σ	0.6σ	0.5σ	1.5σ	0.5σ	1.6σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0.7σ	0.6σ	0.5σ	1.4σ	0.3σ	1.0σ	0.9σ	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0.7σ	0.3σ	0.5σ	1.1σ	0.4σ	0.8σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0.1σ	0.1σ	0.5σ	1.1σ	0.2σ	0.6σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	0.5σ	0.6σ	0.5σ	1.5σ	0.7σ	1.5σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0.1σ	0.1σ	0.5σ	1.8σ	0.5σ	1.6σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	1.2σ	0.1σ	1.5σ	0.2σ	0.7σ	0.9σ	1.5σ	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0.7σ	0.3σ	0.5σ	1.1σ	0.4σ	1.0σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	1.5σ	0.3σ	0.5σ	0.9σ	0.3σ	0.8σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0.1σ	0.3σ	0.5σ	1.8σ	0.7σ	1.8σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	2.1σ	1.1σ	0.5σ	0.7σ	0.3σ	0.6σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	1.2σ	0.3σ	0.5σ	0.2σ	0.5σ	0.8σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0.5σ	0.6σ	1.5σ	1.5σ	0.5σ	1.3σ	1.6σ	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	1.8σ	0.3σ	**3.6σ WARN**	0.2σ	0.5σ	0.4σ	**3.1σ WARN**	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	1.5σ	0.3σ	1.6σ	0.9σ	0.3σ	0.5σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0.5σ	0.1σ	0.5σ	1.3σ	0.7σ	1.5σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	1.8σ	0.1σ	0.5σ	0.0σ	0.7σ	0.9σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0.5σ	0.3σ	1.6σ	0.7σ	2.0σ	0.5σ	1.5σ	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	1.2σ	0.1σ	0.5σ	0.5σ	2.2σ	0.9σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	1.2σ	0.6σ	1.5σ	0.4σ	0.4σ	0.9σ	1.4σ	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0.5σ	1.1σ	0.5σ	0.2σ	2.1σ	0.5σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	1.5σ	0.7σ	0.5σ	0.7σ	0.3σ	0.8σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	1.2σ	1.1σ	1.5σ	0.4σ	0.7σ	0.6σ	1.8σ	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	1.2σ	0.7σ	0.5σ	1.5σ	0.8σ	1.5σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	1.2σ	0.3σ	0.5σ	0.2σ	0.5σ	0.8σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0.5σ	0.3σ	0.5σ	0.2σ	0.7σ	0.8σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	1.2σ	0.1σ	0.5σ	0.2σ	0.5σ	0.6σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0.1σ	**5.7σ ALERT**	0.5σ	0.0σ	0.0σ	0.9σ	0.8σ	ALERT
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0.5σ	1.0σ	0.5σ	1.5σ	0.5σ	1.5σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	0.1σ	0.7σ	0.5σ	0.5σ	2.1σ	0.5σ	0.4σ	PASS

FEMB SN	Test time	I_LArASIC	I_ColdDATA	I_ColdADC	V_LArASIC	V_ColdDATA	V_ColdADC	Total_Power	Worst
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0.7σ	0.1σ	0.5σ	0.9σ	0.2σ	0.9σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0.1σ	0.1σ	0.5σ	0.2σ	2.2σ	0.8σ	0.7σ	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0.1σ	1.5σ	0.5σ	0.2σ	2.1σ	0.8σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0.7σ	0.6σ	0.5σ	1.1σ	0.3σ	0.8σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0.1σ	0.7σ	0.5σ	1.1σ	0.4σ	0.9σ	0.5σ	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	1.5σ	0.1σ	0.5σ	1.4σ	0.5σ	1.0σ	0.9σ	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	1.5σ	0.3σ	0.5σ	0.9σ	0.3σ	0.8σ	0.2σ	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	1.5σ	0.1σ	0.5σ	0.7σ	0.2σ	0.8σ	0.8σ	PASS

Appendix B: Output Files

- `output/criteria_4_2.csv` – mean, σ and 3σ/5σ/7σ limits per parameter
- `output/judgement_4_2.csv` – value, deviation and level per board per parameter
- `output/plot_4_2_*.png` – distribution plots

← Back to summary

FEMB QC Parameter Evaluation - Report Section 4.3: DIFF LDO Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Generated: 2026-10-16 16:37
Script: `eval_4_3_diff_ldo.py`

1. Scope

Section 4.3 of the per-board checkout report lists the outputs of the 11 on-board LDO / monitor points, read in mV by the board monitor ADC while the FEMB is in the DIFF configuration (same configuration as 4.1 / 4.2). It is the second LDO block in the CSV (the first is 3.3, SE OFF). One value per board per point; all 11 are evaluated, using the values as reported in the CSV.

Parameter	Description
`CDVDDA` – COLDATA VDDA	COLDATA analog supply
`CDVDDIO_HALF` – COLDATA VDDIO	COLDATA I/O supply
`ADCRVDDD1P2` – ADC-R VDDD 1.2V	Right-side ColdADC digital 1.2 V supply
`ADCLVDDD1P2` – ADC-L VDDD 1.2V	Left-side ColdADC digital 1.2 V supply
`FERVDDP` – FE-R VDDP	Right-side LArASIC supply
`FELVDDP` – FE-L VDDP	Left-side LArASIC supply
`ADCRP25V_1_HALF` – ADC-R 2.5V #1	Right-side ColdADC 2.5 V supply #1
`ADCLP25V_1_HALF` – ADC-L 2.5V #1	Left-side ColdADC 2.5 V supply #1
`ADCRP25V_HALF` – ADC-R 2.5V	Right-side ColdADC 2.5 V main supply
`ADCLP25V_HALF` – ADC-L 2.5V	Left-side ColdADC 2.5 V main supply
`GND` – Ground	Ground reference reading of the monitor ADC

2. Judgement Method

For each parameter, mean and standard deviation (σ) are computed over all boards; each board is graded by its deviation |x − mean| / σ.

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
ALERT	5σ – 7σ	red
FAIL	> 7σ	dark red

3. Summary

Parameter	Role	Mean	Std	Min	Max	N unique	PASS	WARN	ALERT	FAIL	Flagged boards
CDVDDA	core	1185.1 mV	1.9	1179.0	1189.0	9	43	1	0	0	00051 (WARN 3.2σ)
CDVDDIO_HALF	core	2232.1 mV	5.4	2220.0	2242.0	19	44	0	0	0	–
ADCRVDDD1P2	core	1097.8 mV	1.7	1094.0	1101.0	8	44	0	0	0	–
ADCLVDDD1P2	core	1097.7 mV	1.8	1094.0	1101.0	8	44	0	0	0	–
FERVDDP	core	1665.5 mV	2.4	1657.0	1670.0	10	43	1	0	0	00034 (WARN 3.5σ)
FELVDDP	core	1665.2 mV	2.8	1656.0	1670.0	11	43	1	0	0	00013 (WARN 3.3σ)
ADCRP25V_1_HALF	core	2252.9 mV	5.7	2242.0	2265.0	19	44	0	0	0	–
ADCLP25V_1_HALF	core	2250.6 mV	4.8	2241.0	2260.0	19	44	0	0	0	–
ADCRP25V_HALF	core	2252.0 mV	5.3	2237.0	2263.0	19	44	0	0	0	–
ADCLP25V_HALF	core	2251.8 mV	5.0	2239.0	2265.0	17	44	0	0	0	–
GND	core	33.2 mV	2.9	27.0	38.0	12	44	0	0	0	–

Criteria (limits)

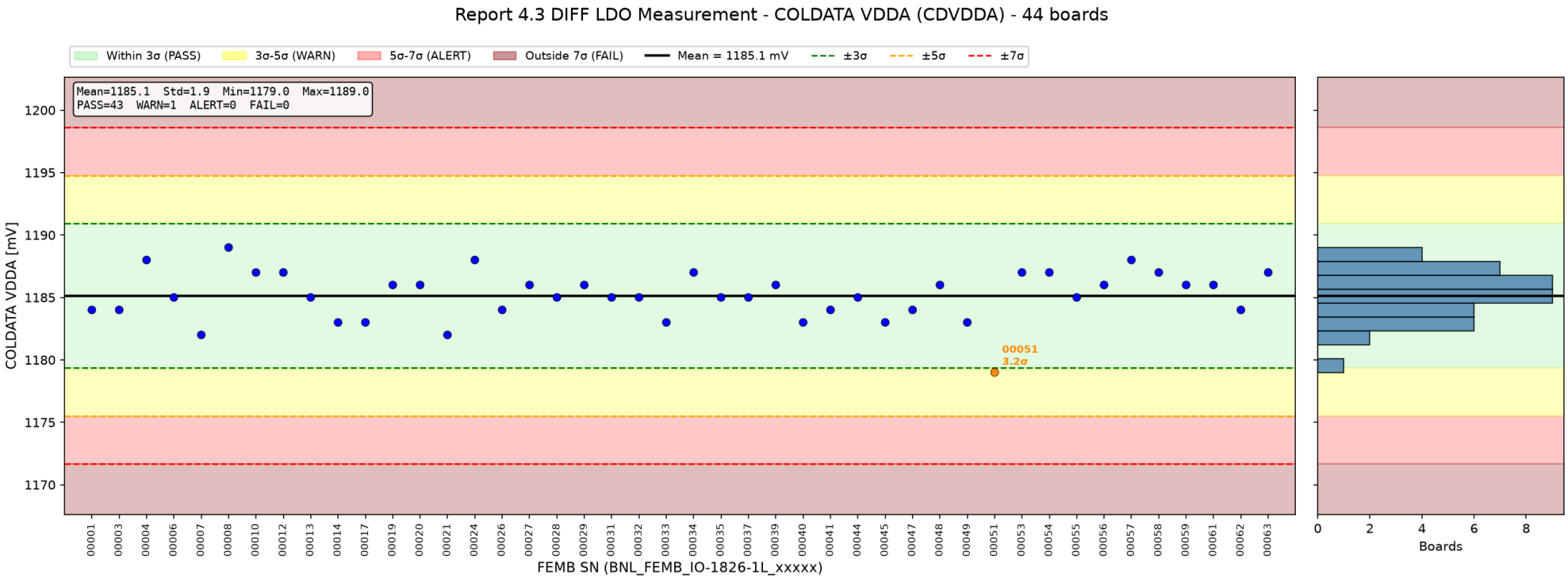
Parameter	Unit	−7σ	−5σ	−3σ	Mean	+3σ	+5σ	+7σ
CDVDDA	mV	1171.7	1175.5	1179.4	1185.1	1190.9	1194.8	1198.6
CDVDDIO_HALF	mV	2194.2	2205.0	2215.8	2232.1	2248.3	2259.1	2270.0
ADCRVDDD1P2	mV	1085.8	1089.2	1092.7	1097.8	1103.0	1106.4	1109.9
ADCLVDDD1P2	mV	1084.9	1088.5	1092.2	1097.7	1103.2	1106.8	1110.5
FERVDDP	mV	1648.4	1653.3	1658.1	1665.5	1672.8	1677.6	1682.5
FELVDDP	mV	1645.8	1651.3	1656.8	1665.2	1673.5	1679.0	1684.6
ADCRP25V_1_HALF	mV	2213.2	2224.5	2235.9	2252.9	2269.9	2281.3	2292.6
ADCLP25V_1_HALF	mV	2216.9	2226.5	2236.2	2250.6	2265.1	2274.7	2284.4
ADCRP25V_HALF	mV	2214.9	2225.5	2236.1	2252.0	2267.9	2278.5	2289.1
ADCLP25V_HALF	mV	2217.1	2227.0	2236.9	2251.8	2266.7	2276.7	2286.6
GND	mV	13.1	18.9	24.6	33.2	41.9	47.6	53.4

4. Parameters

4.1 COLDATA VDDA (`CDVDDA`)

COLDATA analog supply.
Mean = 1185.1 mV, σ = 1.9 mV, range [1179.0, 1189.0], 9 distinct values over 44 boards.

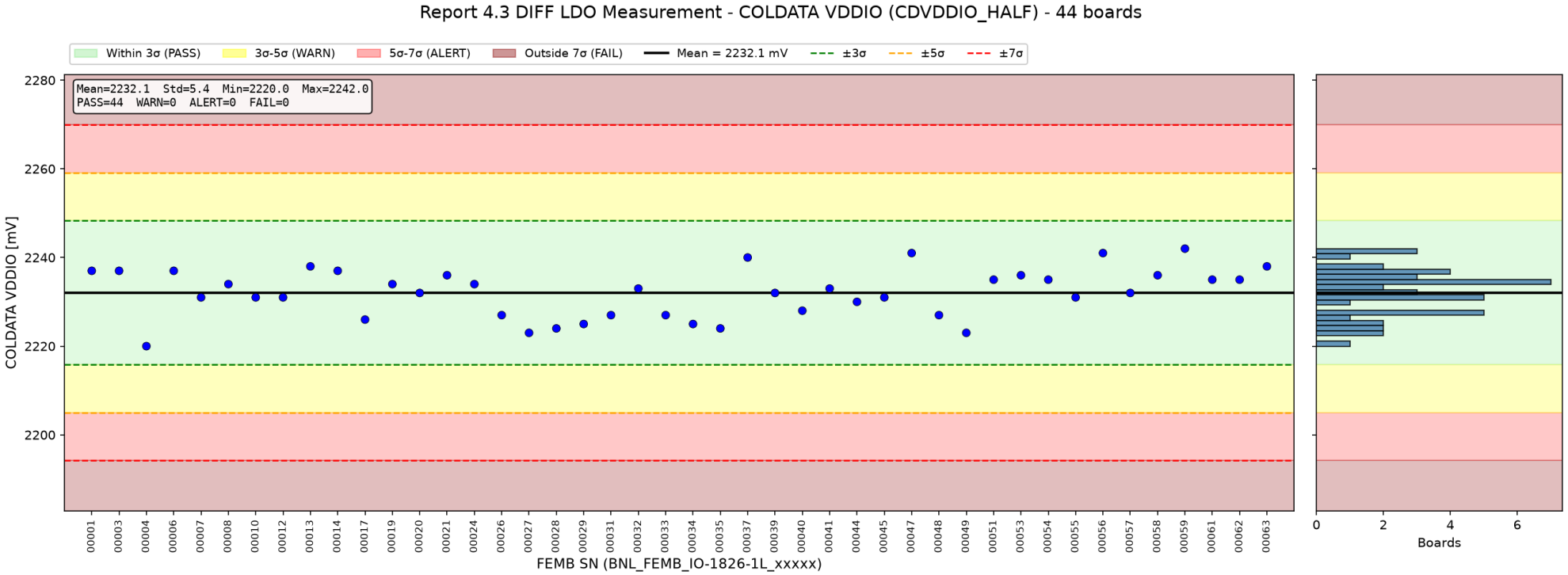
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00051	1179.0 mV	3.2σ	**WARN**



4.2 COLDATA VDDIO (`CDVDDIO_HALF`)

COLDATA I/O supply.
Mean = 2232.1 mV, σ = 5.4 mV, range [2220.0, 2242.0], 19 distinct values over 44 boards.

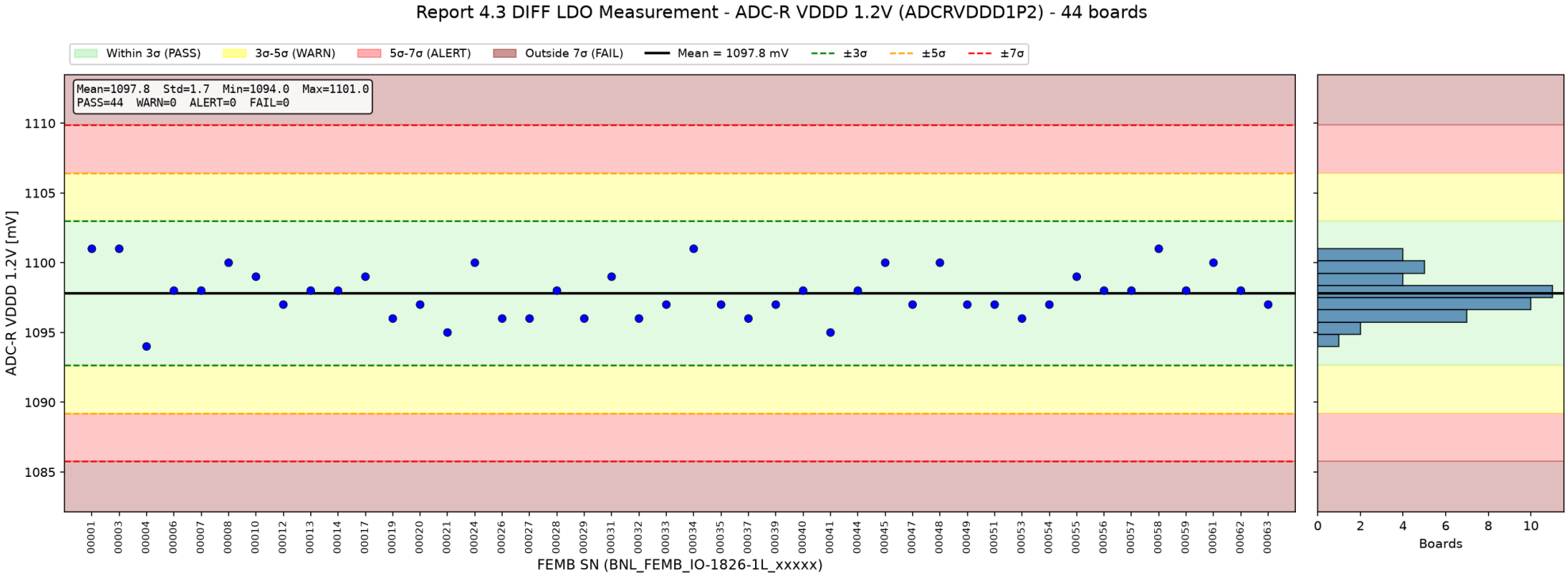
All boards within 3σ.



4.3 ADC-R VDDD 1.2V (`ADCRVDDD1P2`)

Right-side ColdADC digital 1.2 V supply.
Mean = 1097.8 mV, σ = 1.7 mV, range [1094.0, 1101.0], 8 distinct values over 44 boards.

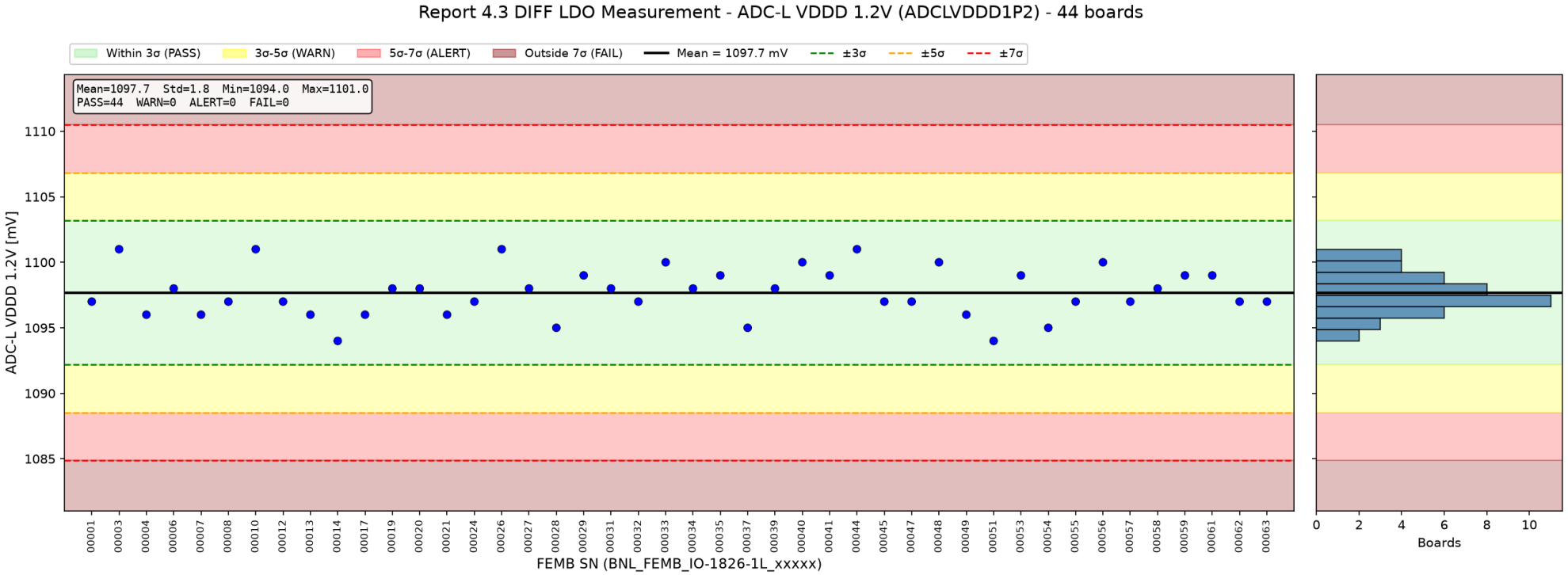
All boards within 3σ.



4.4 ADC-L VDDD 1.2V (`ADCLVDDD1P2`)

Left-side ColdADC digital 1.2 V supply.
Mean = 1097.7 mV, σ = 1.8 mV, range [1094.0, 1101.0], 8 distinct values over 44 boards.

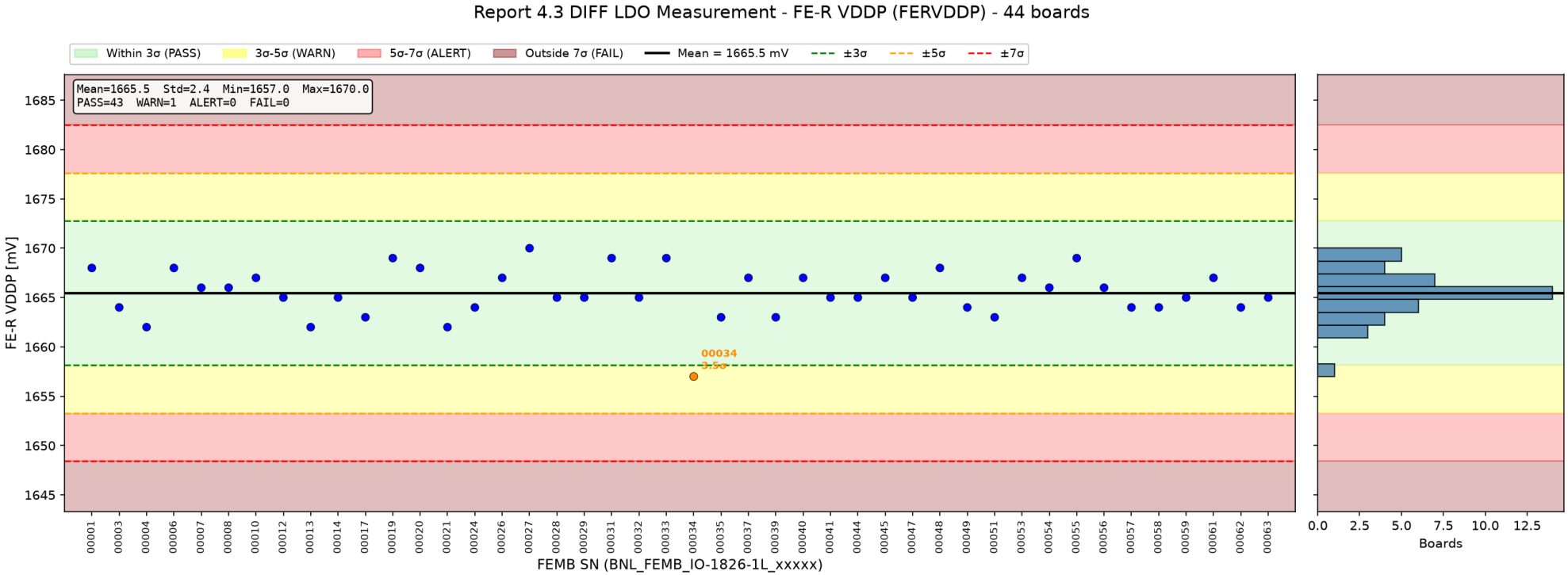
All boards within 3σ.



4.5 FE-R VDDP (`FERVDDP`)

Right-side LArASIC supply.
Mean = 1665.5 mV, σ = 2.4 mV, range [1657.0, 1670.0], 10 distinct values over 44 boards.

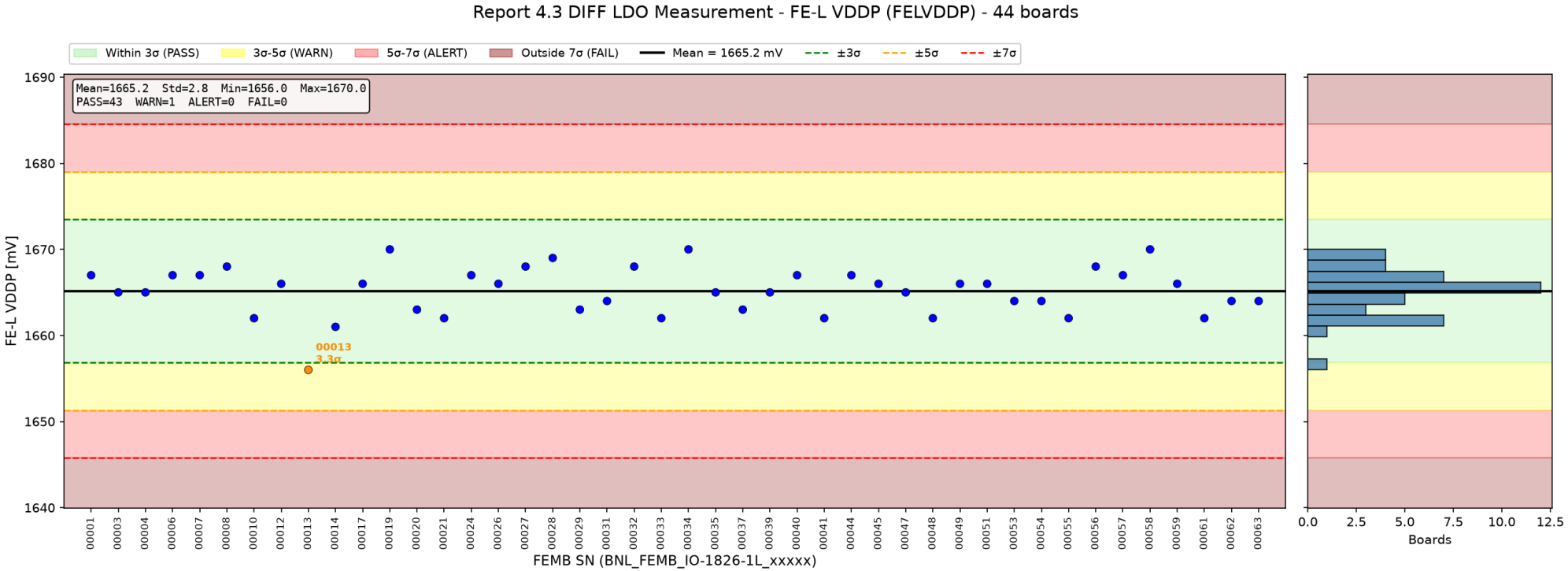
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00034	1657.0 mV	3.5σ	**WARN**



4.6 FE-L VDDP (`FELVDDP`)

Left-side LArASIC supply.
Mean = 1665.2 mV, σ = 2.8 mV, range [1656.0, 1670.0], 11 distinct values over 44 boards.

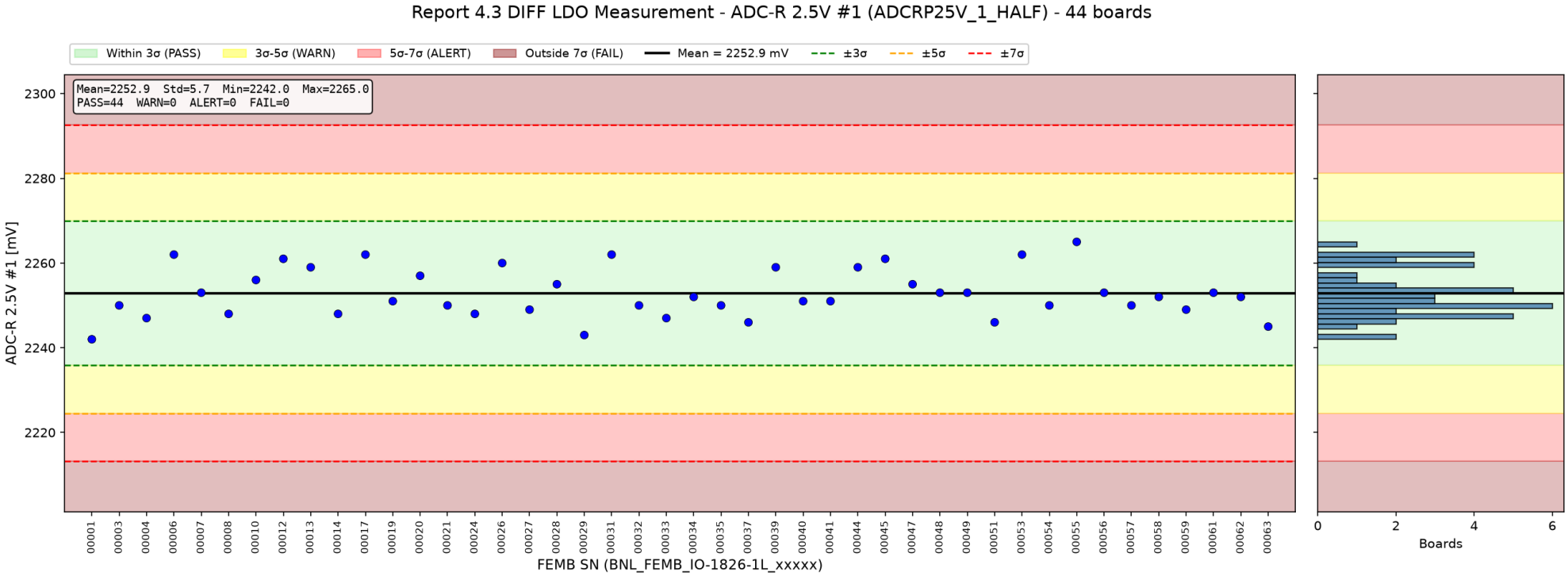
FEMB SN	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00013	1656.0 mV	3.3σ	**WARN**



4.7 ADC-R 2.5V #1 (`ADCRP25V_1_HALF`)

Right-side ColdADC 2.5 V supply #1.
Mean = 2252.9 mV, σ = 5.7 mV, range [2242.0, 2265.0], 19 distinct values over 44 boards.

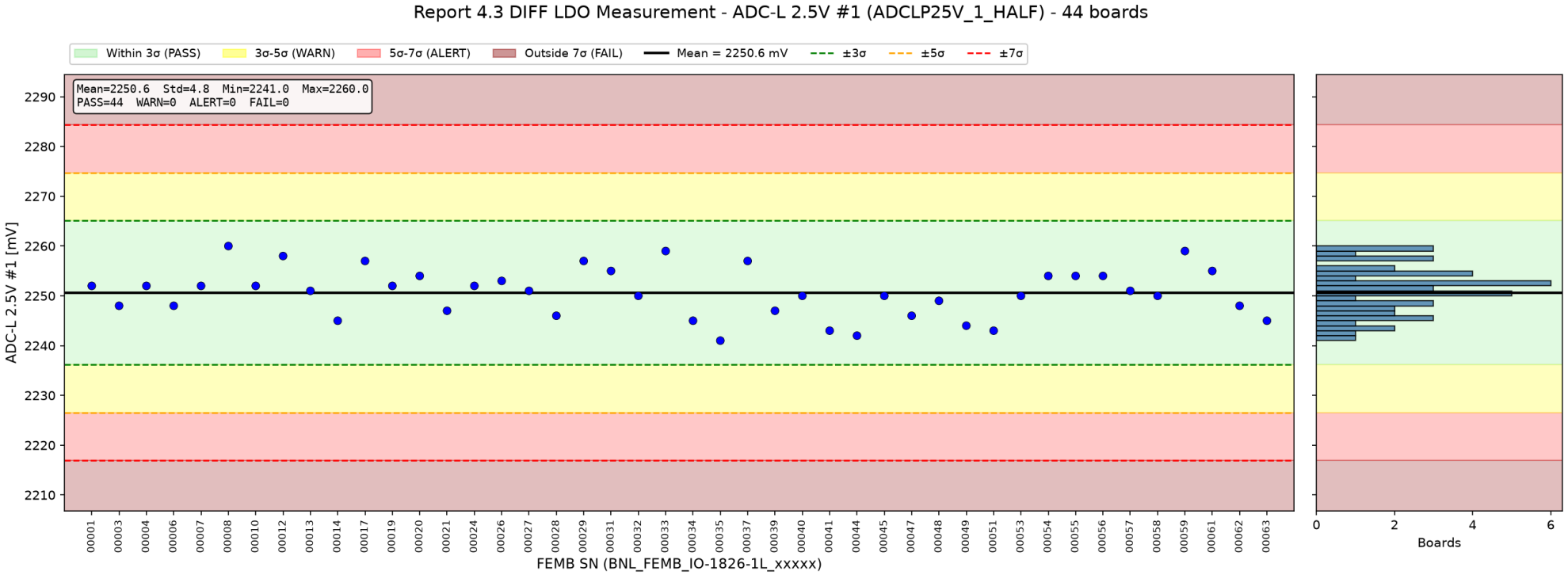
All boards within 3σ.



4.8 ADC-L 2.5V #1 (`ADCLP25V_1_HALF`)

Left-side ColdADC 2.5 V supply #1.
Mean = 2250.6 mV, σ = 4.8 mV, range [2241.0, 2260.0], 19 distinct values over 44 boards.

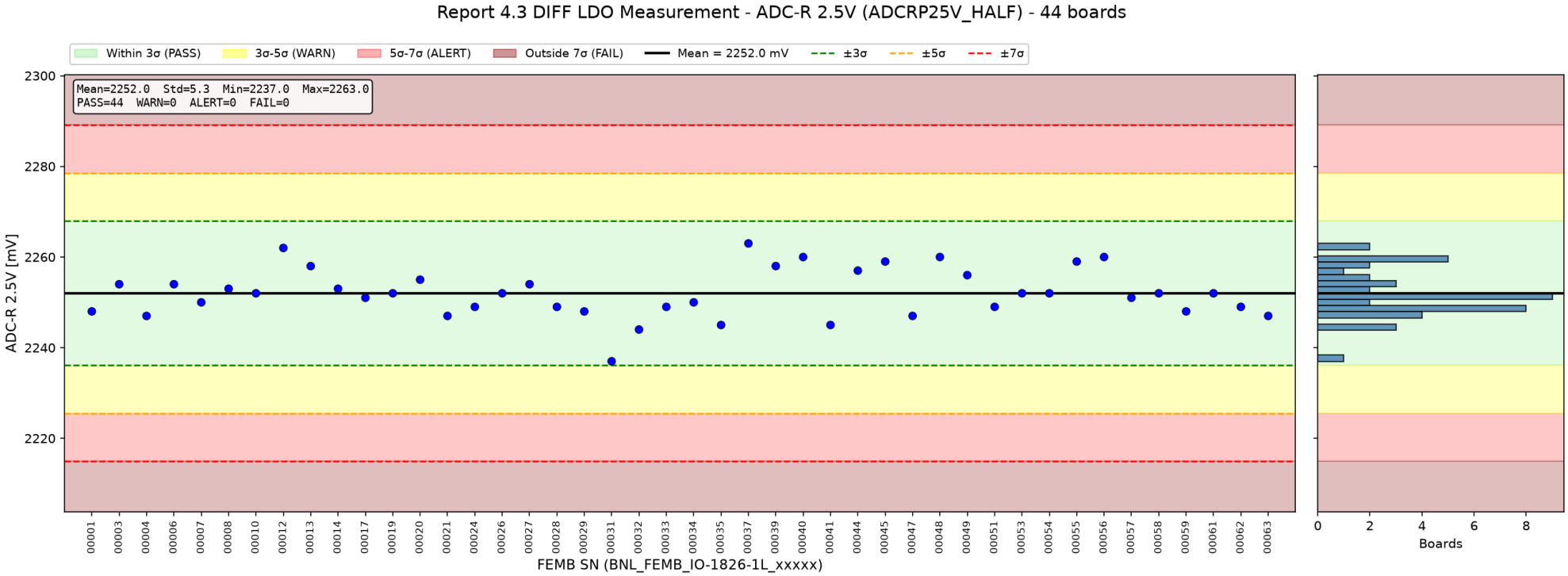
All boards within 3σ.



4.9 ADC-R 2.5V (`ADCRP25V_HALF`)

Right-side ColdADC 2.5 V main supply.
Mean = 2252.0 mV, σ = 5.3 mV, range [2237.0, 2263.0], 19 distinct values over 44 boards.

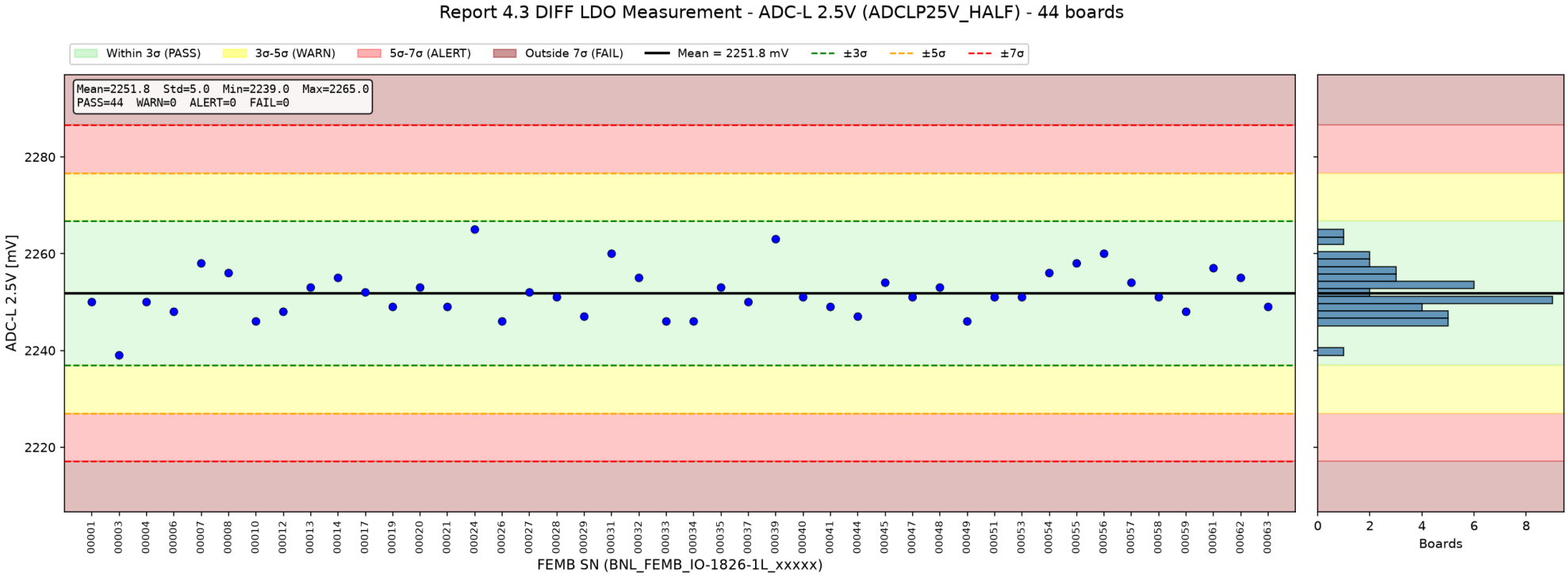
All boards within 3σ.



4.10 ADC-L 2.5V (`ADCLP25V_HALF`)

Left-side ColdADC 2.5 V main supply.
Mean = 2251.8 mV, σ = 5.0 mV, range [2239.0, 2265.0], 17 distinct values over 44 boards.

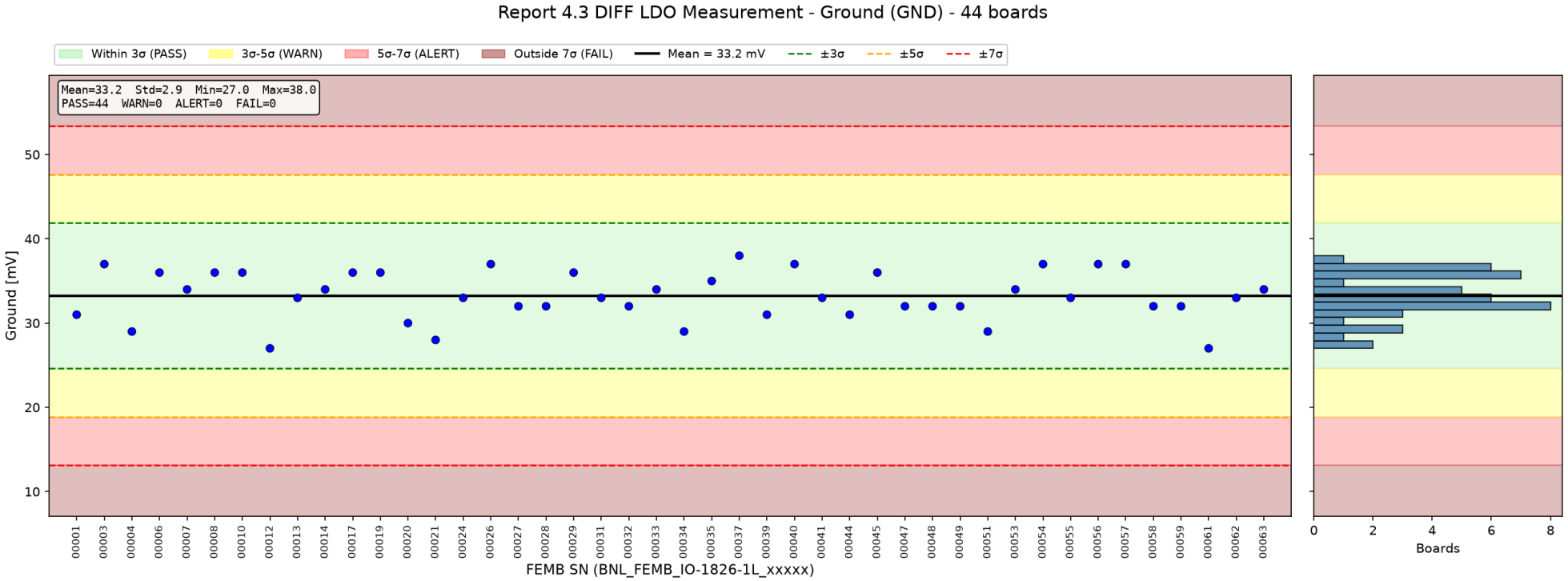
All boards within 3σ.



4.11 Ground (`GND`)

Ground reference reading of the monitor ADC.
Mean = 33.2 mV, σ = 2.9 mV, range [27.0, 38.0], 12 distinct values over 44 boards.

All boards within 3σ.



5. Observations

- **BNL_FEMB_IO-1826-1L_00013**: FELVDDP 3.3σ (WARN)
- **BNL_FEMB_IO-1826-1L_00034**: FERVDDP 3.5σ (WARN)
- **BNL_FEMB_IO-1826-1L_00051**: CDVDDA 3.2σ (WARN)
- Readout resolution is coarse (≤ 10 distinct values) for: CDVDDA, ADCRVDDD1P2, ADCLVDDD1P2, FERVDDP. σ is therefore small and quantized; a minimum tolerance tied to instrument resolution should be considered before using these limits.
- Flagged boards are still included in the statistics and inflate σ; re-deriving limits without them would tighten the criteria.

Appendix A: Per-Board Judgement

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0.6σ	0.9σ	1.8σ	0.4σ	1.0σ	0.7σ	1.9σ	0.3σ	0.8σ	0.4σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0.6σ	0.9σ	1.8σ	1.8σ	0.6σ	0.1σ	0.5σ	0.5σ	0.4σ	2.6σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	1.5σ	2.2σ	2.2σ	0.9σ	1.4σ	0.1σ	1.0σ	0.3σ	0.9σ	0.4σ	1.5σ	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0.1σ	0.9σ	0.1σ	0.2σ	1.0σ	0.7σ	1.6σ	0.5σ	0.4σ	0.8σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	1.6σ	0.2σ	0.1σ	0.9σ	0.2σ	0.7σ	0.0σ	0.3σ	0.4σ	1.2σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	2.0σ	0.4σ	1.3σ	0.4σ	0.2σ	1.0σ	0.9σ	1.9σ	0.2σ	0.8σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	1.0σ	0.2σ	0.7σ	1.8σ	0.6σ	1.1σ	0.5σ	0.3σ	0.0σ	1.2σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	1.0σ	0.2σ	0.5σ	0.4σ	0.2σ	0.3σ	1.4σ	1.5σ	1.9σ	0.8σ	2.2σ	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0.1σ	1.1σ	0.1σ	0.9σ	1.4σ	**3.3σ WARN**	1.1σ	0.1σ	1.1σ	0.2σ	0.1σ	WARN
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	1.1σ	0.9σ	0.1σ	2.0σ	0.2σ	1.5σ	0.9σ	1.2σ	0.2σ	0.6σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	1.1σ	1.1σ	0.7σ	0.9σ	1.0σ	0.3σ	1.6σ	1.3σ	0.2σ	0.0σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0.4σ	0.4σ	1.1σ	0.2σ	1.5σ	1.7σ	0.3σ	0.3σ	0.0σ	0.6σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0.4σ	0.0σ	0.5σ	0.2σ	1.0σ	0.8σ	0.7σ	0.7σ	0.6σ	0.2σ	1.1σ	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	1.6σ	0.7σ	1.6σ	0.9σ	1.4σ	1.1σ	0.5σ	0.8σ	0.9σ	0.6σ	1.8σ	PASS

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	1.5σ	0.4σ	1.3σ	0.4σ	0.6σ	0.7σ	0.9σ	0.3σ	0.6σ	2.6σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0.6σ	0.9σ	1.1σ	1.8σ	0.6σ	0.3σ	1.3σ	0.5σ	0.0σ	1.2σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	0.4σ	1.7σ	1.1σ	0.2σ	1.9σ	1.0σ	0.7σ	0.1σ	0.4σ	0.0σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0.1σ	1.5σ	0.1σ	1.5σ	0.2σ	1.4σ	0.4σ	1.0σ	0.6σ	0.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0.4σ	1.3σ	1.1σ	0.7σ	0.2σ	0.8σ	1.7σ	1.3σ	0.8σ	1.0σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0.1σ	0.9σ	0.7σ	0.2σ	1.5σ	0.4σ	1.6σ	0.9σ	2.8σ	1.6σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0.1σ	0.2σ	1.1σ	0.4σ	0.2σ	1.0σ	0.5σ	0.1σ	1.5σ	0.6σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	1.1σ	0.9σ	0.5σ	1.3σ	1.5σ	1.1σ	1.0σ	1.7σ	0.6σ	1.2σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	1.0σ	1.3σ	1.8σ	0.2σ	**3.5σ WARN**	1.7σ	0.2σ	1.2σ	0.4σ	1.2σ	1.5σ	WARN
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0.1σ	1.5σ	0.5σ	0.7σ	1.0σ	0.1σ	0.5σ	2.0σ	1.3σ	0.2σ	0.6σ	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0.1σ	1.5σ	1.1σ	1.5σ	0.6σ	0.8σ	1.2σ	1.3σ	2.1σ	0.4σ	1.7σ	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0.4σ	0.0σ	0.5σ	0.2σ	1.0σ	0.1σ	1.1σ	0.8σ	1.1σ	2.2σ	0.8σ	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	1.1σ	0.8σ	0.1σ	1.3σ	0.6σ	0.7σ	0.3σ	0.1σ	1.5σ	0.2σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0.6σ	0.2σ	1.6σ	0.7σ	0.2σ	1.1σ	0.3σ	1.6σ	1.3σ	0.6σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	0.1σ	0.4σ	0.1σ	1.8σ	0.2σ	0.7σ	1.1σ	1.8σ	0.9σ	1.0σ	0.8σ	PASS

FEMB SN	Test time	CDVDDA	CDVDDIO_HALF	ADCRVDDD1P2	ADCLVDDD1P2	FERVDDP	FELVDDP	ADCRP25V_1_HALF	ADCLP25V_1_HALF	ADCRP25V_HALF	ADCLP25V_HALF	GND	Worst
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	1.1σ	0.2σ	1.3σ	0.4σ	0.6σ	0.3σ	1.4σ	0.1σ	1.3σ	0.4σ	1.0σ	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	0.6σ	1.7σ	0.5σ	0.4σ	0.2σ	0.1σ	0.4σ	1.0σ	0.9σ	0.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0.4σ	0.9σ	1.3σ	1.3σ	1.0σ	1.1σ	0.0σ	0.3σ	1.5σ	0.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	1.1σ	1.7σ	0.5σ	0.9σ	0.6σ	0.3σ	0.0σ	1.4σ	0.8σ	1.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	**3.2σ WARN**	0.5σ	0.5σ	2.0σ	1.0σ	0.3σ	1.2σ	1.6σ	0.6σ	0.2σ	1.5σ	WARN
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	1.0σ	0.7σ	1.1σ	0.7σ	0.6σ	0.4σ	1.6σ	0.1σ	0.0σ	0.2σ	0.3σ	PASS
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	1.0σ	0.5σ	0.5σ	1.5σ	0.2σ	0.4σ	0.5σ	0.7σ	0.0σ	0.8σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0.1σ	0.2σ	0.7σ	0.4σ	1.5σ	1.1σ	2.1σ	0.7σ	1.3σ	1.2σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0.4σ	1.7σ	0.1σ	1.3σ	0.2σ	1.0σ	0.0σ	0.7σ	1.5σ	1.6σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	1.5σ	0.0σ	0.1σ	0.4σ	0.6σ	0.7σ	0.5σ	0.1σ	0.2σ	0.4σ	1.3σ	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	1.0σ	0.7σ	1.8σ	0.2σ	0.6σ	1.7σ	0.2σ	0.1σ	0.0σ	0.2σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0.4σ	1.8σ	0.1σ	0.7σ	0.2σ	0.3σ	0.7σ	1.7σ	0.8σ	0.8σ	0.4σ	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	0.4σ	0.5σ	1.3σ	0.7σ	0.6σ	1.1σ	0.0σ	0.9σ	0.0σ	1.0σ	2.2σ	PASS
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0.6σ	0.5σ	0.1σ	0.4σ	0.6σ	0.4σ	0.2σ	0.5σ	0.6σ	0.6σ	0.1σ	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	1.0σ	1.1σ	0.5σ	0.4σ	0.2σ	0.4σ	1.4σ	1.2σ	0.9σ	0.6σ	0.3σ	PASS

Appendix B: Output Files

- `output/criteria_4_3.csv` – mean, σ and 3σ/5σ/7σ limits per parameter
- `output/judgement_4_3.csv` – value, deviation and level per board per parameter
- `output/plot_4_3_*.png` – distribution plots

← Back to summary

FEMB QC Parameter Evaluation - Report Section 5: Monitoring Measurement

Board version: FEMB-IO 1826-1L
Boards used: 44 (latest passing test per board; excluded: 00046)
Chips: 44 × 8 = 352
Test period: 2026-07-27 23:48 - 2026-10-15 17:19
Configuration: monitoring path through the FEMB monitor ADC, 200 mV baseline, 14 mV/fC (Mon_200mVBL_14_0mVfC)
Generated: 2026-10-16 16:37
Script: `eval_5_monitoring.py`

1. Scope

Part 05 of the checkout report reads the internal monitoring points of the 8 ASIC pairs on the FEMB (LArASIC + ColdADC, `ASIC #` 0–7) through the board monitor ADC, in mV. Each parameter has one value per chip, i.e. 8 values per board; every chip is judged against the pooled distribution of all chips of all boards.

Parameter	Description
`FE T` – FE Temperature Sensor	LArASIC on-chip temperature sensor output, mV (depends on ambient temperature)
`FE BGP` – FE Bandgap Reference	LArASIC bandgap reference voltage, mV
`ADC_VCMI` – ADC VCMI	ColdADC input common-mode voltage, mV
`ADC_VCMO` – ADC VCMO	ColdADC output common-mode voltage, mV
`ADC_VREFP` – ADC VREFP	ColdADC positive reference voltage, mV
`ADC_VREFN` – ADC VREFN	ColdADC negative reference voltage, mV
`VSSA` – VSSA	ColdADC analog ground (VSSA) as seen by the monitor, mV

2. Judgement Method

For each parameter the mean and σ are computed over all 352 chips pooled together; each chip is graded by |x − mean| / σ. A board takes the worst level among its chips.

Per-ASIC monitoring parameters use **5σ as the limit**: a chip beyond 5σ fails (same as the channel-level noise / pulse parameters; board-level power and LDO parameters keep the wider 7σ limit).

Level	Deviation	Plot background
PASS	≤ 3σ	green
WARN	3σ – 5σ	yellow
FAIL	> 5σ	red

For a Gaussian distribution, about 0.27 % of chips (≈ 1 of 352) are expected beyond 3σ by chance, so isolated WARN chips are normal.

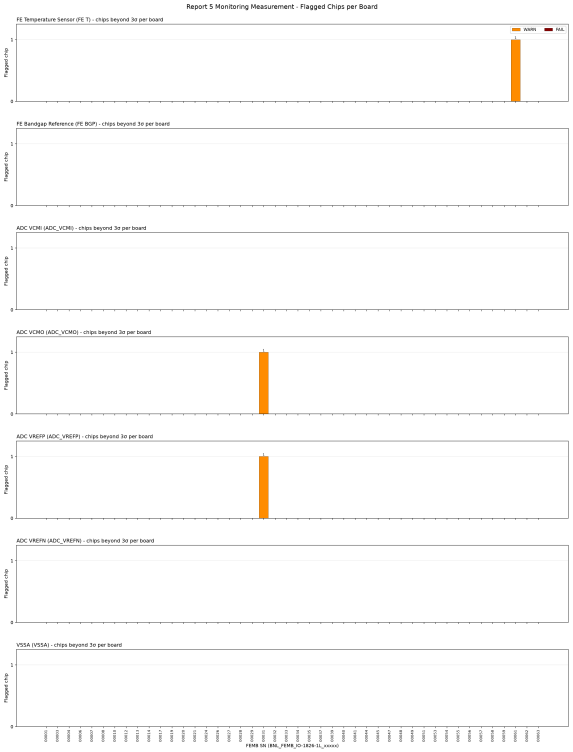
3. Summary

Parameter	Mean	Std	Min	Max	PASS	WARN	FAIL	Boards with ≥1 FAIL
FE_T	902.40	6.31	882.30	918.10	351	1	0	–
FE_BGP	1207.55	3.41	1198.70	1216.50	352	0	0	–
ADC_VCMI	937.28	3.14	929.00	945.60	352	0	0	–
ADC_VCMO	1224.46	3.45	1213.70	1233.40	351	1	0	–
ADC_VREFP	1942.28	4.68	1922.80	1954.40	351	1	0	–
ADC_VREFN	501.51	2.98	494.50	507.80	352	0	0	–
VSSA	67.40	3.10	58.80	74.10	352	0	0	–

Criteria (limits, mV)

Parameter	−5σ (FAIL)	−3σ	Mean	+3σ	+5σ (FAIL)
FE_T	870.86	883.48	902.40	921.32	933.94
FE_BGP	1190.53	1197.34	1207.55	1217.77	1224.58
ADC_VCMI	921.58	927.86	937.28	946.70	952.98
ADC_VCMO	1207.20	1214.10	1224.46	1234.81	1241.71
ADC_VREFP	1918.86	1928.23	1942.28	1956.34	1965.71
ADC_VREFN	486.59	492.56	501.51	510.46	516.42
VSSA	51.89	58.09	67.40	76.71	82.92

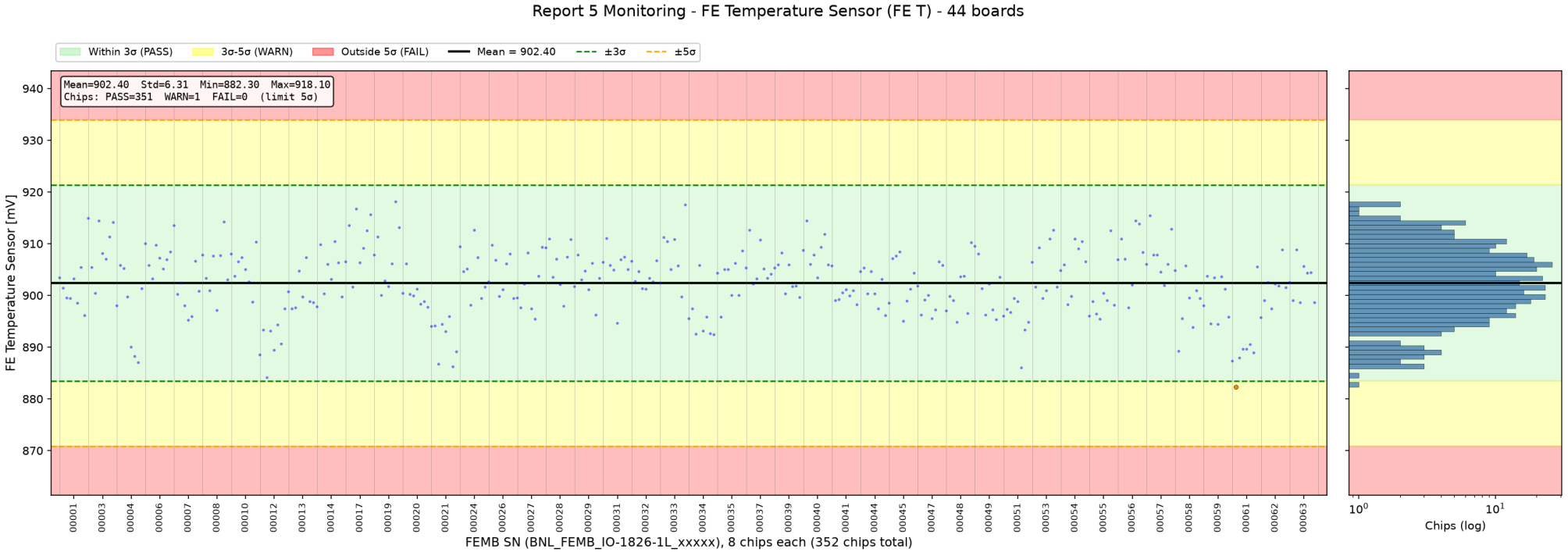
Flagged chips per board



4. Parameters

4.1 FE Temperature Sensor (`FE T`)

Mean = 902.40, σ = 6.31, range [882.30, 918.10] mV. Spread of per-board means: σ = 4.61; spread of per-chip means across boards: σ = 0.90.

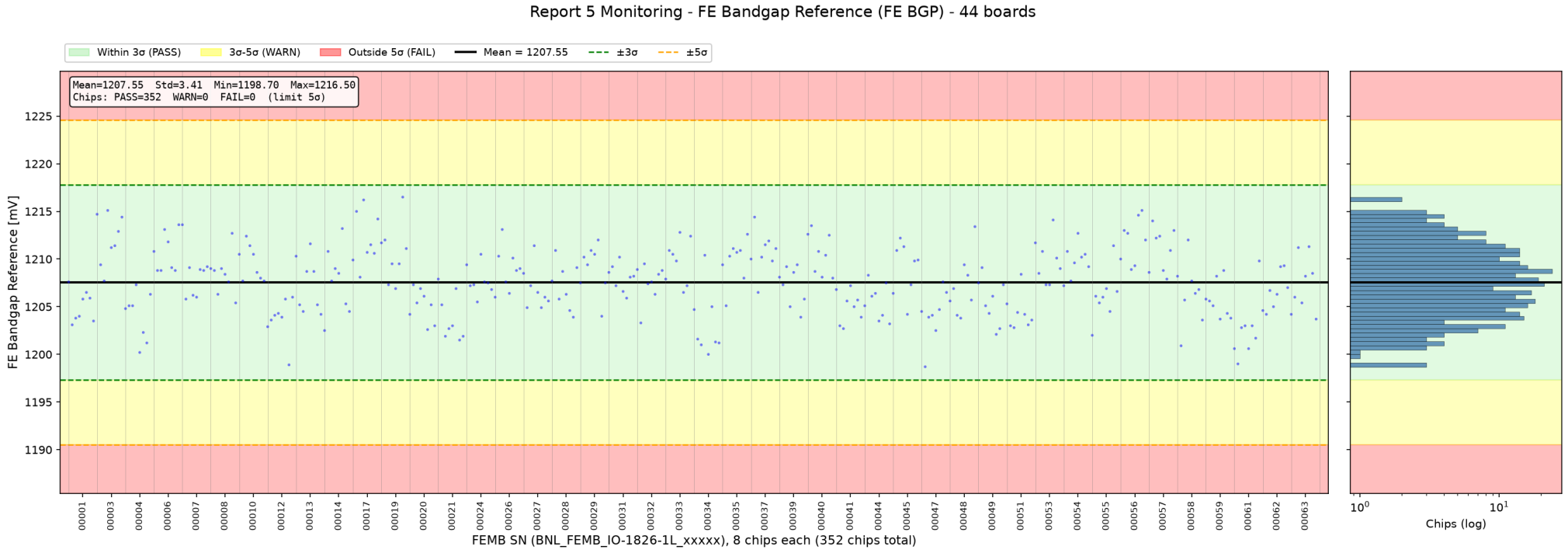


1 chip(s) beyond 3σ:

FEMB SN	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00061	1	882.30	3.2σ	WARN

4.2 FE Bandgap Reference (`FE BGP`)

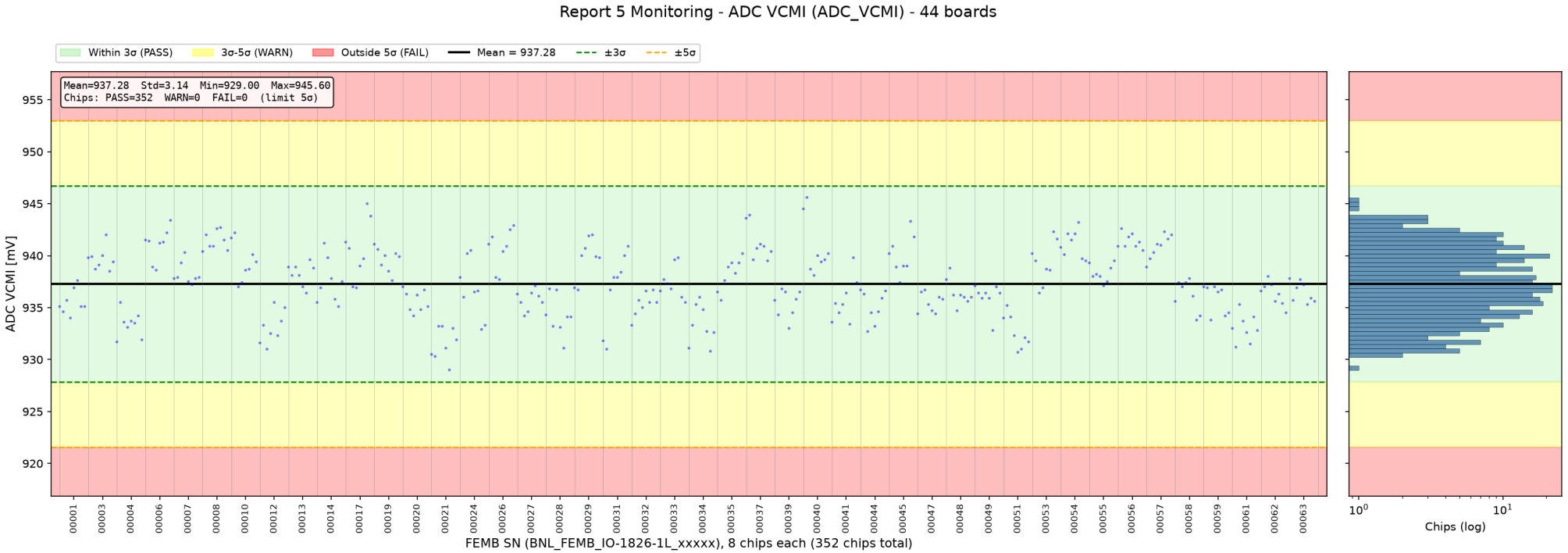
Mean = 1207.55, σ = 3.41, range [1198.70, 1216.50] mV. Spread of per-board means: σ = 2.54; spread of per-chip means across boards: σ = 0.31.



All chips within 3σ.

4.3 ADC VCMI (`ADC_VCMI`)

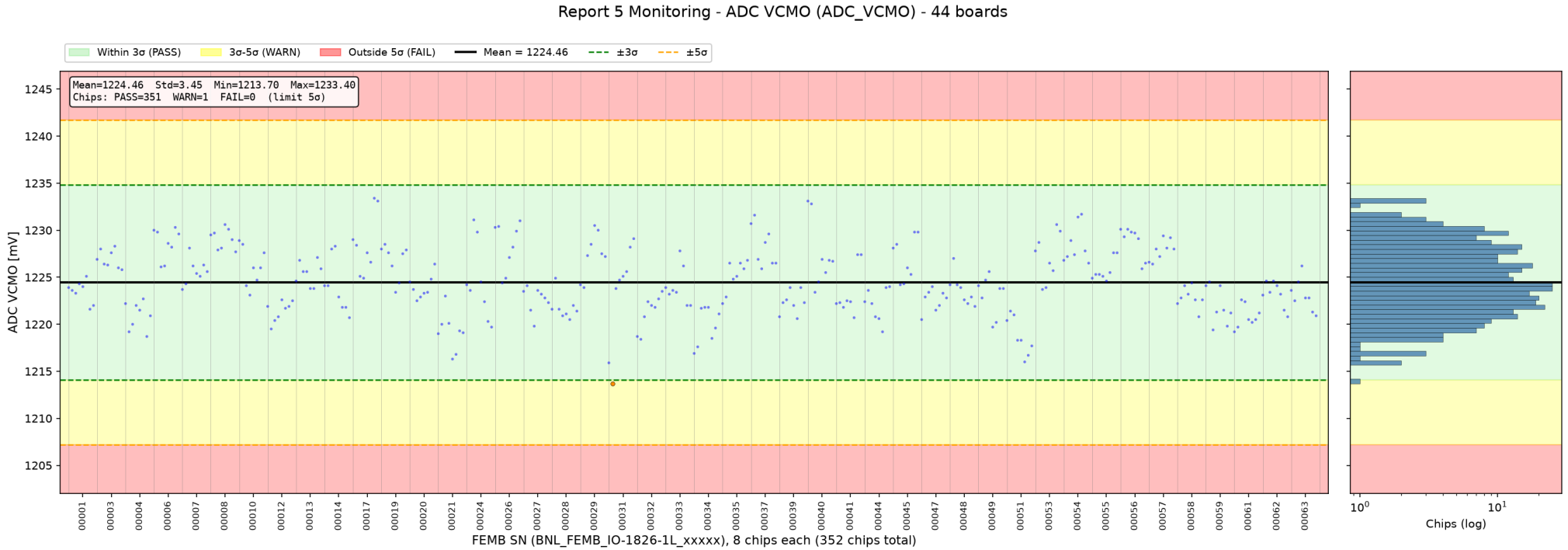
Mean = 937.28, σ = 3.14, range [929.00, 945.60] mV. Spread of per-board means: σ = 2.72; spread of per-chip means across boards: σ = 0.20.



All chips within 3σ.

4.4 ADC VCMO (`ADC_VCMO`)

Mean = 1224.46, σ = 3.45, range [1213.70, 1233.40] mV. Spread of per-board means: σ = 2.82; spread of per-chip means across boards: σ = 0.20.

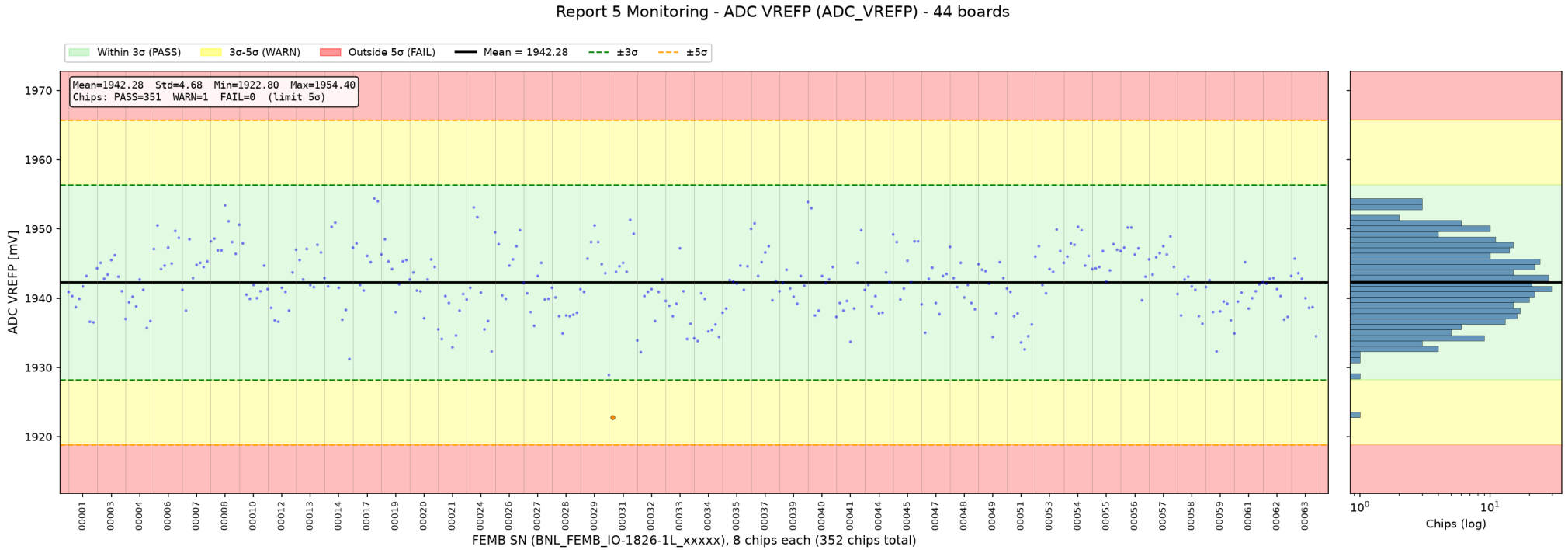


1 chip(s) beyond 3σ:

FEMB SN	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	1	1213.70	3.1σ	WARN

4.5 ADC VREFP (`ADC_VREFP`)

Mean = 1942.28, σ = 4.68, range [1922.80, 1954.40] mV. Spread of per-board means: σ = 3.12; spread of per-chip means across boards: σ = 0.26.

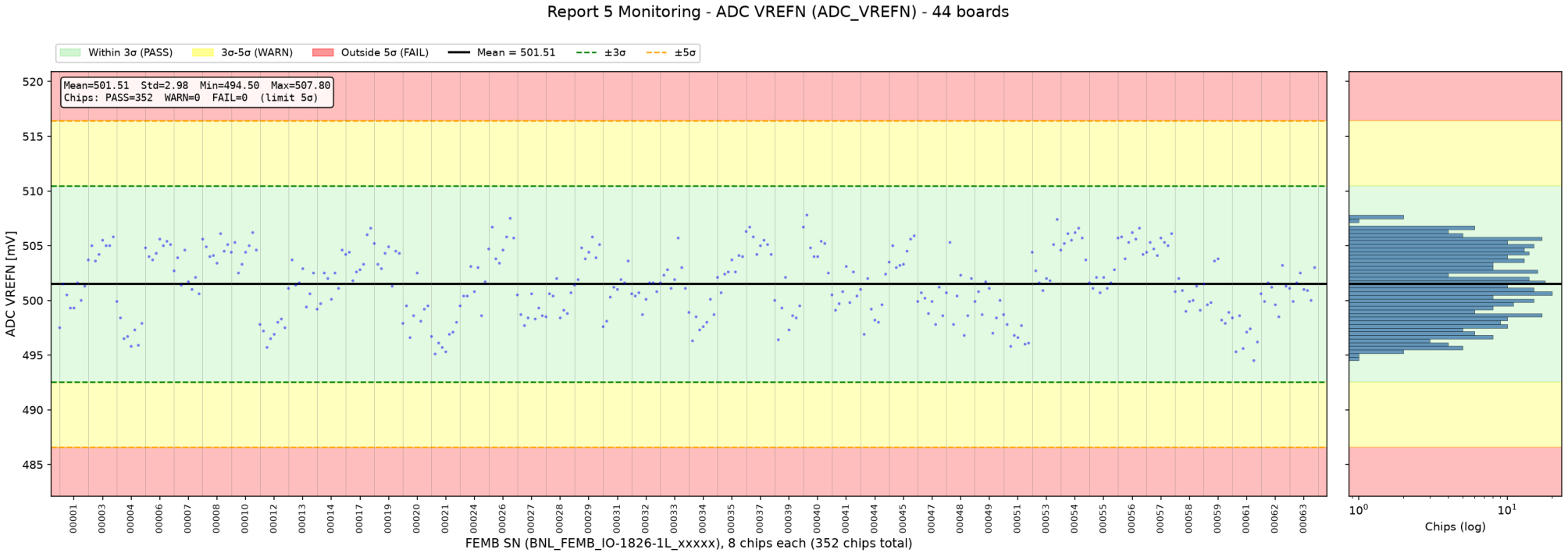


1 chip(s) beyond 3σ:

FEMB SN	Chip	Value	Deviation	Level
BNL_FEMB_IO-1826-1L_00031	1	1922.80	4.2σ	WARN

4.6 ADC VREFN (`ADC_VREFN`)

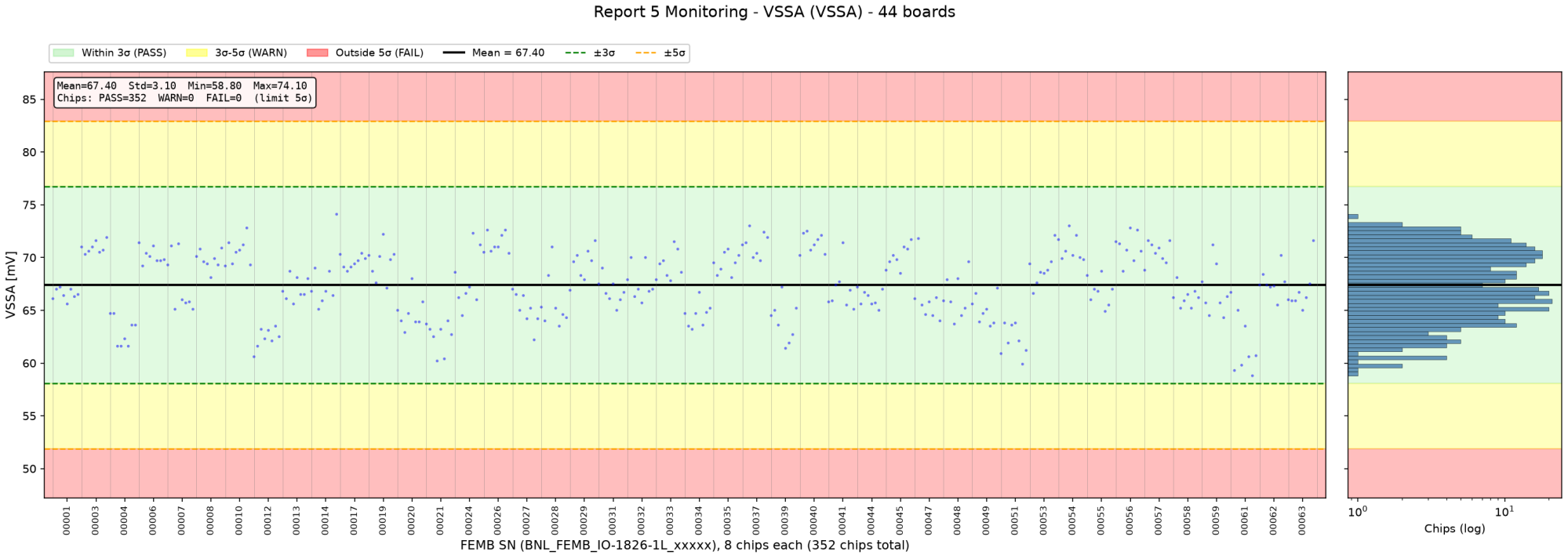
Mean = 501.51, σ = 2.98, range [494.50, 507.80] mV. Spread of per-board means: σ = 2.67; spread of per-chip means across boards: σ = 0.25.



All chips within 3σ.

4.7 VSSA (`VSSA`)

Mean = 67.40, σ = 3.10, range [58.80, 74.10] mV. Spread of per-board means: σ = 2.71; spread of per-chip means across boards: σ = 0.31.



All chips within 3σ.

5. Observations

- **FE T**: no chip beyond 5σ.
- **FE BGP**: no chip beyond 5σ.
- **ADC_VCMI**: no chip beyond 5σ.
- **ADC_VCMO**: no chip beyond 5σ.
- **ADC_VREFP**: no chip beyond 5σ.
- **ADC_VREFN**: no chip beyond 5σ.
- **VSSA**: no chip beyond 5σ.
- `FE T` is a temperature-sensor voltage: board-to-board shifts can come from the room / board temperature at test time rather than from the chip, so a whole-board offset on `FE T` alone is not by itself a chip defect.

Appendix A: Per-Board Judgement

Each cell: number of chips WARN / FAIL, and worst deviation.

FEMB SN	Test time	FE_T	FE_BGP	ADC_VCMI	ADC_VCMO	ADC_VREFP	ADC_VREFN	VSSA	Worst
BNL_FEMB_IO-1826-1L_00001	2026-07-28 15:21	0/0 (1.0σ)	0/0 (1.3σ)	0/0 (1.0σ)	0/0 (0.8σ)	0/0 (1.2σ)	0/0 (1.3σ)	0/0 (0.6σ)	PASS
BNL_FEMB_IO-1826-1L_00003	2026-07-28 15:21	0/0 (2.0σ)	0/0 (2.2σ)	0/0 (1.5σ)	0/0 (1.1σ)	0/0 (0.8σ)	0/0 (1.4σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00004	2026-08-03 22:20	0/0 (2.4σ)	0/0 (2.2σ)	0/0 (1.8σ)	0/0 (1.7σ)	0/0 (1.4σ)	0/0 (1.9σ)	0/0 (1.9σ)	PASS
BNL_FEMB_IO-1826-1L_00006	2026-07-28 20:07	0/0 (1.2σ)	0/0 (1.8σ)	0/0 (1.9σ)	0/0 (1.7σ)	0/0 (1.8σ)	0/0 (1.4σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00007	2026-09-11 18:46	0/0 (1.8σ)	0/0 (1.8σ)	0/0 (1.0σ)	0/0 (1.1σ)	0/0 (1.3σ)	0/0 (1.0σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00008	2026-07-28 20:24	0/0 (1.9σ)	0/0 (1.5σ)	0/0 (1.7σ)	0/0 (1.8σ)	0/0 (2.4σ)	0/0 (1.5σ)	0/0 (1.1σ)	PASS
BNL_FEMB_IO-1826-1L_00010	2026-07-28 00:58	0/0 (1.3σ)	0/0 (1.4σ)	0/0 (1.6σ)	0/0 (1.3σ)	0/0 (1.8σ)	0/0 (1.6σ)	0/0 (1.7σ)	PASS
BNL_FEMB_IO-1826-1L_00012	2026-07-29 22:28	0/0 (2.9σ)	0/0 (2.5σ)	0/0 (2.0σ)	0/0 (1.4σ)	0/0 (1.2σ)	0/0 (1.9σ)	0/0 (2.2σ)	PASS
BNL_FEMB_IO-1826-1L_00013	2026-07-28 14:53	0/0 (0.8σ)	0/0 (1.2σ)	0/0 (0.7σ)	0/0 (0.8σ)	0/0 (1.2σ)	0/0 (0.7σ)	0/0 (0.6σ)	PASS
BNL_FEMB_IO-1826-1L_00014	2026-07-28 20:07	0/0 (1.3σ)	0/0 (1.7σ)	0/0 (1.2σ)	0/0 (1.1σ)	0/0 (2.4σ)	0/0 (1.0σ)	0/0 (2.2σ)	PASS
BNL_FEMB_IO-1826-1L_00017	2026-09-11 19:03	0/0 (2.3σ)	0/0 (2.5σ)	0/0 (2.5σ)	0/0 (2.6σ)	0/0 (2.6σ)	0/0 (1.7σ)	0/0 (1.0σ)	PASS
BNL_FEMB_IO-1826-1L_00019	2026-07-28 15:07	0/0 (2.5σ)	0/0 (2.6σ)	0/0 (1.2σ)	0/0 (1.2σ)	0/0 (1.3σ)	0/0 (1.2σ)	0/0 (1.5σ)	PASS
BNL_FEMB_IO-1826-1L_00020	2026-09-11 19:03	0/0 (0.7σ)	0/0 (1.5σ)	0/0 (1.0σ)	0/0 (0.6σ)	0/0 (1.1σ)	0/0 (1.6σ)	0/0 (1.5σ)	PASS
BNL_FEMB_IO-1826-1L_00021	2026-07-29 22:36	0/0 (2.6σ)	0/0 (1.8σ)	0/0 (2.6σ)	0/0 (2.4σ)	0/0 (2.0σ)	0/0 (2.1σ)	0/0 (2.3σ)	PASS
BNL_FEMB_IO-1826-1L_00024	2026-09-11 19:03	0/0 (1.6σ)	0/0 (0.9σ)	0/0 (1.4σ)	0/0 (1.9σ)	0/0 (2.3σ)	0/0 (1.0σ)	0/0 (1.6σ)	PASS
BNL_FEMB_IO-1826-1L_00026	2026-07-28 15:21	0/0 (1.2σ)	0/0 (1.6σ)	0/0 (1.8σ)	0/0 (1.9σ)	0/0 (1.6σ)	0/0 (2.0σ)	0/0 (1.7σ)	PASS
BNL_FEMB_IO-1826-1L_00027	2026-09-11 18:46	0/0 (1.1σ)	0/0 (1.1σ)	0/0 (1.0σ)	0/0 (1.3σ)	0/0 (1.3σ)	0/0 (1.3σ)	0/0 (1.7σ)	PASS
BNL_FEMB_IO-1826-1L_00028	2026-07-28 00:03	0/0 (1.3σ)	0/0 (1.1σ)	0/0 (2.0σ)	0/0 (1.1σ)	0/0 (1.6σ)	0/0 (1.0σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00029	2026-07-28 19:55	0/0 (1.0σ)	0/0 (1.3σ)	0/0 (1.5σ)	0/0 (1.8σ)	0/0 (1.8σ)	0/0 (1.4σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00031	2026-07-28 00:58	0/0 (1.4σ)	0/0 (0.8σ)	0/0 (2.0σ)	1/0 (3.1σ)	1/0 (4.2σ)	0/0 (1.3σ)	0/0 (0.8σ)	WARN
BNL_FEMB_IO-1826-1L_00032	2026-07-28 15:36	0/0 (0.7σ)	0/0 (1.2σ)	0/0 (1.3σ)	0/0 (1.8σ)	0/0 (2.2σ)	0/0 (0.9σ)	0/0 (0.8σ)	PASS
BNL_FEMB_IO-1826-1L_00033	2026-09-17 22:09	0/0 (2.4σ)	0/0 (1.5σ)	0/0 (0.8σ)	0/0 (1.0σ)	0/0 (1.7σ)	0/0 (1.4σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00034	2026-09-17 22:09	0/0 (1.6σ)	0/0 (2.2σ)	0/0 (2.1σ)	0/0 (2.2σ)	0/0 (1.8σ)	0/0 (1.7σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00035	2026-09-11 18:46	0/0 (1.0σ)	0/0 (1.5σ)	0/0 (0.9σ)	0/0 (0.7σ)	0/0 (0.9σ)	0/0 (0.9σ)	0/0 (1.1σ)	PASS
BNL_FEMB_IO-1826-1L_00037	2026-07-28 00:03	0/0 (1.6σ)	0/0 (2.0σ)	0/0 (2.1σ)	0/0 (2.1σ)	0/0 (1.8σ)	0/0 (1.7σ)	0/0 (1.8σ)	PASS
BNL_FEMB_IO-1826-1L_00039	2026-07-28 20:41	0/0 (0.9σ)	0/0 (1.1σ)	0/0 (1.4σ)	0/0 (1.1σ)	0/0 (0.7σ)	0/0 (1.7σ)	0/0 (1.9σ)	PASS
BNL_FEMB_IO-1826-1L_00040	2026-07-28 20:41	0/0 (1.9σ)	0/0 (1.7σ)	0/0 (2.6σ)	0/0 (2.5σ)	0/0 (2.5σ)	0/0 (2.1σ)	0/0 (1.6σ)	PASS
BNL_FEMB_IO-1826-1L_00041	2026-07-28 20:24	0/0 (0.7σ)	0/0 (1.4σ)	0/0 (1.2σ)	0/0 (1.1σ)	0/0 (1.8σ)	0/0 (0.8σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00044	2026-07-28 15:07	0/0 (1.0σ)	0/0 (1.3σ)	0/0 (1.5σ)	0/0 (1.5σ)	0/0 (1.0σ)	0/0 (1.5σ)	0/0 (0.8σ)	PASS
BNL_FEMB_IO-1826-1L_00045	2026-07-28 14:53	0/0 (1.2σ)	0/0 (1.4σ)	0/0 (1.9σ)	0/0 (1.5σ)	0/0 (1.5σ)	0/0 (1.5σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00047	2026-07-27 23:48	0/0 (1.1σ)	0/0 (2.6σ)	0/0 (0.9σ)	0/0 (1.1σ)	0/0 (1.6σ)	0/0 (1.2σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00048	2026-07-28 14:53	0/0 (1.2σ)	0/0 (1.7σ)	0/0 (0.8σ)	0/0 (0.7σ)	0/0 (1.1σ)	0/0 (1.6σ)	0/0 (1.2σ)	PASS
BNL_FEMB_IO-1826-1L_00049	2026-07-28 19:55	0/0 (1.1σ)	0/0 (1.6σ)	0/0 (1.4σ)	0/0 (1.4σ)	0/0 (1.7σ)	0/0 (1.5σ)	0/0 (1.3σ)	PASS
BNL_FEMB_IO-1826-1L_00051	2026-10-15 17:19	0/0 (2.6σ)	0/0 (1.4σ)	0/0 (2.1σ)	0/0 (2.5σ)	0/0 (2.1σ)	0/0 (1.9σ)	0/0 (2.4σ)	PASS
BNL_FEMB_IO-1826-1L_00053	2026-07-28 15:36	0/0 (1.6σ)	0/0 (1.9σ)	0/0 (1.6σ)	0/0 (1.8σ)	0/0 (1.6σ)	0/0 (2.0σ)	0/0 (1.5σ)	PASS

FEMB SN	Test time	FE_T	FE_BGP	ADC_VCMI	ADC_VCMO	ADC_VREFP	ADC_VREFN	VSSA	Worst
BNL_FEMB_IO-1826-1L_00054	2026-07-28 00:58	0/0 (1.3σ)	0/0 (1.5σ)	0/0 (1.9σ)	0/0 (2.1σ)	0/0 (1.7σ)	0/0 (1.7σ)	0/0 (1.8σ)	PASS
BNL_FEMB_IO-1826-1L_00055	2026-07-28 00:58	0/0 (1.6σ)	0/0 (1.6σ)	0/0 (0.7σ)	0/0 (0.9σ)	0/0 (1.2σ)	0/0 (0.4σ)	0/0 (0.8σ)	PASS
BNL_FEMB_IO-1826-1L_00056	2026-07-28 20:24	0/0 (1.9σ)	0/0 (2.2σ)	0/0 (1.7σ)	0/0 (1.6σ)	0/0 (1.7σ)	0/0 (1.7σ)	0/0 (1.7σ)	PASS
BNL_FEMB_IO-1826-1L_00057	2026-07-28 15:07	0/0 (2.1σ)	0/0 (1.9σ)	0/0 (1.6σ)	0/0 (1.4σ)	0/0 (1.4σ)	0/0 (1.5σ)	0/0 (1.4σ)	PASS
BNL_FEMB_IO-1826-1L_00058	2026-09-17 22:09	0/0 (2.1σ)	0/0 (2.0σ)	0/0 (1.1σ)	0/0 (1.1σ)	0/0 (1.3σ)	0/0 (0.8σ)	0/0 (0.7σ)	PASS
BNL_FEMB_IO-1826-1L_00059	2026-08-03 18:53	0/0 (1.3σ)	0/0 (1.1σ)	0/0 (1.1σ)	0/0 (1.5σ)	0/0 (2.1σ)	0/0 (1.2σ)	0/0 (1.2σ)	PASS
BNL_FEMB_IO-1826-1L_00061	2026-10-09 17:53	1/0 (3.2σ)	0/0 (2.5σ)	0/0 (1.9σ)	0/0 (1.5σ)	0/0 (1.6σ)	0/0 (2.3σ)	0/0 (2.8σ)	WARN
BNL_FEMB_IO-1826-1L_00062	2026-07-27 23:48	0/0 (1.1σ)	0/0 (1.0σ)	0/0 (0.9σ)	0/0 (1.1σ)	0/0 (1.1σ)	0/0 (1.0σ)	0/0 (0.9σ)	PASS
BNL_FEMB_IO-1826-1L_00063	2026-07-28 19:55	0/0 (1.0σ)	0/0 (1.1σ)	0/0 (0.6σ)	0/0 (1.0σ)	0/0 (1.7σ)	0/0 (0.5σ)	0/0 (1.4σ)	PASS

Appendix B: Output Files

- `output/criteria_5.csv` – mean, σ and 3σ/5σ limits per parameter
- `output/judgement_5_chips.csv` – every chip beyond 3σ (board, chip, value, deviation, level)
- `output/judgement_5_boards.csv` – per-board counts and worst level
- `output/plot_5_*.png` – distribution plots and per-board summary